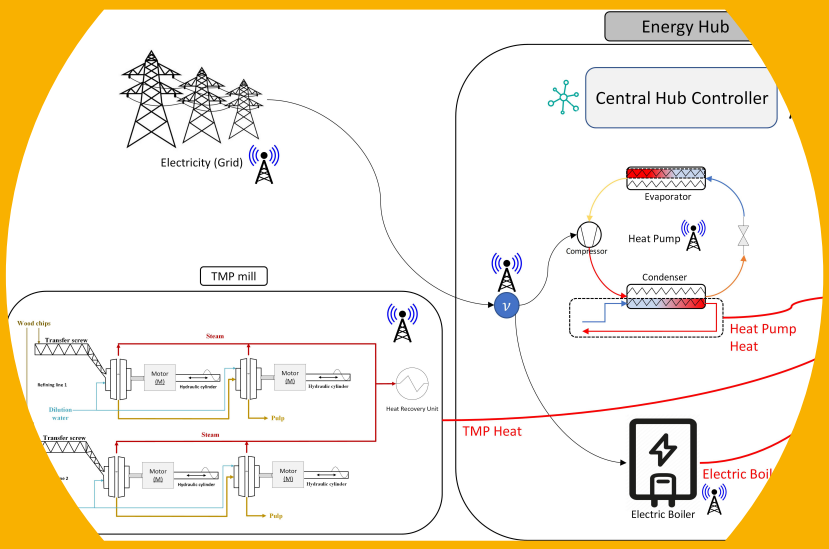
Department of Mechanical Engineering

# Parametric Models for Forest Industry Transformation in Energy Efficiency: Machine Learning Approach

Behnam Talebjedi




Aalto University

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# Parametric Models for Forest Industry Transformation in Energy Efficiency: Machine Learning Approach

**Behnam Talebjedi**

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**Department of Mechanical Engineering**
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# Abstract

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**Abstract**

This thesis is based on industrial projects with Pulp and Paper industry in a Nordic country. The main focus of the thesis is on the energy efficiency development of the thermomechanical pulp (TMP) mill and optimal integration of the TMP mill and paper machine through heat recovery and the concept of an Energy Hub. Advanced statistical approaches and machine learning methods have been employed to develop refining identification models and advanced energy-saving refining optimization methods for the TMP process. Results prove that an accurate refining identification model could be developed through advanced machine learning methods. The refining identification models to predict the refining energy (such as specific energy consumption) and final pulp quality (such as freeness and fiber length) can be further used to develop a refining control and optimization strategy. The developed optimization strategy based on the integration of Machine learning methods and Genetic optimization algorithm confirms an average reduction of 14 % for the total refining-specific energy consumption.

In the following, the optimal integration of the TMP mill and paper machine has been investigated through the Energy Hub (EH) concept. The proposed approach for the cost and energy-efficient design and operation of EH is based on the integration of thermo-economic analysis, reliability and availability analysis, and EH load prediction. The proposed approach was first introduced and evaluated for the energy and cost-efficient design of a combined cooling, heating, and power (CCHP) system that provides the hourly thermal demand of a high-rise residential building. Results prove that by utilizing the proposed method, the system's average total cost could be reduced by 16% during the system's lifespan. As the presented method has shown to be effective in residential EH applications, this method was examined in a second case study (the forest industry) to determine the optimal integration of TMP mill and paper machines. The proposed design method offers a robust design that isn't impacted by penalty rates of unsupplied demand. Depending on the penalty rates, the total system cost could decrease by 14%-28% utilizing the proposed design method.

**Keywords** Pulp & Paper Industry, Thermomechanical Pulping Process, Machine Learning, Energy Hub

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**Tiivistelmä**

Tämä väitöskirja perustuu teollisiin projekteihin sellu- ja paperiteollisuuden kanssa Pohjoismaassa. Työn pääpaino on termomekaanisen sellutehtaan (TMP) energiatehokkuuden kehittämisessä sekä TMP-tehtaan ja paperikoneen optimaalisessa integroinnissa lämmön talteenoton ja Energy Hub -konseptin avulla. Kehittyneitä tilastollisia menetelmiä ja koneoppimismenetelmiä on käytetty kehittämään jauhamisprosessiin ja kehittyneitä energiaa säästäviä jauhatusprosessin optimointimenetelmiä TMP-prosessiin. Tulokset osoittavat, että tarkka jauhatusprosessin identifiointimalli voitaisiin kehittää edistyneillä koneoppimismenetelmillä. Jauhatusenergian (kuten ominaisenergiankulutuksen) ja lopullisen massan laadun (kuten suotavuus ja kuidun pituus) ennustamiseen perustuvia jauhatusmalleja voidaan edelleen käyttää jauhatuksen ohjaus- ja optimointistrategian kehittämiseen. Tässä tutkimuksessa kehitetty optimointistrategia, joka perustuu koneoppimismenetelmien ja geneettisen optimointialgoritmin integrointiin, vahvistaa keskimääräisen 14 % vähennyksen jauhatusprosessin kokonaisenergiankulutuksessa. Seuraavassa TMP-tehtaan ja paperikoneen optimaalista integrointia on selvitetty Energy Hub (EH) -konseptin kautta. Ehdotettu lähestymistapa EH:n kustannus- ja energiatehokkaaseen suunnitteluun ja käyttöön perustuu lämpötaloudellisen analyysin, luotettavuus- ja käytettävyysanalyysin sekä EH-kuormituksen ennusteen yhdistämiseen. Ehdotettu lähestymistapa otettiin käyttöön ja arvioitiin ensimmäisen kerran energia- ja kustannustehokkaassa jäähdytys-, lämmitys- ja sähköjärjestelmän (CCHP) suunnittelussa, joka kattaa kerrostalon asuinrakennuksen tunnittaisen lämmitys- ja jäähdytystarpeen. Tulokset osoittavat, että ehdotettua menetelmää hyödyntämällä järjestelmän keskimääräisiä kokonaiskustannuksia voitaisiin vähentää 16% järjestelmän elinkaaren aikana. Koska esitetty menetelmä on osoittautunut tehokkaaksi asuinrakennusten EH-sovelluksissa, tätä menetelmää tarkasteltiin toisessa tapaustutkimuksessa (metsäteollisuus) TMP-tehtaan ja paperikoneiden optimaalisen integroinnin määrittämiseksi. Ehdotettu suunnittelumenetelmä tarjoaa vankan rakenteen, johon toimittamattoman kysynnän rangaistuskustannukset eivät vaikuta. Rangaistuskustannuksista riippuen järjestelmän kokonaiskustannukset voivat laskea 14–28 % ehdotetulla suunnittelumenetelmällä.

**Avainsanat** sellu- ja paperiteollisuus, termomekaaninen selluprosessi, koneoppiminen, energiakeskus

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Helsinki, 28 June 2023
Behnam Talebjedi

# Abstract

This thesis is based on industrial projects with big Pulp & Paper industries in a Nordic country. The main focus of the thesis is on the energy efficiency development of the thermomechanical pulp (TMP) mill and optimal integration of the TMP mill and paper machine through heat recovery and the concept of an Energy Hub. Advanced statistical approaches and machine learning methods have been employed to develop refining identification models and advanced energy-saving refining optimization methods for the TMP process. Results prove that an accurate refining identification model could be developed through advanced machine learning methods. The refining identification models to predict the refining energy (such as specific energy consumption) and final pulp quality (such as freeness and fiber length) can be further used to develop a refining control and optimization strategy. The developed optimization strategy in this research which is based on the integration of Machine learning methods and Genetic optimization algorithm confirms the average reduction of 14 % for total refining-specific energy consumption.

In the following, the optimal integration of the TMP mill and paper machine has been investigated through the Energy Hub (EH) concept. The proposed approach for the cost and energy-efficient design and operation of EH is based on the integration of thermo-economic analysis, reliability & availability analysis, and EH load prediction. The proposed approach was first introduced and evaluated for the energy and cost-efficient design of a combined cooling, heating, and power (CCHP) system that provides the hourly thermal demand of the high-rise residential building. Results prove that by utilizing the proposed method, the system's average total cost could be reduced by 16% during the system's lifespan. As the presented method has shown to be effective in residential EH applications, this method was examined in a second case study (the forest industry) to determine the optimal integration of TMP mill and paper machines. The proposed design method offers a robust design that isn't impacted by unsupplied demand penalty rates. Depending on the penalty rates, the total system cost could decrease by 14%-28% utilizing the proposed design method.

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# List of Abbreviations and Symbols

| | |
|---|---|
| TMP | Thermo-mechanical pulping |
| MLPNN | Multilayer perceptron neural network |
| ANFIS | Adaptive neuro-fuzzy inference system |
| EH | Energy Hub |
| CCHP | combined cooling, heating, and power system |
| LSTM | Long-short-term memory |
| SSM | State-space method |
| CSF | Canadian Standard Freeness |
| FL | Fiber length |
| RNN | Recurrent Neural Network |
| MAPE | Mean absolute percentage error |
| MPC | Model predictive control |
| RNN | Recurrent Neural Networks |
| GA | Genetic Algorithm |
| PSO | Particle swarm optimization |
| TLBO | Teaching Learning-Based Optimization Algorithm |
| ACO | Ant Colony Optimization Algorithm |
| DE | Differential Evolution Algorithm |
| BBO | Biogeography-Based Optimization Algorithm |
| RMSE | Root mean square error |
| MSE | mean square error |
| MAPE | Mean absolute percentage error |

# List of Publications

This doctoral dissertation consists of a summary of the following publications which are referred to in the text by their numerals

**1.** Talebjedi, B.; Khosravi, A.; Laukkanen, T.; Holmberg, H.; Vakkilainen, E.; Syri, S. Energy Modeling of a Refiner in Thermo-Mechanical Pulping Process Using ANFIS Method. Energies, 2020, 13, 5113. https://doi.org/10.3390/en13195113

**2.** Talebjedi, B.; Laukkanen, T.; Holmberg, H.; Vakkilainen, E.; Syri, S. Energy Efficiency Analysis of the Refining Unit in Thermo-Mechanical Pulp Mill. Energies, 2021, 14, 1664. https://doi.org/10.3390/en14061664

**3.** Talebjedi, B.; Laukkanen, T.; Holmberg, H.; Vakkilainen, E.; Syri, S. Energy simulation and variable analysis of refining process in thermo-mechanical pulp mill using machine learning approach. *Mathematical and Computer Modelling of Dynamical Systems,* 2021, 27(1), 562-585. https://doi.org/10.1080/13873954.2021.1990967

**4.** Talebjedi, B.; Laukkanen, T.; Holmberg, H.; Vakkilainen, E.; Syri, S. Advanced Energy-Saving Optimization Strategy in Thermo-mechanical Pulping by Machine Learning Approach. *Nordic Pulp & Paper Research Journal,* 2022, 37 (3), 434-452. https://doi.org/10.1515/npprj-2022-0013

**5.** Talebjedi, B.; Behbahaninia, A. Availability analysis of an Energy Hub with CCHP system for economical design in terms of Energy Hub operator. *Journal of Building Engineering*. 2021, 33, 101564. https://doi.org/10.1016/j.jobe.2020.101564

**6.** Talebjedi, B.; Laukkanen, T.; Holmberg, H.; Syri, S. 2023. Advanced design and operation of Energy Hub for forest industry using reliability assessment. *Applied Thermal Engineering, 2023, 230, 120751.* https://doi.org/10.1016/j.applthermaleng.2023.120751

# Author's Contribution

**Publication 1:** Energy Modeling of a Refiner in Thermo-Mechanical Pulping Process Using ANFIS Method.

B. Talebjedi: Conceptualization, methodology, software, writing—original draft, formal analysis, writing—review & editing. A. Khosravi: Conceptualization, methodology, software, writing—original draft.T. Laukkanen and H. Holmberg: data curation, funding acquisition, project administration, supervision, writing—review & editing. E. Vakkilainen: funding acquisition, writing—review & editing. S. Syri: funding acquisition, supervision, project administration, writing—review & editing.

**Publication 2:** Energy Efficiency Analysis of the Refining Unit in Thermo-Mechanical Pulp Mill.

B. Talebjedi: Conceptualization, methodology, software, writing—original draft, formal analysis, writing—review & editing. T. Laukkanen and H. Holmberg: data curation, funding acquisition, project administration, supervision, writing—review & editing. E. Vakkilainen: funding acquisition, writing—review & editing. S. Syri: funding acquisition, supervision, project administration, writing—review & editing.

**Publication 3:** Energy simulation and variable analysis of refining process in thermo-mechanical pulp mill using machine learning approach.

B. Talebjedi: Conceptualization, methodology, software, writing—original draft, formal analysis, writing—review & editing. T. Laukkanen and H. Holmberg: data curation, funding acquisition, project administration, supervision, writing—review & editing. E. Vakkilainen: funding acquisition, writing—review & editing. S. Syri: funding acquisition, supervision, project administration, writing—review & editing.

**Publication 4:** Advanced Energy-Saving Optimization Strategy in Thermo-mechanical Pulping by Machine Learning Approach.

B. Talebjedi: Conceptualization, methodology, software, writing—original draft, formal analysis, writing—review & editing. T. Laukkanen and H. Holmberg: data curation, funding acquisition, project administration, supervision, writing—review & editing. E. Vakkilainen: funding acquisition, writing—review & editing. S. Syri: funding acquisition, supervision, project administration, writing—review & editing.

**Publication 5:** Availability analysis of an Energy Hub with CCHP system for economical design in terms of Energy Hub operator.

B. Talebjedi: Investigation, Formal analysis, Methodology, Software, Validation, Writing - original draft. A. Behbahaninia: Conceptualization, Writing - review & editing, Supervision.

**Publication 6:** Advanced design and operation of Energy Hub for forest industry using reliability assessment.

B. Talebjedi: Conceptualization, methodology, software, writing—original draft, formal analysis, writing—review & editing. T. Laukkanen and H. Holmberg: data curation, funding acquisition, project administration, supervision, writing—review & editing. S. Syri: funding acquisition, supervision, project administration, writing—review & editing.

# 1. Introduction

## 1.1 Background

Energy efficiency analysis and improvement are the most important issues for energy-dependent industries for the transition toward a more sustainable and energy-efficient industry. Most often, the level of production and profitability of the industries are strongly affected by the reliability of energy supply as well as fluctuations in energy prices. The pulp and paper industry is in fourth place as the most extensive industrial energy consumer worldwide. The pulp and paper industry contributes to approximately 6% and 2% percent of final industrial energy consumption and carbon dioxide emissions, respectively [1]. Today, the most dominant mechanical pulping process is the Thermo-mechanical pulping process (TMP). Fig. 1 shows a typical flow diagram of a TMP plant. This process is one of the most energy-consuming processes in the pulp and paper industry, with an energy efficiency of 10–15%. Due to the recent increase in the electricity cost in some countries, the TMP process profitability has been heavily affected because of the massive electrical energy consumption. Moreover, environmental considerations and carbon emissions taxes have obligated pulp mills to reduce their energy consumption and carbon footprint. Furthermore, more improved and uniform pulp quality is required for pulp mills to keep their position in the market share. Refining is the main process in the thermo-mechanical pulp mill in which the fiber is treated to achieve the desired quality for the stocks sent to the paper mill. Different approaches to increase refining energy efficiency include optimizing the chip pre-heating condition, improving heat integration, using more appropriate refining plates by optimizing plate patterns geometry, advanced process optimization, and control techniques [2].

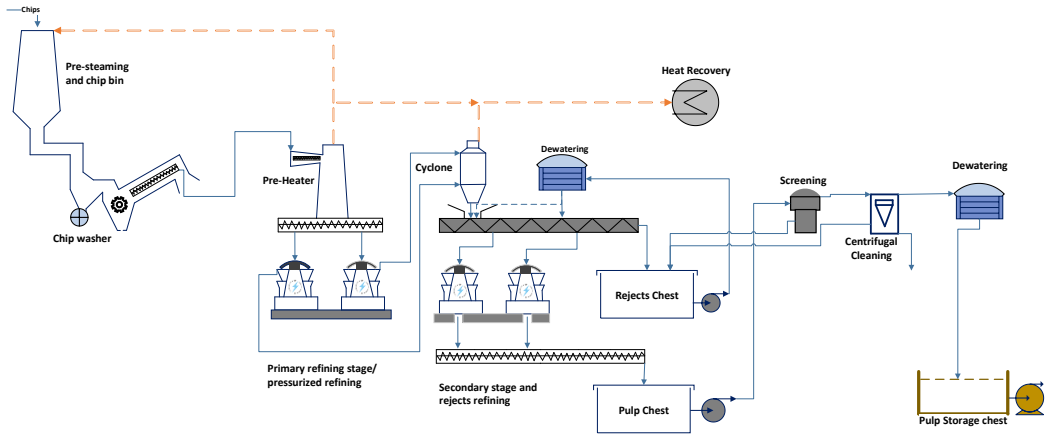



**Fig. 1.** Sample layout of the Thermo-mechanical pulp (TMP) mill.

The key to developing an energy-saving optimization strategy in the thermomechanical pulping process is constructing the refining identification model. Since it is not clear how the energy transforms into pulp in the chip refiner, developing an accurate refining identification model is challenging. For example, it is demanding to estimate the impact of refining disturbance variables such as plate condition and the quality of the input wood chips on the refining process's performance. The quality of the wood chips, including the chip species, dramatically affects the pulping process and end-product characteristics [3]. Several studies address the seasonal variations in wood chips properties and their effect on final pulp quality and energy-saving opportunities [4]–[6]. The artificial neural network (ANN) concept is a powerful data-driven method for developing process identification models and pattern recognition in variable relationships when the hidden pattern is complex and hard to present. Several researches [7], [8] address the superiority of the neural network over simple regression models. These models can be utilized to develop accurate refining identification models in the thermo-mechanical pulping process.

This thesis investigates developing refining identification models by mechanistic approach as well as advanced machine learning methods. To maximize TMP mill energy efficiency while maintaining the quality of the final pulp, identification models are utilized to develop an advanced energy-saving refining optimization strategy. Different machine learning methods such as Adaptive neuro-fuzzy inference system (ANFIS) and Multi-layer perceptron neural network (MLPNN) are employed to predict the refining motor load, specific energy consumption, and final pulp quality based on the refining variables. The MATLAB Simulink library model for the thermomechanical pulping process which is based on the combination of the mechanistic and data-driven methods is developed in this thesis for longer-term refining motor load prediction.

Energy efficiency development in industrial processes can have a great effect on reducing costs, greenhouse gas emissions, and primary energy consumption. Heat recovery, advanced control strategy, and sustainable energy network are examples of methods to increase the efficiency of industrial processes. Multi-carrier energy systems can be optimized with the help of an Energy Hub (EH), which is a new and promising concept. Several energy carriers are converted, stored, and supplied through the EH system for multiple energy types, meeting challenges such as energy consumption and resources, multiple energy infrastructure, and flexibility. Research work on the EH has begun over the last few years following its introduction in 2007 [9]. EHs facilitate the move towards sustainable multi-energy systems and achieve the realization of energy system models [10]. Furthermore, the development and optimization of process control systems can contribute greatly to increasing productivity, reducing production costs, and high-quality final products. The Energy Hub could be a good option to integrate a pulp mill and paper machine. The paper machine requires heat for the process and in a TMP mill, most of the supplied electrical energy is converted to heat during the refining process.

In this thesis, the advanced design and operation of EH have been introduced for a reliable, cost and energy-efficient heat supply system for both residential (high-rise residential building) and industrial (forest industry) sectors. The suggested method involves integrating the analysis of thermodynamics and economics, reliability and availability, and predictions of EH load. Combining economics and thermodynamics is crucial for allocating energy system resources, and reliability assessment consider the availability of system components during the design phase, resulting in more accurate modeling of real-life operating conditions. Load profile prediction estimates the energy demand for the upcoming hour, which aids in optimizing Energy Hub operations. For the residential case study, the application of EH consisting of a combined cooling, heating, and power (CCHP) system is investigated to supply the heating and cooling demand of a high-rise residential building and in the other study (industrial case study), the application of Energy Hub has been examined for the optimal integration of the TMP mill and paper machine through heat recovery.

To sum up, this thesis studies energy efficiency development strategies in the pulp & paper industry by focusing on the TMP process. In the first part of the thesis, the advanced energy-saving refining optimization strategy is developed based on the integration of advanced machine learning methods and heuristic optimization algorithms such as Genetic Algorithms. The refining identification models are necessary to develop refining optimization and control strategies. Therefore, in the first part of the thesis, the study of developing a refining identification model has been made. The second part of the thesis was about the energy and cost-efficient design and operation of EH for the optimal integration of the pulp mill and paper machine through heat recovery. The concept of EH was used for this purpose. The proposed approach for the energy and cost-efficient design and operation of EH is based on the integration of thermo-economic analysis, reliability & availability analysis, and EH load prediction.

## 1.2 Research Objective and thesis structure

This thesis investigates energy efficiency improvement and sustainable production in the pulp & paper industry. The main focus of this research can be divided into two parts. First, a detailed understanding of the refining process in a thermo-mechanical pulp mill will be achieved, followed by the development of a machine learning-based energy-saving optimization strategy. Developing an energy-saving refining optimization strategy requires an advanced system identification model. Since refining is a time-varying process with an intricate and multivariate nature, refining energy simulation is challenging [11]. Several factors, such as rotational speed, crossing angle, refining gap, bar dimensions, and pulp consistency, influence pulp and paper properties. In this thesis, refining energy and pulp quality identification models are developed through advanced machine learning methods. However, other refining identification models that are based on the combination of mechanistic and data-driven approaches have also been examined.

The second part of the thesis consists of the optimal integration of pulp mill and paper machine by heat recovery for cost-efficient, energy-efficient, and sustainable paper production. The concept of an EH has been introduced and utilized to design and operate a reliable and efficient heat supply system for the paper machine. The proposed approach for energy and cost-efficient design and operation of EH is based on the integration of thermo-economic analysis, reliability & availability analysis, and EH load prediction. The thermo-economic analysis combines economics and thermodynamics, which is necessary for energy system resource allocation. Reliability assessment will lead to more accurate modeling of real-life system operating conditions since system components' availability is considered in the design process. Load profile prediction estimates the energy demand for the next hour, which helps with the optimal operation of the EH.

The proposed approach for the advanced design and operation of the EH is first introduced and implemented for the residential application where the EH consists of a combined cooling, heating, and power (CCHP) system to supply the heating and cooling demand of the high-rise residential building (case study 1). As a result of its good performance, the recommended method has been used in the forest industry to integrate pulp mills and paper machines through heat recovery. The heat supply system in this case study (case study 2) comprises the TMP mill and EH which consists of the electric boiler and heat pump. Through the use of energy hubs, the paper machine's heating demand can be met stably and optimally.

The main research objective and contribution are summarised in the following

- Developing refining identification models by advanced machine learning methods.
- Developing advanced energy-saving optimization strategy in the Thermo-mechanical pulping process by integration of machine learning methods and heuristic optimization algorithms such as Genetic Algorithm (GA).

- Energy and cost-efficient integration of pulp mill and paper machine utilizing the Energy Hub concept and combination of thermo-economic analysis, reliability & availability analysis, and EH load prediction.

# 2. Literature Review and dissertation contribution

This chapter introduces past studies and research on the application of advanced machine learning methods in the Pulp & Paper industry as well as the application of Energy Hub for the sustainable, energy-efficient, and cost-efficient integration of TMP mill and paper machine. However, the literature on the application of computer science and machine learning methods in the pulp and paper industry is limited and mostly focused on chemical pulp. The literature review in this thesis, as explained in the introduction, is divided into two parts: 1- the application of advanced machine learning, data-driven, and mechanistic methods in the pulp and paper industry, and 2- the energy and cost-efficient design and operation of the EH for the optimal integration of the TMP mill and paper machine through heat recovery.

## 2.1 Advanced Energy-saving Optimization Strategy in the thermo-mechanical pulping process

Energy efficiency is one of the most critical issues in energy-intensive industries, such as the pulp and paper sector, which stands as the fourth largest industrial energy user worldwide [12], [13]. Pulp mills are trying to achieve a more sustainable production process by increasing energy efficiency and reducing energy costs to remain competitive [14]. 20 to 25% of the world's pulp production is mechanical pulp, and this figure is increasing because of the high yield ratio of the mechanical pulping process and the limitation in fiber resources [15]. In addition, a higher quality of pulp is required for pulp mills to retain their position in the competitive market [16]. Refining is the main process in the thermo-mechanical pulp (TMP) mill in which the fiber is treated to achieve the desired quality for the stocks sent to the paper mill [17]. The refiner is the core and backbone of the thermo-mechanical pulping process. In these hydraulic machines, the rotor and stator crush and beat the pulp fibers in a repeated cycle. Refiners have the same principle for operation; however, they are different in terms of configurations. Wood chips are inserted between two metal discs where one of those is rotating (single-disc refiner), and rotary energy is transferred to the pulp by patterned fragments of the plates [18]. Since the refining process is highly affected by different refining variables which are not controllable in some cases such as disturbance variables, it is not an easy task to develop an optimal refining control strategy [11]. In general, pulp and paper production is highly complicated and integrated. The process consists of several interconnected operations and recycling loops. Moreover, there are many constraints such as fuel and power prices, environmental issues, availability, the performance of equipment, production schedules, and process variability that extremely harden the process of energy optimization in these units [19].

There are several factors such as rotational speed, crossing angle, refining gap, bar dimensions, pulp consistency, etc. influencing the pulp and paper properties. Implementing all these factors into theoretical or thermodynamic equations for defining the refining process (system identification) is approximately impossible. A mathematical simulation model (identification model), which summarizes a physical process into mathematical equations, can be mechanistic, empirical, or a combina-

tion of both for a complicated process. A mechanistic model relies on physical principles and is applicable to a wide range of system operating conditions. The major problem with mechanistic models is that they often become complicated and time-consuming to develop for some processes. Empirical models are mostly based on fitting a mathematical equation to the experimental data. They are easier to produce but approximate the process over limited operating conditions. The other drawback of empirical models is that performing experimental tests is often impossible or expensive for some processes. Harianth et al. [20] developed predictive optimal control for thermo-mechanical pulping processes with multi-stage low-consistency refining. They used the MATLAB thermomechanical pulping toolbox (which is a combination of mechanistic and data-driven methods) as the refining identification model where the model limitation is discussed in this thesis. Artificial intelligence (AI) is an alternative powerful data-driven method for energy modeling complicated systems and processes. An accurate simulation and prediction of the refining process are achievable by utilizing AI methods.

The majority of the previous studies in the pulp industry have been conducted on the use of analytical methods for the TMP process analysis. Analytical methods like thermodynamic analysis and numerical methods are complicated, time-consuming, and sometimes with low accuracy and hard to employ in many practical applications for complex systems or processes. However, for the initial design of a system where no data is accessible, numerical analysis and experimental models are of high power and value. Khokhar [21] utilized numerical analysis (CFD simulation) to model a simple groove refiner. For the simulation, Khokhar considered pulp as a single-phase Newtonian fluid having 100 times more viscosity compared to water. Ping et al. [22] used ANSYS CFX to simulate a high-consistency refiner for the purpose of reducing refining energy consumption. The findings show an impressive ability of numerical analysis to investigate the torque and pressure of the refiner discs and also calculate the refining motor load. Huhtanen and Karvinen [23] studied the performance of non-Newtonian fluid dynamics in papermaking. The purpose of their research is to investigate the application of the non-Newtonian fluid models for simulating the pulp suspension in a turbulent and laminar flow. By studying inter-fiber refining and the energy consumption distribution in the refiner plate gap, Illikainen [18] obtained new information about mechanisms of thermo-mechanical pulp refining in the inner area of a refiner disc utilizing an experimental test. Elahimehr et al. [24] introduced an experimental structure to analyze the effect of manipulated refining variables such as the refiner plate gap, dilution water, plate pattern, and feeder screw speed on the LC refiner performance. Mithrush [25] experimentally investigated the transport phenomena inside a low-consistency disc refiner. The main purpose of his research is to examine the effects of the operating parameters of a refiner on the flow field. Results show a linear relationship between stator backflow velocities and the pressure differential across the refiner.

On the other hand, with the development of AI methods, analyzing complex systems with high accuracy is facilitated [26]. Machine learning is a subdivision of artificial intelligence that is currently undergoing growth and is a very active field in computer science [27]. The purpose of machine learning is to build computer systems learning from experiences and are able to adapt to their surroundings just by relying on data patterns instead. In many cases, theoretical modeling of a mechanical system is time-consuming and needs vast knowledge about the detailed processes happening in the system. On the other hand, the validation of theoretical analyzes is always controversial, and in many cases, requires costly experimental tests. Several researches [7], [8] address the superiority of the neural network to the regression models. These models can be utilized to develop accurate refining identification models in the thermo-mechanical pulping process. Ciesielski and Olejnik [28] applied the neural network approach to predict the paper properties of refined chemical pulp. Their findings confirm a good correlation between the experimental and simulated data. Musavi and Qiao [29] suggested a radial basis function (RBF) neural network for the prediction of pulp quality. Their results show significant accuracy in predicting the pulp quality factor ($k\#$) in the digester. Dufour et al. [30]

trained a feedforward neural network with a two-layer perceptron for pulp digester to appraise the impact of changes in feedstock properties.

There has been limited research in the past on the application of machine learning techniques within the Pulp & Paper industries. Particularly, in the previous studies, there has been no research regarding the use of advanced machine learning methods to develop refining identification models and advanced refining optimization strategies in the thermo-mechanical pulping process.

## 2.2 Energy and cost-efficient design and operation of Energy Hub for the TMP Mill and Paper Machine Integration

It was common practice to plan and manage different energy systems independently in the past. Nevertheless, as technology advances, such as the development of efficient multi-generation systems, and integrated energy infrastructures such as electricity, natural gas, and district heating (DH) networks become increasingly advantageous, the movement towards multi-energy systems (MES) is accelerating. Such systems interact in a synergistic way between different energy carriers and systems. Nevertheless, the application of such a concept requires a management tool that integrates the components of the system. EHs, which can be described as places where different energy carriers are produced, converted, stored, and consumed, are promising for MES integration [31]. The concept of Energy Hub can be used for the optimal integration of the TMP mill and paper machine. The optimum design and operation of EH are highly regarded nowadays. Distributed energy production units such as EH, which produce different types of energy simultaneously, enhance the efficiency and stability of the energy systems [32]. The advanced proposed approach in this thesis for the energy-efficient and cost-efficient design and operation of the EH is based on the integration of thermo-economic analysis, reliability & availability analysis, and EH load prediction. Thermoeconomic analysis combines economics and thermodynamics, which is necessary for energy system resource allocation. Reliability assessment will lead to more accurate modeling of real-life system operating conditions since system components' availability is considered in the design process. Load profile prediction estimates the energy demand for the next hour, which helps with the optimal operation of the Energy Hub. The cost-effective and secure operation of EH depends on accurate load profile predictions. Operators can complete unit commitments and assess system stability with the help of load forecasting.

Time series models are the best way to estimate the future value of a time-dependent process. In time series forecasting, future values are predicted based on the history of the previous values. Long-short-term memory (LSTM) networks are well suited to processing, prediction, and classification in analyzing time series data. LSTMs have recently received increased attention due to their ability to deal with highly complex, non-linear datasets. An LSTM algorithm is tested on a publicly available dataset of residential meters in reference [33]. The results indicated that LSTM outperformed rival machine learning (ML) algorithms at forecasting residential loads in the short term. In reference [34], a dynamic model was trained using the LSTM algorithm to generate predictions for impulsive loads. LSTM is discussed in multiple configurations in reference [35], and the best architecture and optimum hyperparameters are proposed. In the other research, Taheri et al. [36] studied electricity demand time series forecasting based on a hybrid prediction model by integrating LSTM with empirical mode decomposition (EMD). Intrinsic mode functions (IMFs) are subseries of load time series data that are broken down by the EMD algorithm. Different LSTM models are developed for each IMF derived. For the final step of energy demand prediction, the outputs from each LSTM learner are fed into a meta-learner. In experiments on the California ISO dataset, their proposed hybrid model outperforms single LSTM, Logistic Regression (LR), and XGBoost algorithms. However, forest industry research is lacking when it comes to predicting electricity consumption and heat production. This thesis utilizes different LSTM structures to predict the TMP mill heat production.

The analyzed LSTM structures are single LSTM with hidden dense layer, stacked LSTM with hidden dense layer, and Bidirectional LSTM. Results show that all the LSTM structures perform effectively for the TMP heat prediction.

In the past, research on the thermo-economic analysis, load prediction, and reliability assessment of EH has been made separately. Given the importance of the system sustainability in the system total cost, reliability & risk assessment plays a vital role in designing a sustainable system. Based on 200 articles published from 2007 to 2017, Maroufmashat et al. [37] conducted a systematic review of energy hubs. An overview of the articles is provided on the basis of modeling methodology, planning and operation, economic and environmental aspects, and applications of energy hubs. Reliability modeling has only been presented in a few reports for cogeneration systems. Haghifam and Manbachi [38] made a reliability and availability modeling of combined heat and power (CHP) systems. They classified combined heat and power subsystems and modeled the mean-time-to-failure, availability, and reliability from the production unit to end-use. Their suggested CHP availability and reliability model is based on the state space and the continuous Markov method with heat-generation, fuel distribution, and electricity-generation subsystems. As a result of this study, it appears that improving the gas distribution network, the water distribution network, and the water delivery network to the CHP, as well as optimizing the failure and repair rates of CHP systems, can enhance the reliability of a complete integrated system and play a major role in determining whether CHP systems are economically and technically feasible. A cooperative approach to energy management was presented by Tiwari and Singh [39] for multi-carrier energy hubs. The proposed scheme facilitates energy exchange between energy hubs, resulting in greater economic and environmental benefits. Results show that networked systems are more reliable and have lower operating costs and greenhouse gas emissions. Parisio et al. [40] proposed a robust solution to the EH operation scheduling problem. First, they made an integer dynamic model to establish a general EH modeling framework. Next, a short-term operating schedule was proposed in a deterministic environment and uncertain scenario. They got a robust schedule of input power flows, which is less sensitive to uncertain converter efficiencies than the nominal schedule. To schedule an EH including a tri-generative advanced adiabatic compressed air energy storage (AA-CAES), Garmroodi et al. [41] proposed a model-free, safe deep reinforcement learning (DRL) method using primal-dual optimization and imitation learning. The authors first modeled the operation of AA-CAES under off-design conditions and linearized it using mixed-integer linear programming (MILP). In the next step, they presented a safe DRL approach that includes training and testing steps. According to their results, the proposed approach reduces operational costs and meets operational constraints.

This thesis investigates the cost-efficient design and operation of residential (high-rise residential building) and industrial (forest industry) applications of EH based on the integration of thermo-economic analysis, reliability & availability analysis, and EH load prediction. The proposed method has not been implemented in other studies in the literature and is the author's novelty. The proposed method was first introduced and evaluated for the residential application of EH. Due to the good performance, the proposed method implemented for the energy and cost-efficient design and operation of EH to integrate the TMP mill and paper machine through a heat recovery system.

# 3. Thermo-mechanical pulping process (TMP)

The primary target of pulping is to make fibers free from the lignin and suspend the fibers in water. Lignin is a natural glue that binds the fibers together in wood. In mechanical pulping, fibers are separated and developed mainly through mechanical forces in the refining process. The typical mechanical pulping process has a yield ratio of 97% for Norway spruce, while the mentioned ratio is 45%-50% in chemical pulping [42]. A higher yield ratio implies that more paper could be produced out of limited raw material resources, which is beneficial from the ecological and national economic point of view. In addition, mechanical pulping is attractive to pulp producers due to its low investment cost and simplicity compared to chemical pulping. However, mechanical pulping also has drawbacks. The mechanical pulping process requires high-quality raw wood input materials, and electricity consumption per unit of pulp production is high. Nowadays, thermo-mechanical pulping (TMP) is the most predominant type of mechanical pulping process [43]. TMP has found widespread growth since the early 1970s. Today, TMP is used in most newsprint and lightweight magazine paper mills throughout the world. A high yield ratio, good optical properties, and adequate strength make TMP a dominant player in the market. A major drawback of TMP is high electrical energy consumption. Based on the species being pulped and the degree of refinement, the TMP system can consume 2200-2800 kWh per ton of pulp. Waste heat makes up a large percentage of energy consumption. The importance of waste heat recovery, therefore, lies in its economic potential. Table 1 shows the potential use of TMP and chemi-thermomechanical pulping (CTMP) in different end products [44].

Refining is the most energy-intensive process in thermo-mechanical pulping, which consumes almost 80% of the mill's electricity consumption[17]. Refiners are mechanical devices that use electric motor actuators to transfer mechanical energy to the wood chips, reduce wood chips to individual fibers, and correct wood fiber morphology. The refiner plate grooves soften the lignin to separate fibers and break the wood chips. Wood chips after pretreatment (e.g., washing and steaming) are supplied to the primary refiner by a screw feeder. The second-stage refiner contributes to the further development of wood quality. High-consistency refiners operate at a pulp consistency of 20-50%. A cyclone follows each refiner to separate steam flow from the pulp and avoid pulp brightness reduction by cooling the pulp. Fig. 2 shows a typical flow diagram of a TMP plant [15].

**Table 1.** Potential use of TMP and CTMP in different end products

| Product | Softwood TMP (%) | Softwood CTMP (%) | Hardwood CTMP (%) |
|---|---|---|---|
| Newsprint | 95–100 | 80-100 | 70-80 |
| SC paper | 65-80 | 65-80 | 70-90 |
| Fine paper | Not Significant | 50-80 | 50-60 |
| Paperboard | - | - | - |
| Tissue | Not Significant | 20-40 | 20-40 |

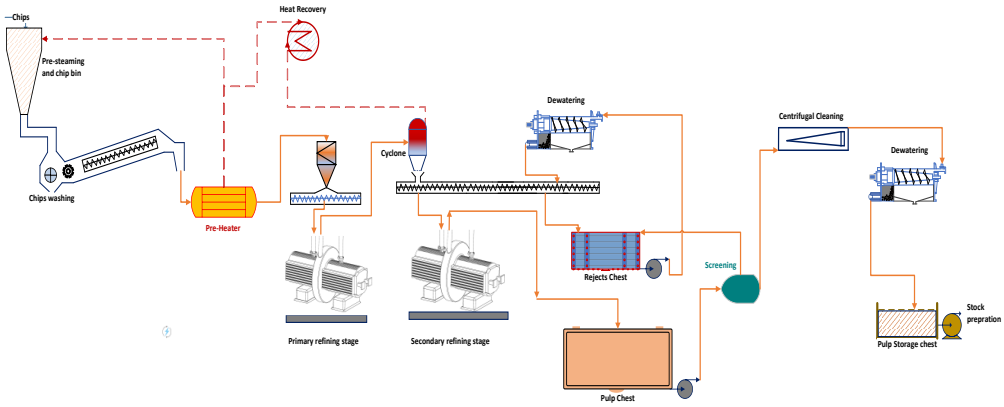



**Fig. 2.** Sample layout of the thermo-mechanical pulp mill

Due to the small distance between the refiner plates and the strong frictional force, a large part of the consumed electricity by the electric motor causes heat. The generated heat evaporates the refining diluting water and the moisture along with the wood chips, which turns the water-chip suspension into a fiber-steam suspension [45]. Exhaust steam can be recovered and used as a secondary heat source. Exhaust steam from the refiners is often contaminated and contains entrained air. For this reason, clean steam must be produced before it can be used in certain processes. Clean steam can be produced in a reboiler. This secondary heat can be used to replace primary heat. The pressure of the steam may need to be increased by a thermo-compressor for some applications, such as paper machine dryers. Exhaust steam can also be used to heat water in a heat exchanger. To further increase heat recovery, the chip steaming vessel and the refiner can be isolated from each other. This will allow a higher pressure in the refiner, which maximizes heat recovery. In addition to wood chip pre-steaming, the refining generated heat can be used for district heating, paper machine dryer, black liquor evaporation, and water or stock heating.

There are single or double-disc configurations of wood chips refiner. The single-disc setting has one rotating disc and one stationary disc, and a double-disc setup has two rotating discs in opposite directions. Fig. 3 shows a schematic of a single-disc refiner. Refiners have a typical rotational speed of $1200 - 1800\ r/min$ which is induced by a grand electrical motor. The power and plate diameter of the new refiners reach 18000 $Hp$ (13400 $kW$) and 1.8 meters, respectively [46]. High-capacity refineries require more power and higher plate diameter. The refining plate gap is being controlled by a hydraulic cylinder at a distance of $0.5 - 2\ mm$, intently [47].

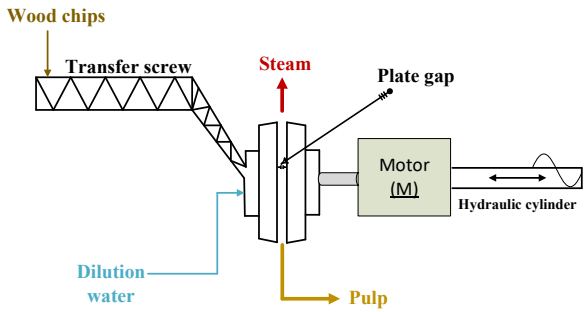



**Fig. 3.** Schematic of a single-disc refiner

# 4. Long Short-Term Memory (LSTM)

In this thesis, the time series prediction models such as different LSTM methods were used for the Energy Hub (EH) load prediction in forest industry applications. Load prediction is essential for the optimal operation of the EH. The EH is utilized in forest industry applications as an optimal heat integration tool between the TMP mill and the paper machine. Since the paper mill's heat demand is relatively constant and does not change over time, the load of the EH will be mostly dependent on the estimated heat production of the pulp mill.

## 4.1 LSTM fundamentals

Classical neural networks cannot catch the dataset variables' temporal dependencies. Recurrent Neural Networks (RNNs) are developed to address this problem. As presented in Fig. 4, a recursive structure of the RNNs gives the ability to predict a series of potential values using observed historical inputs while keeping knowledge over a multitude of time horizons.

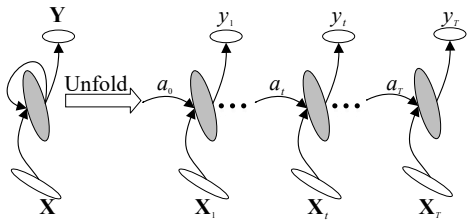



**Fig. 4.** Standard RNN structure: packed and unfolded in time.

Recurrent Neural Networks (RNNs) are capable of using contextual information in the process of mapping the input and output sequence. However, the context range for the architecture of standard RNNs is limited. The effect of the input variables on the hidden and output layer deteriorates rapidly as it cycles around the network's recurrent links. This phenomenon is repeatedly mentioned in the literature as the vanishing gradient problem. Therefore, typical RNNs are unqualified for learning long-term temporal dependencies because of the vanishing gradients problem phenomenon. Reference [48] further explains and discusses this phenomenon. In 1997, Hochreiter and Schmidhuber [49] introduced the Long-Short-Term Memory (LSTM) architecture to address the problem of vanishing gradients for RNNs, shifting from normal to deep recurrent neural networks. The LSTM training approach is based on keeping the information regarding the previous states and can be qualified for state or memory recognition tasks. Utilizing the mentioned method, LSTM is capable of solving gradient bursting and vanishing problems. The LSTM cell architecture is presented in Fig. 5, where the memory cell mode block controls the signal through the forget gate, input gate, and output gate. Each LSTM gate has a specific function, and the following Equations show the procedure of computing each vector at time t.

$$f_t = \sigma(W_f x_t + W_f m_{t-1} + b_f) \quad (1)$$
$$i_t = \sigma(W_i x_t + W_i m_{t-1} + b_i) \quad (2)$$
$$o_t = \sigma(W_o x_t + W_o m_{t-1} + b_0) \quad (3)$$
$$a_t = \tanh(W_a x_t + W_a m_{t-1} + b_a) \quad (4)$$
$$c_t = c_{t-1} \otimes f_t + i_t \otimes a_t \quad (5)$$
$$m_t = o_t \otimes \tanh(c_t) \quad (6)$$

The $\otimes$ operator refers to the element-wise product. H is the activation function (tanh), $\sigma$ denotes the logistic sigmoid function, which is the recurrent step activation function. $x_t$ is the input at time $t$, $m_t$ is the output at time $t$. $i_t$, $f_t$ , and $o_t$ are the input, forget, and output gates at time $t$. $a_t$ is the candidate memory at time $t$. $c_t$ is the current memory at time $t$, and $W$ and $b$ are weight matrices and bias vectors that are learned during training.

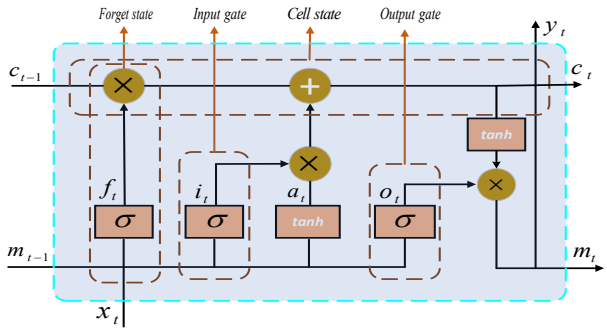



**Fig. 5.** LSTM cell principal structure

Fig. 6 shows the single LSTM layer, including the LSTM cells. Each layer in LSTM is composed of the same number of cells as the window size. The LSTM layer learns how time series data and sequence data are related over time. The layer enhances gradient flow over long sequences during training through additive interactions. Different LSTM structures have been examined in this study to achieve the best heat prediction accuracy. In this regard, three main LSTM structures are studied and employed: 1- Single LSTM with hidden dense layer, 2- Stacked LSTM with hidden dense layer, and 3- Bidirectional LSTM.

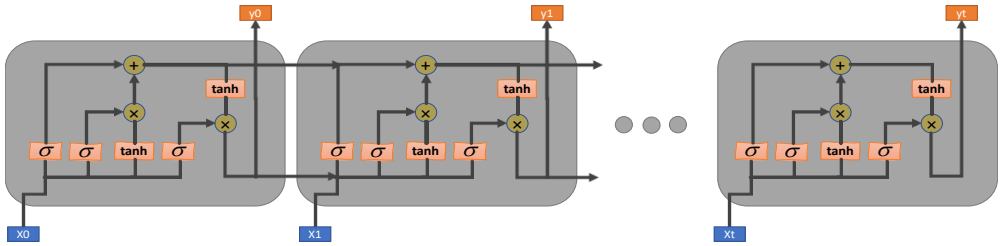



**Fig. 6.** Unfolded Single LSTM Layer.

## 4.2 Single LSTM with a hidden dense layer

The principal LSTM network is formed of a single hidden LSTM layer followed by a standard feed-forward output layer. The LSTM structure and principal concept have been discussed before. The structure of the employed single LSTM followed by a hidden dense layer is depicted in Fig. 7.

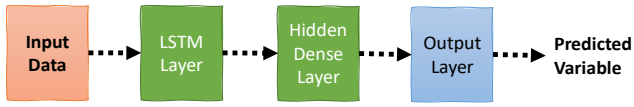



**Fig. 7.** Overview of the Single LSTM with a hidden dense layer

### 4.3 Stacked LSTM with a hidden dense layer

A stacked LSTM architecture refers to a model comprised of multiple layers of LSTM. The stacked LSTM is an extension of the principal LSTM model, which consists of a single hidden LSTM layer followed by a standard feedforward output layer. This model has multiple hidden LSTM layers, where each layer includes several memory cells. The stacked LSTM hidden layers make the model deeper by increasing the hidden layers and LSTM units. In contrast to single value output, an above LSTM layer provides a series of values instead of a single value to the below LSTM layer. They produce one output per input time step instead of having a single output for all input time steps. The objective of the different proposed LSTM network structures is to predict the next total heat production from the pulp mill for different time horizons. Fig. 8 illustrates the general structure of stacked LSTM with a hidden dense layer.

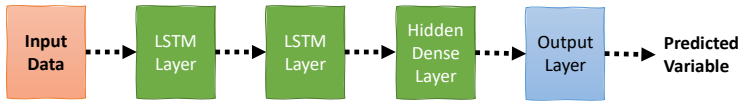



**Fig. 8.** Overview of the Stacked LSTM with a hidden dense layer

### 4.4 Bidirectional LSTM

The bidirectional LSTM is an augmentation of the standard LSTMs to enhance a model's prediction performance by utilizing the sequence of information in both forward and backward directions. In this regard, two LSTM networks are trained based on the original and a reversed replica of the data. This approach adds more meaning to the network to improve its performance. The bidirectional LSTM principal concept is simple. The observed data in its current format is assigned to the first recurrent layer input data. In contrast, the repeated replica of the observed data is considered an input to the copy of the first recurrent layer. Fig. 9 shows the principal concept of Bidirectional LSTM [50]. Based on Fig. 9, the backward hidden layer ($\overleftarrow{h_t}$) forward hidden layer ($\overrightarrow{h_t}$), and the output ($y$) can be calculated as [51]:

$$\overrightarrow{h_t} = H(w_{x\vec{h}}x_t + w_{\vec{h}\vec{h}}\vec{h}_{t-1} + b_h) \tag{7}$$

$$\overleftarrow{h_t} = H(w_{x\overleftarrow{h}}x_t + w_{\overleftarrow{h}\overleftarrow{h}}\vec{h}_{t+1} + b_{\overleftarrow{h}}) \tag{8}$$

$$y_t = w_{\vec{h}y}\vec{h}_t + w_{\overleftarrow{h}y}\overleftarrow{h}_t + b_y \tag{9}$$

Where $w_{x\vec{h}}$ and $w_{x\overleftarrow{h}}$ are the feedforward and backward hidden weight matrices, respectively. $b_h$ and $b_{\overleftarrow{h}}$ are the bias vectors, and $H$ indicates the hidden layer.

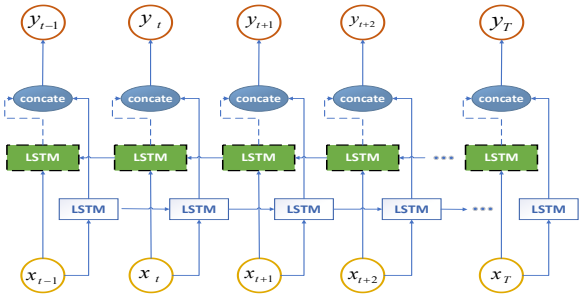



**Fig. 9.** Architecture of bidirectional LSTM

## 4.5 Hyperparameter tuning

Model hyperparameter tuning is applied to the experiments to achieve a time-efficient and accurate prediction model. A good performance can be achieved with the LSTM network by choosing the optimum hyperparameters. Depending on hyperparameters, deep learning algorithms differ in performance, time cost, and memory usage. A hyperparameter selection can influence the performance of machine learning algorithms from state-of-the-art to mediocre [52]. While there are some general guidelines regarding the appropriate values of these hyperparameters, hyperparameter tuning is still necessary because the optimal values depend on the data sets, the performance criteria, and several other variables [53]. The hyperparameter tuning in this research has been made through the RandomizedSearchCV module from the Scikit-learn (Sklearn) library in Python. The parameter values are not all tested as in GridSearchCV, but instead, a fixed number of parameter settings are sampled from the specified distributions.

In addition to the learning rate, the batch size is an important hyperparameter. As the main hyperparameter of the training algorithm, the learning rate determines how quickly the weights are updated during each training cycle. Using a smaller learning rate reduces loss but results in slower convergence. Training speed can be increased with a larger batch size, but the convergence speed may be lower, and with an increased batch size, generalization ability may decrease. In the literature [54], an extensive discussion is made of the processes of LSTM hyperparameter tuning. The hyperparameter dictionary and optimal values for the RandomizedSearchCV are given in Fig. 10.

**a)**

```
[1101] hyperparams_dict={'momentum' : [0.1, 0.5],
                         'lr':[0.0001, 0.001, 0.01],
                         'batch_size':[64, 128, 256, 512],
                         'LSTM_units':[35, 45, 65],
                         'Dense_layer_units':[30, 40, 50],
       }
     print(hyperparams_dict)

{'momentum': 0.1, 'lr': 0.001, 'batch_size': 256,
 'LSTM_units': 35, 'Dense_layer_units': 30}
```

**b)**

```
[93] hyperparams_dict={'momentum' : [0.1, 0.5],
                       'lr':[0.0001, 0.001, 0.01],
                       'batch_size':[64, 128, 256, 512],
                       'LSTM_units':[35, 45, 65],
                       'Dense_layer_units':[30, 40, 50],
       }
     print(hyperparams_dict)

{'momentum': 0.5, 'lr': 0.001, 'batch_size': 128,
 'LSTM_units': 35, 'Dense_layer_units': 30}
```

**c)**

```
[131] hyperparams_dict={'momentum' : [0.1, 0.5],
                        'lr':[0.0001, 0.001, 0.01],
                        'batch_size':[64, 128, 256, 512],
                        'LSTM_units':[35, 45, 65],
          }
     print(hyperparams_dict)

{'momentum': 0.1, 'lr': 0.0001, 'batch_size': 64, 'LSTM_units': 45}
```

**Fig. 10.** The hyperparameter dictionary and optimal values for the RandomizedSearchCV: a)Single LSTM with a hidden dense layer, b) stacked LSTM with a hidden dense layer, c) Bidirectional LSTM.

# 5. Refining Identification Model Development

Fig. 11 shows the refining principle by considering a refiner having a single rotating disc and one stationary disc. Wood chips are fed to the refiner stator and are beaten by plate segments and patterned grooves. The refiner output is pulp mixed by generated steam from moisture content in the wood chips and evaporation of refining dilution water. The pulp is separated from steam by the following cyclone. The generated steam is often utilized to supply the paper machine heating demand in the drying section.

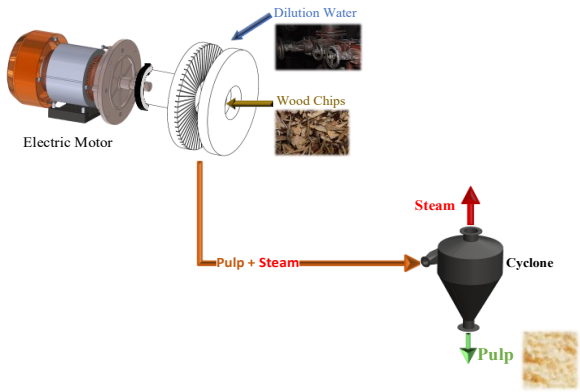



**Fig. 11.** Mechanical refining principle.

Several statistical regression approaches create refining energy and pulp quality identification models from measured refining data. The general objective of these methods, such as linear regression models, nonlinear regression models, and discriminant analysis (which primarily work by least-squares error or maximum likelihood to measure the quality of an estimator) is to find a mathematical relationship (such as linear or polynomial) between the predictor variables and model outputs (operational variables). However, these models' efficiency decreases significantly with the increasing number of variables and the complexity of the problem [55].

Artificial intelligence is an alternative powerful data-based method to develop refining energy and quality identification models. Refining is a complicated process with a large number of variables potentially affecting the products. Based on this thesis outcome, accurate refining identification models are achievable by utilizing advanced AI methods. Understanding and controlling the refining process is paramount in the thermo-mechanical pulping process. In order to control the refining-specific energy consumption and pulp quality, it is essential to recognize the non-linear hidden pattern between refining variables [20]. For the advanced energy-saving optimization strategy, mathematical models (identification models) are required to model the refining specific energy consumption and final pulp quality based on system-manipulated variables. The models for the refining (first and second stages) energy simulation form the optimization objective function, and the models for the final pulp quality simulation contribute to constructing the optimization constraints.

None of the previous studies in the literature have comprehensively analyzed the parameters affecting the refining-specific energy consumption and final pulp quality in TMP mills. On the other hand, no accurate refining identification model is proposed to be incorporated into the energy-saving optimization strategy. This thesis proposes an advanced energy-saving refining optimization strategy by integration of ANN and heuristic optimization algorithm such as Genetic algorithm. Refining identification models are developed utilizing state-of-the-art machine learning methods. For this purpose, data regarding the refining process are collected from two different TMP mills in a Nordic country. Refining disturbance variables are discovered to be very time-variant. The effect of disturbance variables such as refining plate erosion and variations in wood chips quality such as chip bulk and slurry density are considered via the variable time.

## 5.1 Refining variables

Refining variables can be classified into three categories: manipulated variables, operating variables, and disturbance variables. Prominent manipulated variables are dilution water, plate gap, and chip transfer screw speed. Refining specific energy consumption, pulp production rate, pulp consistency, and final pulp quality such as fiber length and freeness value are the most important operation variables dependent on manipulated and disturbance variables. In the two studied mills, the pulp production rate (which is given as oven-dry ton of pulp) is calculated by multiplying a constant coefficient by the chip transfer screw speed. This simplification causes an error in calculating the pulp production rate. Since pulp production is a significant variable for energy simulation of the refining process, an error in the pulp production rate could lead to a decrease in the accuracy of the refining energy simulation.

Variation in the quality of the input wood chips leads to changes in the chip's bulk density and must be considered for the accurate calculation of the pulp production rate. Pulp production rate can be also calculated from pulp consistency, But there is no reliable online device for determining pulp consistency, and accurate measurement requires laboratory tests. Our measurements show a 20% difference in the amount of production rate calculated based on the chip's screw speed and consistency measurement device.

Disturbance variables can be divided into two general categories: refiner plate erosion and the quality of wood chips. Wood species, wood density, chip size, chip moisture, seasonal variations, and bark content are among the most important variables regarding wood chip quality disturbances. Fig. 12 shows a schematic of the refining variables and their relation. More information regarding the refining variables is provided in the following.

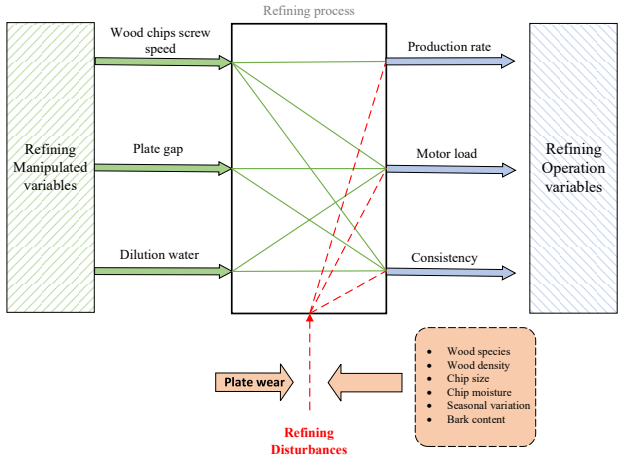



**Fig. 12.** Schematic of the refining variables.

### 5.1.1 Refining manipulated variables

*Chip transfer screw speed:* It is the responsibility of the screw feeder to control the flow of wood chips in the refining line. The pulp production rate, which is dry fiber flow, could be reached to the desired value by controlling the screw feeder rotational speed.

*Dilution water flow rate:* Refining pulp consistency depends on the dilution water flow rate. One of the vital refining variables for estimating pulp quality is the consistency of the pulp in the refining zone. Variations in the pulp consistency for a given refining specific energy consumption alter refining intensity, refining specific energy consumption, and quality of pulp. The high flow rate of dilution water leads to a rise in refining plate vibration and process instability. It is essential to control the refining dilution water to keep the pulp consistency at the desired value.

*Plate gap:* The gap between refiner plates is controlled by either a hydraulic or electro-mechanical loading system. The refiner disc gap should be measured and optimized for efficient refining operations and to prevent plate clashing [56]. Changes in the refiner plate gap alter the applied mechanical forces to the wood material and affect the refining-specific energy consumption for a given pulp production rate.

### 5.1.2 Refining disturbance variables

Generally, disturbance variables are the type of variables that cannot be controlled and measured in many situations. Refiner plates operating conditions and inlet wood chips uniformity are the most dominant disturbances affecting the refining energy consumption and pulp quality.

*Refiner plate erosion:* The nature of the refining process does not follow a stationary behavior because of the refiner plate erosion. Most often, refiner plates erode gradually based on the plate operation time and need to be renewed after hundreds of hours. However, in the case of plate clashing, plates cannot work anymore, and they should be changed immediately [57]. Generally speaking, new plates produce pulp with higher quality and freeness values for a given refining-specific energy consumption. Preventing plates from eroding is impossible, but the erosion effect could be compensated by advanced manipulating the refining control variables to some extent.

*Raw material quality:* Chip bulk density, wood species, and chips' solid content are mentioned as the prominent disturbances for the raw material quality. Meanwhile, chip bulk density is figured as the main disturbance for the dry wood mass flow or production rate. Wood chips density variation straightly changes the pulp quality and specific energy consumption for a given motor load [58].

### 5.1.3 Refining operating variables

*Refining motor load:* An electrical motor is always integrated into the mechanical refiners to actuate the refiner discs. It is known that refining specific energy consumption has a great correlation to the pulp quality, and for a given production rate, the motor load is proportional to the specific energy consumption. Motor load variation is a function of pulp consistency, plate gap, production rate, and wood chips properties.

*Refining consistency:* Refining consistency changes the ratio of the refining-specific energy consumption and pulp quality. Refining dilution water flow rate is the most simple and dominant variable to control the refining consistency, although other variables such as motor load and production rate affect the consistency as well. For a given refining specific energy, variations in refining consistency lead to different pulp quality [3].

*Production rate:* Refiner throughput or pulp production rate is the oven-dry flow rate of wood fed into the refiner. The production rate is proportional to the chip-transfer screw speed and could be easily computed by multiplying the chip-transfer screw speed by a constant value. Chip bulk density and chip solid content are the main disturbance variables affecting the production rate. Eq. 10 calculates the production rate (ton/hour) by taking into account the variations in wood chips properties [20].

$$P(R) = 0.06 k_p s_c d_c R \tag{10}$$

Where $k_p(\frac{m^3}{rev})$ is a proportional constant, $s_c(\%)$ is chip solid content, $d_c(\frac{Kg}{m^3})$ is chip bulk density, and $R\ (rpm)$ is chip-transfer screw speed, respectively.

*Refining specific energy consumption:* Refining specific energy consumption refers to the amount of energy required to refine a specific quantity of wood pulp. Refining specific energy consumption can be calculated by dividing the refining motor load by the production rate. typically measured in units such as kilowatt-hours per metric ton (kWh/t). This can be a useful metric for evaluating the energy efficiency of a refining process and identifying opportunities for improvement.

*Refining intensity:* Wood pulp refining intensity refers to the degree to which wood pulp fibers are mechanically treated and broken down during the refining process. Wood pulp refining is a crucial step in the production of paper products, as it allows the fibers to be separated and refined to achieve the desired characteristics of the final product, such as strength, smoothness, and printability. The refining intensity can be controlled by adjusting the refining conditions, such as the consistency of the pulp, the type of refining equipment used, and the refining time. Higher refining intensity generally results in shorter and more fragmented fibers, which can increase the strength and density of the paper but may also reduce its opacity and printing quality.

*Pulp freeness:* The measurement of the water portion in the pulp suspension (Freeness measurement) is a common method to observe stock preparation corrections. Freeness value could represent the refining impacts on pulp properties since it has a massive correlation with fines formation and fiber fibrillation. Therefore, freeness could indicate the pulp and fiber characteristics [59].

The Canadian Standard Freeness (CSF-test) is mainly used to measure the drainability of a dilute pulp suspension under specific conditions according to ISO 5267-2 (suspension of 3 g of pulp in 1 L of water), especially in the laboratory scale. It is proven that some pulp properties such as the distribution of fiber length in the short and medium fractions, conditions of the surface, and fiber compression affect the pulp freeness. Drainability or freeness composes a helpful index of the intensity of mechanical treatment to which the pulp has been subjected [4]. CSF decreases with refining and is sensitive to the quality of water and fines. CSF is one of the most well-known indicators to determine the pulp's quality and refining level. CSF is utilized in this study to define the quality of the final produced pulp.

To better understand the effect of refining on freeness value, the theory of drainage ability in paper manufacturing could be helpful. Refining makes the fibers more flexible and softer by swelling the fibers with mechanical forces. In such a case, firmly entangled fibers make a web in the drainage test. Moreover, secondary fines are generated during the refining process by external fibrillation or fiber shortening. These fines are not bound to the fibers and can easily move through the pulp suspension while eventually getting stuck between the fibers' pores. This phenomenon decreases the drainage ability since fines block the water flow path and increase the drainage time. Therefore, the drainage ability is deteriorated by the refining, which is not desirable in the paper-making process [60]–[62]. Fig. 13 gives more information about the fibers and fines formation before and after refining [63].

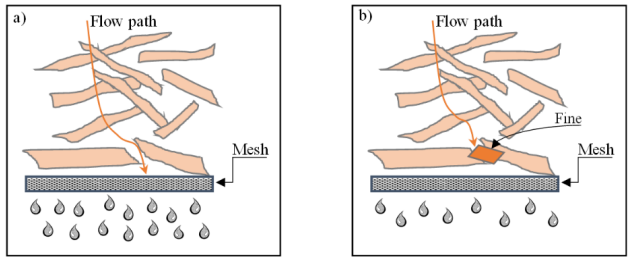



**Fig. 13.** Pulp fibers characteristic and drainage ability a) before and b) after refining [63].

Table 2 shows the typical ranges of the energy consumption and fiber length for different mechanical pulping subdivisions such as thermo-mechanical pulping (TMP), chemi-thermomechanical pulping (CTMP), refiner mechanical pulping (RMP), groundwood (GW), and pressure groundwood (PGW) pulping [63].

**Table 2.** Typical ranges of the pulp freeness and energy consumption in mechanical pulping.

| Mechanical pulping subdivisions | Canadian Standard Freeness (ml) | Specific energy consumption (kWh/ton of pulp) |
|---|---|---|
| Thermomechanical (TMP) | 400-30 | 800-3600 |
| Chemi-thermomechanical (CTMP) | 700-30 | 1000-4300 |
| Refiner mechanical (RMP) | 350-30 | 1600-3000 |
| Groundwood (GW) | 350-30 | 1100-2200 |
| Pressure groundwood (PGW) | 350-30 | 1110-2300 |

*Fiber length:* Fiber shortening is an unwanted change in fiber quality due to fiber beating (refining) [64]. Fiber breaking could occur by sufficient stress on the fiber or when fibers get chopped through shearing forces during the passages of the refiner bars or in case of being pulled from other fibers [65]. Also, fines are generated during the refining as a consequence of extreme force throughout external fibrillation. Therefore, there is a clear correlation between fines generation and fiber cutting

[66]. More information regarding the effect of fiber length on paper characteristics has been provided and investigated in [67], [68]. Table 3 demonstrates the pulping properties of US Softwoods and Hardwoods [15].

**Table 3.** Average Fiber Length of the US Softwoods and Hardwoods.

| Species | Scientific Name | Average Fiber Length (mm) |
|---|---|---|
| **Softwoods** | | |
| Jack (pine) | Pinus banksiana | 3.5 |
| Loblolly (pine) | Pinus taeda | 4.00 |
| Lodgepole (pine) | Pinus contorta | 3.5 |
| Monterey (pine) | Pinus radiata | 2.60 |
| Black (spruce) | Picea mariana | 3.5 |
| Blue (spruce) | Picea pungens | 2.8 |
| **Hardwoods** | | |
| Ailanthus | Ailanthus, altissima | 1.20 |
| Ash, green | Fraxinus pennsylvanica | 1.05 |
| Maple, silver | Acer saccharinum | 1.75 |

## 5.2 Neural Networks Performance Functions

Model performance has been investigated by the correlation coefficient ($R$), coefficient of determination ($R^2$), mean absolute percentage error ($MAPE$), mean square error($MSE$), and root mean square error ($RMSE$). The determination coefficient designates the ratio of the total variation in the dependent variable that the prediction model could explain. A higher determination and correlation coefficient indicate better accuracy and performance of the model. The lower error indicates the higher performance of the identification model [69].

$$R = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}} \tag{11}$$

$$R^2 = \frac{[\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})]^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{12}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\frac{|x_i - y_i|}{|x_i|} \tag{13}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(x_i - y_i)^2 \tag{14}$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(x_i - y_i)^2}{n}} \tag{15}$$

Where $x_i, y_i, \bar{x}, \bar{y}$ and $n$ are real data (actual value), predicted data, mean of real data, mean of predicted data, and the number of data, respectively.

## 5.3 Combination of empirical and mechanistic methods for refining identification model development

*Basic MATLAB Simulink model:* MATLAB Simulink model (presented in Fig. 14) is a powerful tool to model a two-stage thermo-mechanical pulping refining process. Model inputs are feeder rotational speed, dilution water flow rate, and plate gap, while model outputs are refining motor load, generated steam, pulp consistency, and refiner vibration for each refining stage. Fig. 15 gives a better

description of the model for calculating first-stage pulp consistency, motor load, and refining evaporated water. The primary limitation of this model is its failure to take into account how the condition of the plate impacts refining performance. Changing the refiner plate's condition creates disturbances that affect refining energy consumption and pulp quality. This might happen as a consequence of plate erosion. Plate operation time has a minor effect on refining behavior for short-term refining simulation after plate change. When refining plates are new and in good condition, the basic MATLAB Simulink model could be able to predict refining motor load with high accuracy. However, for long-term simulation, the basic MATLAB Simulink model works poorly in predicting refining characteristics because of ignoring the effect of refining plate conditions on refining performance.

For this reason, the influence of plate segment operation time on refining behavior is modeled utilizing the real measured data from the thermo-mechanical pulp mill in a Nordic country. Pearson correlation coefficient reveals the high correlation between refiner plate operation time and refining motor load (-0.6). More time to use plates results in faster plate erosion and, consequently, lower motor load and pulp quality. The Pearson correlation indicates the extent of the linear relation between two variables. This number has the upper and lower bound of +1 and -1, respectively. A correlation of -1 means the linear descending link of data points in a scatter plot. In this case, variables are quite negatively linearly related. A correlation of 0 shows the zero degrees of the linear relationship between variables. Though, there might be a non-linear relation between the two variables. The correlation coefficient of 1 determines the perfect linear relation between two variables.

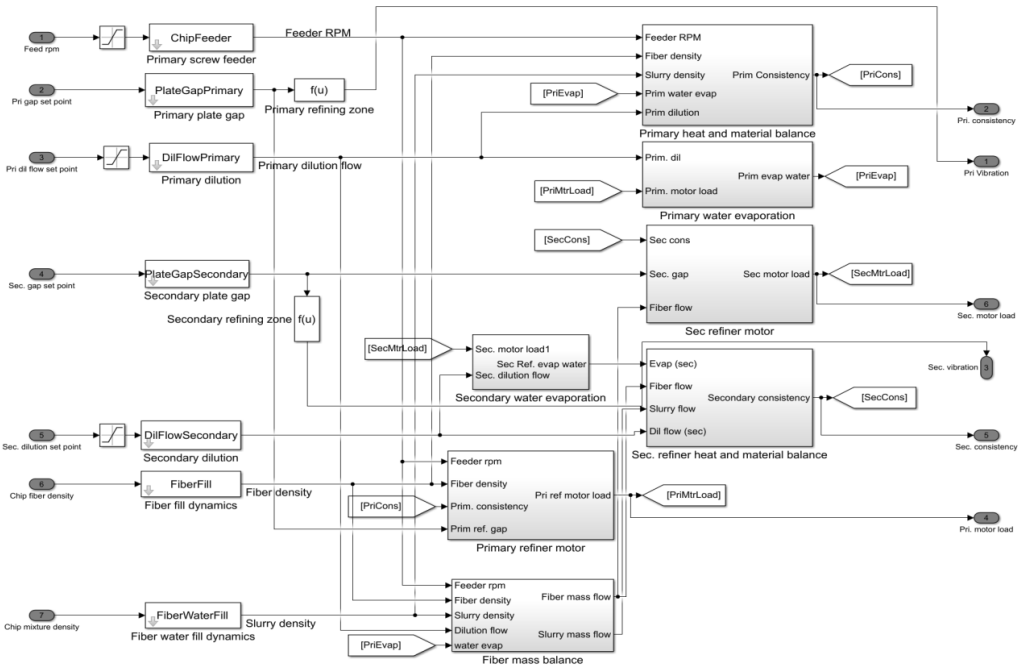



**Fig. 14**. MATLAB Simulink toolbox model for the two-stage Thermo-mechanical pulping process[1].

[1] This model has been used in the case study of MPC Supervisory Control of a Two Stage Thermo-Mechanical Pulping Process, Model Predictive Control Toolbox, MATLAB R2019a.

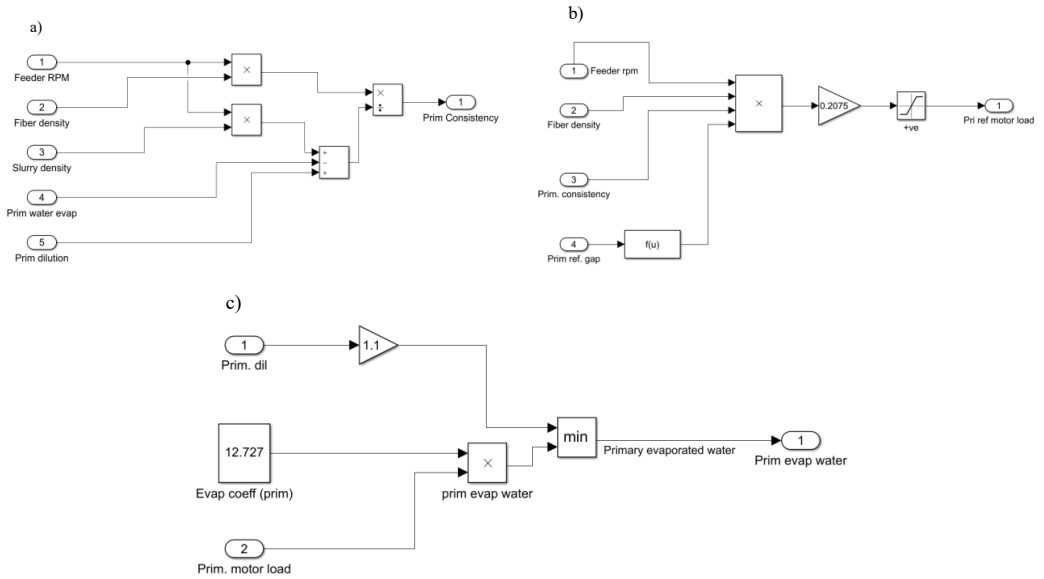



**Fig. 15.** Relationship between refining variables (MATLAB Thermo-mechanical pulping Simulink toolbox model).

*Developed Simulink model:* The developed refining Simulink model is a modified version of the basic MATLAB thermo-mechanical pulping Simulink toolbox model. The primary purpose of this model is to predict the refining energy variables for longer-term energy analysis. For a longer-term study, the refining plate condition plays an important role in refining behavior and should be taken into account. As stated earlier, the plate condition is working as a strong disturbance for refining simulation. As plate operation time increases, plates become more eroded, and there would be more variation in refining motor load for a given production rate.

In order to model the refining plate erosion, a block diagram related to the refining motor load (in Fig. 14) should be replaced by the new one, which introduces the motor load as a function of the plate operating time. Fig. 16 is the modified version of the Simulink block diagram regarding the motor load prediction (Fig. 15-b). The new block diagram (deterioration function) has four coefficients to implement the effect of the refiner plate deterioration on the refining motor load. Since the amount of produced steam is calculated from the motor load, modifying the motor power also indirectly corrects the amount of produced steam. Fig. 16 illustrates the new block diagram, where inside the mentioned black box is the plate deterioration function. The deterioration function coefficients are calculated by running an optimization model for the best motor load fit where the optimization error criterion is the mean least square error ($\varepsilon_1$). The rest of this chapter is devoted to explaining the optimization method. The refining motor load in the deterioration function (Fig. 16) is corrected by Eq. 16.

$$ML^{Corrected} = C_1 + C_2 \times ML^{Old} + C_3 \times POT + C_4 \times ML^{Old} \times POT \quad (16)$$

Where $C_i$ are model coefficients, $ML^{Old}$ is the predicted refining motor load by the Basic MATLAB Simulink model, $ML^{Corrected}$ is the corrected value of motor load by the deterioration function, and $POT$ is the refiner plate operation time. Model coefficients in Eq. 16 are calculated by minimizing the model's least square error, which is provided in Eq. 17. It is possible to calculate the minimum of $\varepsilon_2$ instead of $\varepsilon_1$ since they behave in the same way for the minimization purpose. In order to obtain the minimum value of $\varepsilon_1$, derivatives with respect to the model coefficients must be set to zero as given in Eq. 20-Eq. 23. By rearranging Eq. 20-Eq. 23 in an implicit form, everything comes down to solving a 4 by 4 linear system (AX=B) given in Eq. 24, where there are four equations and four unknowns. It

should be noted that since error (the difference between measured and predicted value) is convex, setting the derivatives to zero will find the minimum of the error function and not the maximum of it.

$$\varepsilon_1 = \sqrt{\frac{1}{n}\sum_{k=1}^{n}\left|ML_k^{Corrected} - ML_k^{Measured}\right|^2} \tag{17}$$

$$\varepsilon_2 = \sum_{k=1}^{n}\left|ML_k^{Corrected} - ML_k^{Measured}\right|^2 \tag{18}$$

Where $ML^{Measured}$ is the measured value of refining motor load. Substituting Eq. 16 in Eq. 18 gives:

$$\varepsilon_2 = \sum_{k=1}^{n}\left|C_1 + C_2 \times ML_k^{Old} + C_3 \times POT_k + C_4 \times ML_k^{Old} \times POT_k - ML_k^{measured}\right|^2 \tag{19}$$

Eq. 20-Eq. 23 provides derivatives taken from $\varepsilon_2$ with respect to model coefficients ($C_i$):

$$\frac{\partial \varepsilon_2}{C_1} = 0 \quad \rightarrow \quad \sum_{k=1}^{n} 2 \times \left|C_1 + C_2 \times ML_k^{Old} + C_3 \times POT_k + C_4 \times ML_k^{Old} \times POT_k - ML_k^{measured}\right| \times 1 = 0 \tag{20}$$

$$\frac{\partial \varepsilon_2}{C_2} = 0 \quad \rightarrow \quad \sum_{k=1}^{n} 2 \times \left|C_1 + C_2 \times ML_k^{Old} + C_3 \times POT_k + C_4 \times ML_k^{Old} \times POT_k - ML_k^{measured}\right| \times ML_k^{Old} = 0 \tag{21}$$

$$\frac{\partial \varepsilon_2}{C_3} = 0 \quad \rightarrow \quad \sum_{k=1}^{n} 2 \times \left|C_1 + C_2 \times ML_k^{Old} + C_3 \times POT_k + C_4 \times ML_k^{Old} \times POT_k - ML_k^{measured}\right| \times POT_k = 0 \tag{22}$$

$$\frac{\partial \varepsilon_2}{C_4} = 0 \quad \rightarrow \quad \sum_{k=1}^{n} 2 \times \left|C_1 + C_2 \times ML_k^{Old} + C_3 \times POT_k + C_4 \times ML_k^{Old} \times POT_k - ML_k^{measured}\right| \times ML_k^{Old} \times POT_k = 0 \tag{23}$$

Writing Eq. 20-Eq. 23 in an implicit form, gives:

$$\begin{pmatrix} n & \sum_{k=1}^{n} Ml_k^{\,old} & \sum_{k=1}^{n} POT_k & \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k \\ \sum_{k=1}^{n} Ml_k^{\,old} & \sum_{k=1}^{n} Ml_k^{\,old^2} & \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k & \sum_{k=1}^{n} Ml_k^{\,old^2} \times POT_k \\ \sum_{k=1}^{n} POT_k & \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k & \sum_{k=1}^{n} POT_k^{\,2} & \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k^{\,2} \\ \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k & \sum_{k=1}^{n} Ml_k^{\,old^2} \times POT_k & \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k^{\,2} & \sum_{k=1}^{n} Ml_k^{\,old^2} \times POT_k^{\,2} \end{pmatrix} \times \begin{pmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{pmatrix} = \begin{pmatrix} \sum_{k=1}^{n} Ml_k^{\,measured} \\ \sum_{k=1}^{n} Ml_k^{\,old} \times Ml_k^{\,measured} \\ \sum_{k=1}^{n} POT_k \times Ml_k^{\,measured} \\ \sum_{k=1}^{n} POT_k \times Ml_k^{\,old} \times Ml_k^{\,measured} \end{pmatrix} \tag{24}$$

Therefore model coefficients in Eq. 16 can be calculated from Eq. 25:

$$\begin{pmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{pmatrix} = \begin{pmatrix} n & \sum_{k=1}^{n} Ml_k^{\,old} & \sum_{k=1}^{n} POT_k & \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k \\ \sum_{k=1}^{n} Ml_k^{\,old} & \sum_{k=1}^{n} Ml_k^{\,old^2} & \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k & \sum_{k=1}^{n} Ml_k^{\,old^2} \times POT_k \\ \sum_{k=1}^{n} POT_k & \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k & \sum_{k=1}^{n} POT_k^{\,2} & \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k^{\,2} \\ \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k & \sum_{k=1}^{n} Ml_k^{\,old^2} \times POT_k & \sum_{k=1}^{n} Ml_k^{\,old} \times POT_k^{\,2} & \sum_{k=1}^{n} Ml_k^{\,old^2} \times POT_k^{\,2} \end{pmatrix}^{-1} \times \begin{pmatrix} \sum_{k=1}^{n} Ml_k^{\,measured} \\ \sum_{k=1}^{n} Ml_k^{\,old} \times Ml_k^{\,measured} \\ \sum_{k=1}^{n} POT_k \times Ml_k^{\,measured} \\ \sum_{k=1}^{n} POT_k \times Ml_k^{\,old} \times Ml_k^{\,measured} \end{pmatrix} \tag{25}$$

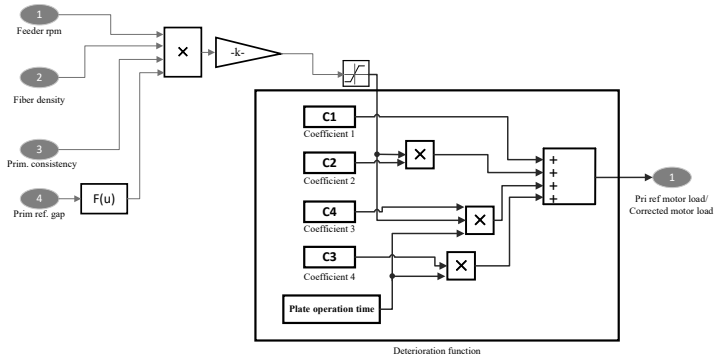



**Fig. 16.** New motor load formulation.

### 5.4 ANFIS method to develop Refining Identification Model.

An adaptive neuro-fuzzy inference system or adaptive network-based fuzzy inference system (ANFIS) is a kind of artificial neural network that is based on the Takagi–Sugeno fuzzy inference system. ANFIS method consists of neural networks and fuzzy systems and defines patterns between the input and output vector by benefiting from the hybrid learning method to determine the optimal distribution of the membership functions [70]. The ANFIS method is constructed by fuzzification, inference engine, and defuzzification [71]. ANFIS method integrates both neural networks and fuzzy logic principle and inherits the advantages of both in a single framework. This method is addressed as a neuro-fuzzy system. Five layers are considered to create the ANFIS inference system. Each ANFIS layer consists of multifold nodes and has been characterized by the node function. The inputs of each layer are determined by the output of the prior layer.

The Sugeno-Takaki model is commonly used to reduce computational costs as well as pinpoint differences with other models. Two rules for the first-order two-rule Sugeno fuzzy inference system are provided in Eq. 26 and Eq. 27. The neural network and fuzzy system are combined by the Radial basis function and Sugeno-Takaki fuzzy inference. Fig. 17 illustrates a simplified version of an ANFIS formation, with only two inputs and one output. The adaptive nodes (which are tunable nodes) and fixed nodes are marked as square and circle in Fig. 17, respectively [70].

$$If\ x\ is\ P_1\ and\ y\ is\ Q_1 \rightarrow f_1 = r_1 x + t_1 y + c_1 \tag{26}$$

$$If\ x\ is\ P_2\ and\ y\ is\ Q_2 \rightarrow f_2 = r_2 x + t_2 y + c_2 \tag{27}$$

Where $A$ and $B$ are the fuzzy sets, $x$ and $y$ are the ANFIS inputs, $f_i(x.y)$ is a first-order polynomial regression.

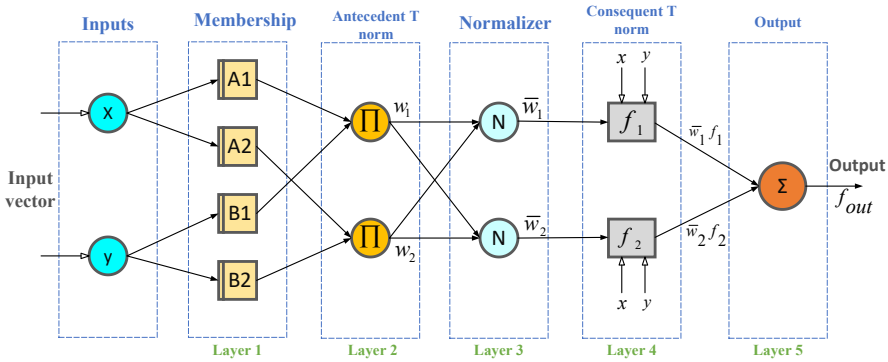



**Fig. 17**. The structure of the adaptive neuro-fuzzy inference system (ANFIS) method consists of two rules, two model inputs, and one model output.

The description of the node function for ANFIS layers (presented in Fig. 17) is provided in the following. Equations 28 and 29 provide the calculation term for the node's output in layer 1, which is the fuzzification layer [70].

$$LO_1^i = \mu_{A_j}(x), \qquad for\ j = 1,2 \tag{28}$$

$$LO_1^i = \mu_{B_{j-2}}(y), \qquad for\ j = 3,4 \tag{29}$$

where A and B are the semantic tags, $x$ and $y$ are the input nodes, $\mu_x$ and $\mu_y$ are membership functions. The Gaussian membership functions are limited from 0 to 1 and presented in Eq. 30 or in a different form in Eq. 31.

$$\mu_{(x)} = \frac{1}{1+\left(\frac{x-c_j}{a_j}\right)^{2b_j}} \tag{30}$$

$$\mu_{(x)} = \exp\left\{-\left(\frac{x-c_j}{a_j}\right)^2\right\} \tag{31}$$

In which $x$ is the model input and $\{a_j, b_j, c_j\}$ are premise parameters. Based on the values of the premise parameters, the membership function's shape differs. Nodes in layer 2, which is a rule layer, are fixed and determine the firing strength $(w_j)$ of a rule. Node output is the multiplication of the node input signal (Eq. 32).

$$LO_2^i = w_j = \mu_{A_j}(x)\ \mu_{B_j}(y), \qquad for\ j = 1,2 \tag{32}$$

Layer 3 nodes are fixed node that computes the firing strength of the jth rule to all rules. Eq. 33 provides the output of the jth node in layer 3.

$$LO_3^j = \bar{w}_j = \frac{w_j}{w_1 + w_2}, \qquad for\ j = 1,2 \tag{33}$$

The Defuzzification layer is layer 4; Eq. 34 shows the function of each node in layer 4, consisting of adaptive nodes.

$$LO_4^j = \bar{w}_j f_j = \bar{w}_j (r_j x + t_j y + c_j), \qquad for\ j = 1,2 \tag{34}$$

In which $\bar{w}_i$ is the layer 3 output, $\{p_i . q_i . r_i\}$ are the resultant parameters, and $f_1$ and $f_2$ are the fuzzy if-then rules (Equations 26 and 27).

The final output layer is layer 5, where the node sums the receiving signals from layer 4, which is explained in Eq. 35.

$$LO_5^j = \sum \bar{w}_j f_j = \frac{\sum_j w_j f_j}{\sum_j w_j} = overall\ output, \qquad for\ j = 1,2 \tag{35}$$

A linear combination of the resulting parameters gives the final output of the ANFIS, which is explained by Eq. 36.

$$\begin{aligned} f_{out} &= \bar{w}_1 f_1 + \bar{w}_2 f_2 = \frac{w_1}{w_1 + w_2} f_1 + \frac{w_2}{w_1 + w_2} f_2 \\ &= (\bar{w}_1 x) p_1 + (\bar{w}_1 y) q_1 + (\bar{w}_1) r_1 + (\bar{w}_2 x) p_2 + (\bar{w}_2 y) q_2 + (\bar{w}_2) r_2 \end{aligned} \tag{36}$$

### 5.5 Biogeography-Based Optimization Algorithm (BBO)

An evolutionary optimization algorithm named Biogeography-based optimization algorithm is inducted from the geographical distribution of biological organisms and is based on the geographical distribution of biological organisms. This optimization algorithm has common features with other biology-based optimization algorithms like particle swarm and genetic algorithm. Therefore, BBO can be used as an alternative solution for the same problems where PSO and GA are employed. For choosing the best solution, the efficiency of these algorithms can be examined. Mathematical models of biogeography explain the procedure of migration of one species from one island to another, how new species are generated, and how they extinct. The term "island" refers to any habitat that is geographically isolated from other habitats.

The main parameters of BBO are as follows: habitat suitability index (HIS), the geographical areas that are well-matched for biological species concerning the residence, have a high habitat suitability index (HSI), suitability index variables (SIVs), the immigration rate ($\lambda$), the emigration rate ($\mu$), the maximum possible immigration rate to the habitat ($I$), the capacity of the habitat to support the species ($S_{max}$), the maximum emigration rate ($E$), the balance number of species where immigration and emigration rates are equal ($S_0$). Fig. 18 depicts a model of species abundance in a single habitat [72]. To simplify the species model in a single habitat, immigration and emigration curves are considered to be a straight line, whereas they could be involved curves. Due to the graph, $\lambda$ and $\mu$ are functions of the number of species in the habitat. The main steps for implementing the BBO algorithm are presented in Fig. 19 (further details in [73]).

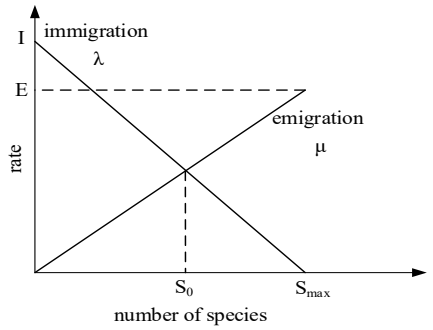



**Fig. 18.** Single habitat species model.

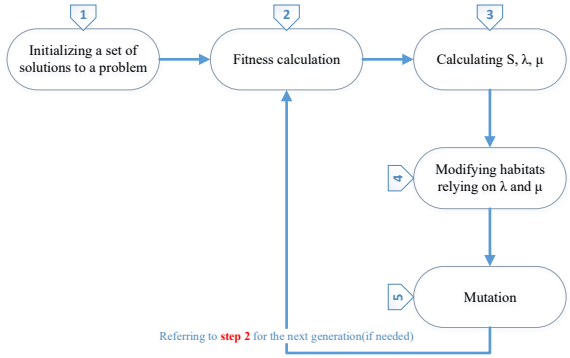



**Fig. 19.** The main steps for implementing a biogeography-based optimization algorithm (BBO) [73].

## 5.6 Teaching Learning-Based Optimization Algorithm (TLBO)

The TLBO algorithm was developed by Rao et al. and is a type of evolutionary optimization algorithm [74]. This population-based algorithm is inspired by the effect of a teacher on learners. This method utilizes a population of solutions to obtain a global solution. The individual members of the TLBO algorithm are represented as a group of learners. The algorithm of TLBO is divided into the "Teacher Phase" and "Student Phase" part. The process of the TLBO optimization algorithm is outlined in Fig. 20. The concept of the Teacher Phase originates from the training from the teacher, and the Student Phase is generated by means of learning from the interplay between learners. A desirable teacher is one who could be able to increase the knowledge of their learners at the same level as themselves. Trainees enhance their knowledge within both the teacher and the learner's interaction. In [40,41], further details about TLBO can be found.

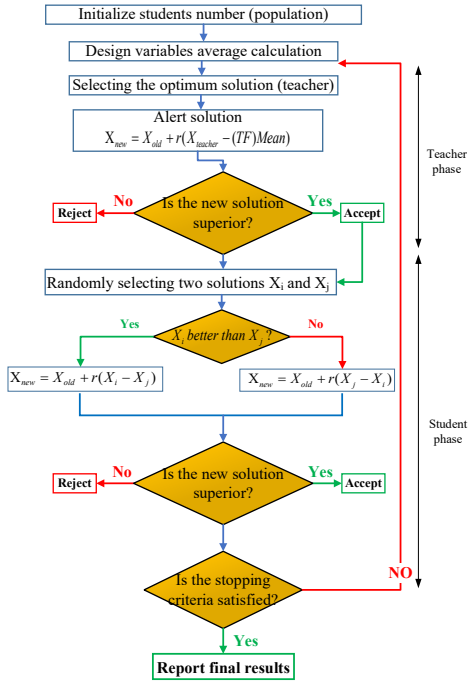



**Fig. 20.** A schematic diagram of the teaching learning-based optimization algorithm (TLBO) algorithm [74].

## 5.7 Particle Swarm Optimization (PSO)

Particle swarm optimization (PSO) is a heuristic optimization technique that was basically put forward by Kennedy and Eberhart in 1995 [76]. The inspiration of the PSO algorithm is relying on the social behavior of birds, insects, and fish and is developed from swarm intelligence. To find the foremost solution, several congested particles (candidate solutions) are generated and moved around the search space. The algorithm is launched by producing primitive particles and assessing their preliminary speeds. The movement arrangement of each particle is established on the particle's experience and the received experience from other particle displacements. The guidance reaches the swarm when better sites are identified. PSO construction is stated in Fig. 21 [77].

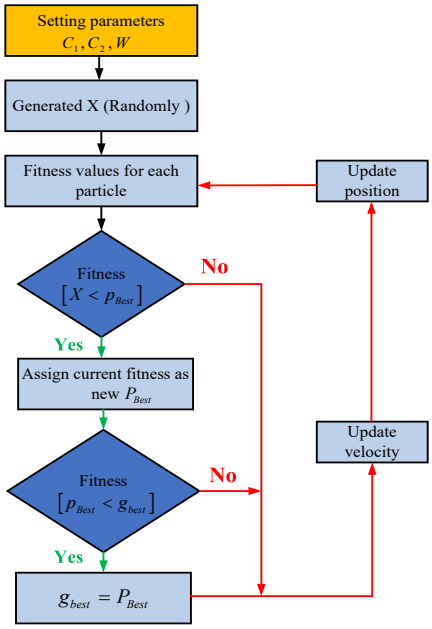



**Fig. 21.** Schematic diagram of the particle swarm optimization process [76].

## 5.8 Genetic Algorithm (GA)

Genetic Algorithm (GA) is a heuristic algorithm commonly used for search and optimization problems, relying on the biological evolution process. In 1960, Holland introduced genetic algorithms based on the evolutionary theory of Darwin. "Natural Selection" and "Genetic Inheritance" are the basic concept for the development of the genetic algorithm [78]. The process begins with a random initial generation of the population and chromosomes as abstract presentations of solution candidates. Population members compete for the right to reproduction. Those who better act in terms of the reduction of the fitness function will be chosen for the next generation. The selection process is organized in a way that only fitter individuals could survive and are selected for the next generation. At the end of each iteration, selected individuals are permitted to generate a progeny population for the next iteration of the GA. This iterative behavior is summarized in Fig. 22. The optimum solution is obtained after some iterations, where the stopping criteria are satisfied, and the program converges [79].

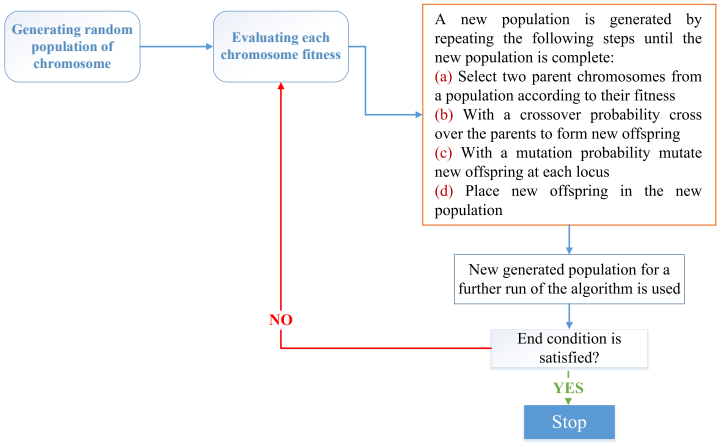



**Fig. 22.** The design structure of the genetic algorithm.

## 5.9 Ant Colony Optimization Algorithm (ACO)

There have been many studies for solving intricate optimization problems by inspiring complex models of the social habits of animals. Ant colony theory was first introduced by Dorigo in 1992. By the inspiration of the behavior of ant colonies, he proposed an optimization algorithm [80]. The foundation of the ACO algorithm is to detect the most diminutive pathway between the ant den and food. In the food discovery process, some amount of pheromone trail is left by every aunt to facilitate root identification by others. The probability of a trail to be followed by ants is conditioned on the amount of leaving pheromone. Therefore, the likelihood of choosing pathways with more pheromones is higher. During this process, ants determine the shortest route from the colony to the food sources. The structure of the ACO optimization algorithm is provided in Fig. 23.

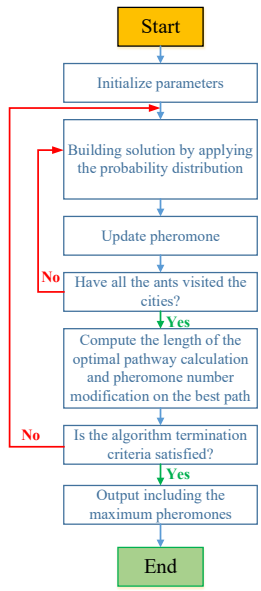



**Fig. 23.** Structure of ant colony (ACO) optimization algorithm [80].

## 5.10 Differential Evolution (DE)

Differential evolution (DE) is a sort of metaheuristics method practice to solve optimization problems (trying to improve the fit solutions), iteratively. This population-based algorithm (presented by Storn in 1995) has three foremost advantages: independent of the initial parameter values, operating with few control parameters, and fast convergence [81]. DE is a stochastic optimization method and is greatly effective for optimization problems with non-linear constraints [82]. The main feature that distinguishes DE from other evolutionary algorithms like the genetic algorithm is the mutation arrangement and selection process that forms self-adaptive DE [83]. Because of the robustness, stability, and ability for global search, DE propelled into one of the most powerful optimization algorithms. The iteration of the algorithm starts with four-part: initialization, mutation, crossover, and selection. Different termination criteria like the maximum iteration number, the maximum number of function appraisals, or the maximum fulfilling time could be set to stop the process for the optimal solution. Fig. 24 provides a flowchart of the DE algorithm [84].

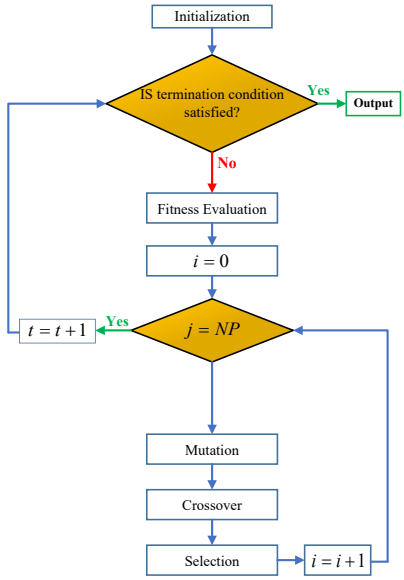



**Fig. 24.** Differential evolution algorithm flowchart [84].

## 5.11 The proposed Hybrid method to develop refining identification models

For the least variation in pulp quality, three input variables could be regulated to control the motor load, pulp consistency, and chip transfer screw speed [3]. The production rate could be simply calculated by multiplying the transfer screw speed ($R$) by a proportionality coefficient [85]. Variation in chip's density and chip dry solid contents are of the external disturbances affecting the TMP refining process [86].

In this study, to identify the model for predicting the refining motor load and steam generation with the available data sets, four main effective refining variables are taken into account as input parameters of the model. The model input and target variables are provided in Table 4. The AI methods based on the integration of the ANFIS and six different heuristic algorithms are examined to obtain the most accurate system identification. In the first stage of the proposed method, the whole data set is reduced to two-part by creating Train and Test data sets to train the AI network and evaluate the

network accuracy, respectively. Training and testing datasets are created to train the data with 70% of the dataset (2952 samples) and test the efficiency of the artificial neural network (ANN) by the remaining 30% of the dataset (1265 samples). From stage 2 to stage 4, the ANFIS model is created, trained, and optimized for higher accuracy by optimizing the model parameters utilizing six different dominant heuristic optimization algorithms. The final results are evaluated in step 5 to achieve the optimal simulation model. The optimal combination of the ANFIS model and optimization algorithm that has higher accuracy and efficiency is selected as the final model and output of the proposed method. Fig. 25 illustrates the proposed data-driven based intelligent simulation approach of the refining process in the thermo-mechanical pulping industry. Descriptions of the performance of each optimization method are provided in previous chapters.

As displayed in Table 4, The network is constructed based on the ambient temperature (℃), dilution water (kg/s), plate gap (mm), and chip transfer screw speed (rpm) as input variables (predictive variables), where, except for the ambient temperature, other input variables could be considered manipulated variables. The mentioned variables are used to determine the refining motor load (MW) and refining produced steam (kg/s). A comprehensive dataset with a wide range of variation is provided for this investigation.

**Table 4.** Artificial neural network (ANN) model input and target variables.

| | Refining Plate Gap | Refining Dilution Water | Site Ambient Temperature | Feeder Screw Speed | Refining Motor Load | Refining Generated Steam |
|---|---|---|---|---|---|---|
| predictor variables | ✓ | ✓ | ✓ | ✓ | | |
| Target variable | | | | | ✓ | ✓ |

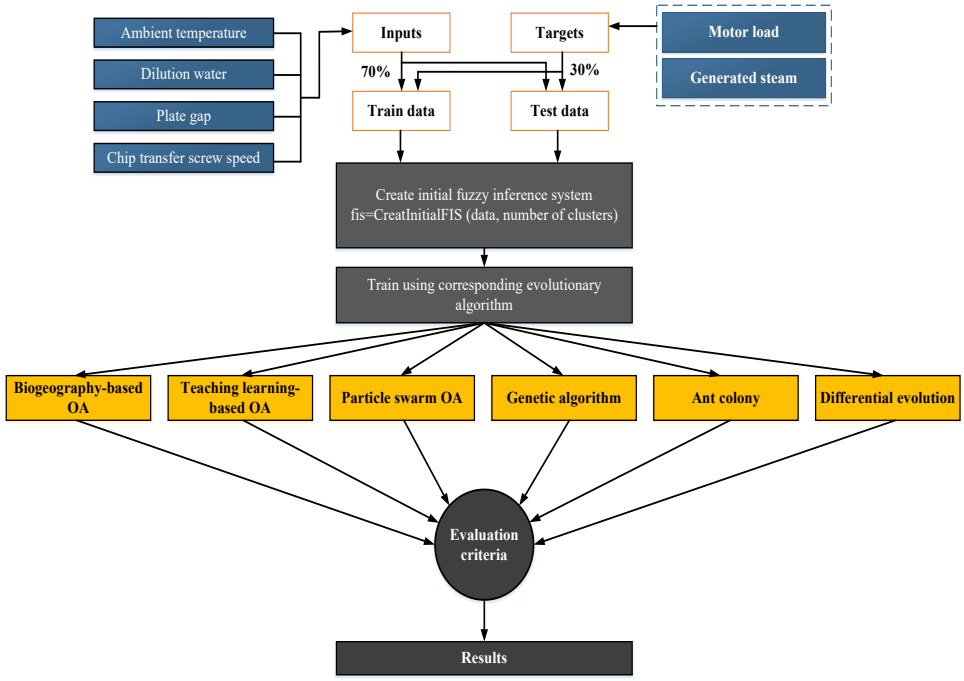



**Fig. 25.** The structure of the proposed method for refining energy simulation.

The “CreateInitialFIS” function is employed to build an initial fuzzy inference system (FIS) for the model. The fuzzy inference system is constructed based on genfis3 in the MATLAB software library (Fig. 26-a). The concept of fuzzy c-means (FCM) clustering is practiced to model the data behavior where a set of rules are cited. Khosravi et al. [87] compared three types of ANFIS, including ANFIS-FCM (fuzzy c-means), ANFIS-SC (subtracting clustering), and ANFIS-GP (grid partitioning). They reported that ANFIS-FCM performs better than the ANFIS based on SC and GP. For this reason, in this study, the ANFIS model was created based on genfis3 (FCM) in Matlab Software with a number of clusters of 10.

The number of clusters and input data is the input argument of the created function. The AI model is trained by employing different evolutionary algorithms (step 4 in Fig. 25). For instance, Fig. 26-b presents the function of Train Using PSO. The objective of the process is to define the optimum FIS parameters. $p_1^0, p_2^0, \dots, p_n^0$ are the primary values of FIS parameters, while $p_1^*, p_2^*, \dots, p_n^*$ introduce the optimal parameters. Eq. 37 describes the optimization problem. The best $x$ value is specified utilizing an evolutionary optimization algorithm (Fig. 26-c).

$$p_i^* = x_i p_i^0 \tag{37}$$

a)

```
function fis=CreateInitialFIS(data,nCluster)
Statements...
fis=genfis3(x,t,'sugeno',nCluster,fcm_options);
end
```

b)

```
function bestfis=TrainUsingPSO(fis,data)
    % Problem Definition
    p0=GetFISParams(fis);
    Problem.CostFunction=@(x)TrainFISCost(x,fis,data);
    Statements...
    %% Run PSO
    results=RunPSO(Problem,Params);
    %% Get Results
    p=results.BestSol.Position.*p0;
    bestfis=SetFISParams(fis,p);
end
```

c)

```
function results=RunPSO(Problem,Params)
% Problem Definition
    CostFunction=Problem.CostFunction; % Cost Function
    Statements...
%% PSO Parameters
Statements...
    MaxIt=Params.MaxIt;
    nPop=Params.nPop;
    w=1;              % Inertia Weight
    wdamp=0.99;       % Inertia Weight Damping Ratio
    c1=1;             % Personal Learning Coefficient
    c2=2;             % Global Learning Coefficient
%% Initialization
Statements...
%% PSO Main Loop
for  it=1:MaxIt
    for i=1:nPop
Statements…
    end
    BestCosts(it) = BestSol.Cost;
    Statements...
end
    results.BestSol=BestSol;
    results.BestCost=BestCost;
end
```

**Fig. 26.** (a) creating an initial fuzzy inference system, (b) the function of the particle swarm optimization algorithm (PSO) algorithm for training the ANFIS, and (c) PSO structure.

### 5.12 Multi-layer Perceptron Neural Network (MLPNN) for Refining Identification Model

In this thesis, the MLPNN method has been utilized to generate the refining identification model to predict the refining motor load (publication 3). In addition, this method has been used in other research (publication 4) to generate refining energy (refining specific energy consumption) and pulp quality (final pulp fiber length and freeness) identification models to develop an advanced refining optimization strategy which is discussed in chapter 6.

Deep learning methods and intelligent techniques are potent approaches to modeling complex systems as a black box [88]. Widely deep learning is referred to as neural networks with high sequential (deep) layers. The most prevalent feedforward neural network is Multilayer perceptron neural networks (MLPNNs). These models are easy to implement and require a smaller training data set [89]. Three sequential layers (which are input, hidden, and output layers) form the multilayer perceptron neural networks. Data and information from the input layer convey to the output layer via the hidden layer. Fig. 27 depicts the mathematical model of a single neuron. The product of multiplying weights ($w_i$) to the input vector ($x_i$) is summed up with the bias ($b$). Ultimately, it passes through the activation function ($f$) and gives the neuron output ($y$). Eq. 38 explains the relation between the input and output of each neuron. In this study, the Hyperbolic tangent (tanh) activation function has been

used. The activation function expression has been provided in Eq. 39, where $x$ and $f(x)$ are the activation function's input and output.

$$y = f(\sum_{i=1}^{n} w_i x_i + b) \tag{38}$$

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{39}$$

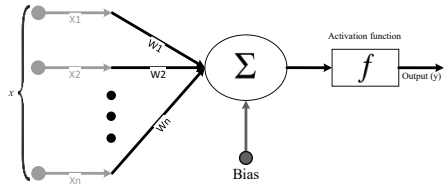



**Fig. 27.** Schematic of an artificial neuron.

Selecting the correct number of hidden layers and the number of neurons, and the learning rate is essential to avoid overfitting and a wrong generalization. Overfitting is one of the errors that could occur while training an artificial neural network. In this case, the error on the trained data can be very small, but when the network is tested with new data, the model error greatly increases. The overfitted model has memorized the training samples and has not learned to predict the new data samples. In general, the optimal performance of MLP requires the optimal selection of hyper-parameters of the number of hidden layers and neurons. Due to the high computational cost of the grid search method to optimize hyperparameters, in this thesis, the random search method is used to develop refining energy and final pulp quality identification models, which has a lower computational load [90]. Therefore, in this research, neural networks with up to 4 hidden layers and a maximum of 20 neurons in each layer were examined. For the dataset that includes all refining predictor variables, models are trained for different hyperparameters settings. Optimal results show that a Multilayer perceptron neural network with four hidden layers, ten neurons in the first and fourth layers, and fifteen neurons in the second and third layers with a learning rate of 0.09 gives better performance compared to the other settings. The Neural Network optimum structure to develop refining identification models is illustrated in Fig. 28. The neural network performance function is a mean square error (MSE) which is discussed in Chapter 5.2.

In order to build a feedforward neural network, 'trainlm' network training function is utilized in MATLAB software. The network bias and weights are being updated according to Levenberg-Marquardt optimization. Although 'trainlm' requires more memory to run, it is still considered the best option for supervised learning as it is the fastest algorithm for backpropagation in the MATLAB toolbox. However, this model has some limitations. The 'trainlm' function employs the Jacobian for computations, which supposes that performance is a mean or sum of squared errors (MSE). Accordingly, networks with this training function must use MSE as a performance function.
The Levenberg-Marquardt algorithm is developed to increase the training speed without the need to compute the Hessian matrix, similar to the Gauss-Newton technique. The Hessian matrix can be approximated by Eq. 40 for the ordinary feedforward network performance function with a sum of squares formation.

$$H = J^T J \tag{40}$$

and the gradient can be computed as:

$$g = J^T e \tag{41}$$

Where $e$ is a vector of network errors and $J$ is the Jacobian matrix including the first derivatives of the network errors with respect to the biases and weights. The Jacobian matrix can be calculated via a standard backpropagation approach which induces less computational load and complexity than calculating the Hessian matrix. Eq. 42 is the Marquardt-Levenberg modification to the Gauss-Newton method:

$$x_{k+1} = x_k - [J^T J + \mu I] J^T e \tag{42}$$

$I$ is the identity matrix. The algorithm is similar to the Gauss-Newton method for small $\mu$. For the large $\mu$, the algorithm becomes gradient descent with a low step length. Newton's method has a higher speed around the minimum error, so it is determined to move toward Newton's method as fast as possible. The $\mu$ decreases after each successful step which reduces the performance function, whereas if an unconfirmed step increases the performance function, it causes to increase $\mu$. Thus, the algorithm is designed in such a way that the performance function is always reduced along with it. A comprehensive description of the Levenberg-Marquardt algorithm is provided in [91]. In addition, the application of the Levenberg-Marquardt algorithm to the neural network concept is discussed in [92]. This algorithm has been shown to have the highest speed in training medium-sized neural networks that can cover up to several hundredweights. Since MATLAB has an internal function for matrix equation solutions, this algorithm shows high performance in MATLAB software, and its features are more prominent in MATLAB.

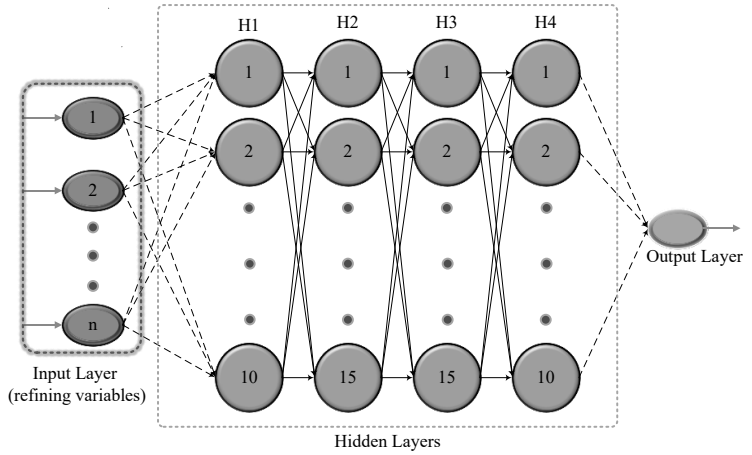



**Fig. 28.** The architecture representation of the optimum ANN consists of four hidden layers

### 5.12.1 MLPNN to develop Refining Motor Load Identification Model

In publication 4, the MLPNN has been utilized to develop a refining motor load identification model. Neural network model inputs are different combinations of measured refining feeder screw speed, dilution water, plate gap, and time, while the only model output is the refining motor load. The number of the measured data points is 8000 in mill 1 and 4300 in mill 2, collected from the mills using online monitoring systems. Fig. 28 shows the topology of the employed multilayer perceptron neural network. A comprehensive dataset with a wide range of variations is provided for this investigation. The dataset is divided into three subsets, namely, train, validation, and test data. The learning model is trained with 70% of the dataset, and half of the remained data, which is 15% of the total data set, is employed to test the network and the remaining 15% for validation. With the use of a training

subset, the gradient is calculated, and the neural network biases and weights are updated. More information regarding the feature selection, parametric study, and results are discussed in section 8.3.

### 5.12.2 MLPNN method to develop refining Energy and Pulp Quality Identification Models

The refining identification model describes the hidden correlation between refining variables. A mathematical model is required to calculate refining-specific energy consumption and pulp quality (fiber length and CSF) based on the refining variables. In this thesis (publication 3) the MLPNN model has been utilized to develop refining specific energy consumption and final pulp quality (fiber length and CSF) identification models. Four neural networks are established and trained to predict the first and second stages of refining specific energy consumption, final pulp fiber length, and final pulp freeness (CSF value). Model input features and outputs for each neural network are explained in section 6.1.

Identification model data sets consist of 3800 measured data from the TMP mill, randomly divided into three parts; 70% as training data set, 15% for validation, and the remaining 15% are dedicated to the testing data set. The training subset contributes to updating the network biases, and weights and calculating gradients. The validation data set evaluates the model fit on the training data set while tuning the model parameters. Validation error determines the running stopping point to avoid overfitting to the training data set. When the model is entirely trained, the test data set is used to evaluate the model's accuracy by the performance function of mean squared error (MSE).

# 6. Advanced Energy-Saving Optimization Strategy in Thermo-mechanical Pulping by Machine Learning Approach

The raw data for this study is derived from refining data collected at a thermomechanical pulp mill in the Nordic region. The studied mill has multiple parallel refining lines with the same technical details. This research considers the first refining line (main refining line) as the case study. The main refining line's schematic and representative measured process data are presented in Fig. 29. Process measured data are refining plate gap, refining dilution water, refining motor load for the first and second refining stages, feeder screw speed, final pulp freeness (CSF), and pulp fiber length. Wood chips from the wood handling unit are pre-heated and fed to the first-stage refiner. Cyclone in the following of each refiner separates the pulp from steam. The first cyclone in the refining line separates backflow steam (which is undesirable). The final pulp is directed to the screening unit after the pulper for post-refining treatment and probable third-stage refining in the reject refiner.

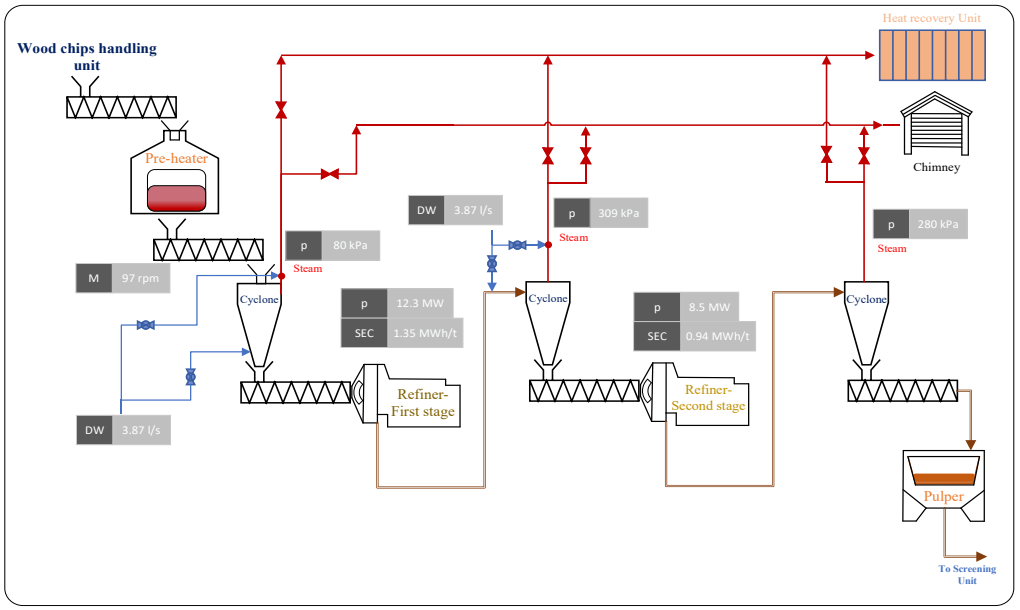



**Fig. 29.** Schematic of the main refining line with sample measured data.

### 6.1 Refining variables and ANN feature selection

Refining variables could be categorized into refining manipulated, disturbance, and operating variables. Manipulated variables are independent variables that influence refining operating variables. Refining operating variables such as production rate, refining specific energy consumption, pulp freeness, and fiber length can be predicted by the combination selection of the manipulated variables. The main two-stage refining process manipulated variables are chip transfer screw speed, primary and secondary refining plate gap, and dilution water [93]. Disturbance variables are basically uncontrollable and disrupt the optimal control of the process by changing the system's operating conditions. The main refining disturbance variables are seasonal changes in input wood chips quality and refiners plate condition. Changes in the quality of wood chips, such as moisture content, affect the pulp consistency and production rate. It should be noted that there is no accurate sensor for measuring pulp consistency, and the consistency measurement includes undesirable errors (in the studied mill, the error could be up to 20%).

On the other hand, the exact rate of pulp production must be calculated based on pulp consistency, but due to errors in measuring the pulp consistency, pulp mills usually calculate the production rate according to the empirical equations based on screw feeder speed. This calculation method also has an error, but according to experience, it seems more reliable than other consistency-based calculation methods. On the other hand, with the operation of refiner plates over time, these plates wear out due to the harsh conditions inside the refining zone, which causes changes in the system's operating condition. Reference [3] reviews the effect of the main variables of wood chips qualities and the refining process on the final quality of the TMP product. To apply the effect of disturbance variables that affect the performance of the refining process, publication 3 suggests adding the time variable as one of the predictor variables of the ANN refining identification model.

Three data sets are prepared to predict refining energy and quality operating variables by training four neural network models. Each dataset includes selected feature combinations as ANN model input variables for the particular refining operating variable prediction. Data set 1 is responsible for creating the first-stage refining specific energy consumption (RSEC) identification model. The second data set (data set 2) is used as the set of selected model features to simulate the second stage refining specific energy consumption, while the third data set (data set 3) is dedicated to final pulp freeness (CSF) and fiber length prediction. The total RSEC forms the objective function of the optimization model. Table 5 presents the data sets details which construct the neural networks' input feature combinations.
Table 6 illustrates the different refining neural network identification models' input (predictor) and output (target) variables.

**Table 5.** Details of each neural network data set

| Input data combinations | Refining plate gap – 1st stage | Refining dilution water – 1st stage | Refining plate gap – 2nd stage | Refining dilution water – 2nd stage | Feeder screw speed | Time |
|---|---|---|---|---|---|---|
| Data set 1 | ✓ | ✓ | ✗ | ✗ | ✓ | ✓ |
| Data set 2 | ✗ | ✗ | ✓ | ✓ | ✓ | ✓ |
| Data set 3 | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

**Table 6.** Refining neural networks identification models input (predictor) and output (target) variables.

| $ANN^1$ (Refining energy identification model) | | $ANN^2$ (Refining energy identification model) | | $ANN^3$ (Pulp quality identification model) | | $ANN^4$ (Pulp quality identification model) | |
|---|---|---|---|---|---|---|---|
| Predictor variables | Target variable | Predictor variables | Target variable | Predictor variables | Target variable | Predictor variables | Target variable |
| Data set 1 | RSEC-First stage | Data set 2 | RSEC-Second stage | Data set 3 | Fiber length | Data set 3 | Freeness (CSF) |

## 6.2 Refining energy-saving two-phase optimization plan

The simplified construction of the two-phase optimization plan is presented in Fig. 30. The optimization process starts with phase 1, where generating refining energy and pulp quality identification models (neural networks 1 to 4). As mentioned earlier, four datasets have been used to model refining energy consumption and final refined pulp quality, the details of which are given in Table 5 and
Table 6. After the training phase (phase 1), neural networks are used to form the GA optimization constraints and objective function in phase 2. The optimization method is limited to keep the paper's quality (CSF value and fiber length) at the expected level (current operating condition) while minimizing the model objective function, which is the summation of the first and second stage refining specific energy consumption (RSEC). Other optimization constraints define the boundaries of the decision variables based on the real system operating condition and the component characteristic. The optimization decision variables are the first and second-stage refining dilution water and plate gap. Once the optimization is completed, the model reports the optimal values of the primary and secondary refining dilution water, plate gap, and refining-specific energy consumptions as the final results. The obtained total refining-specific energy consumption from the optimization plan is compared to the measured values of the total refining-specific energy consumption from the mill (current system operating condition) to evaluate the performance of the proposed optimization method to increase the refining energy efficiency. The following gives more information about the optimization objective function and constraints.

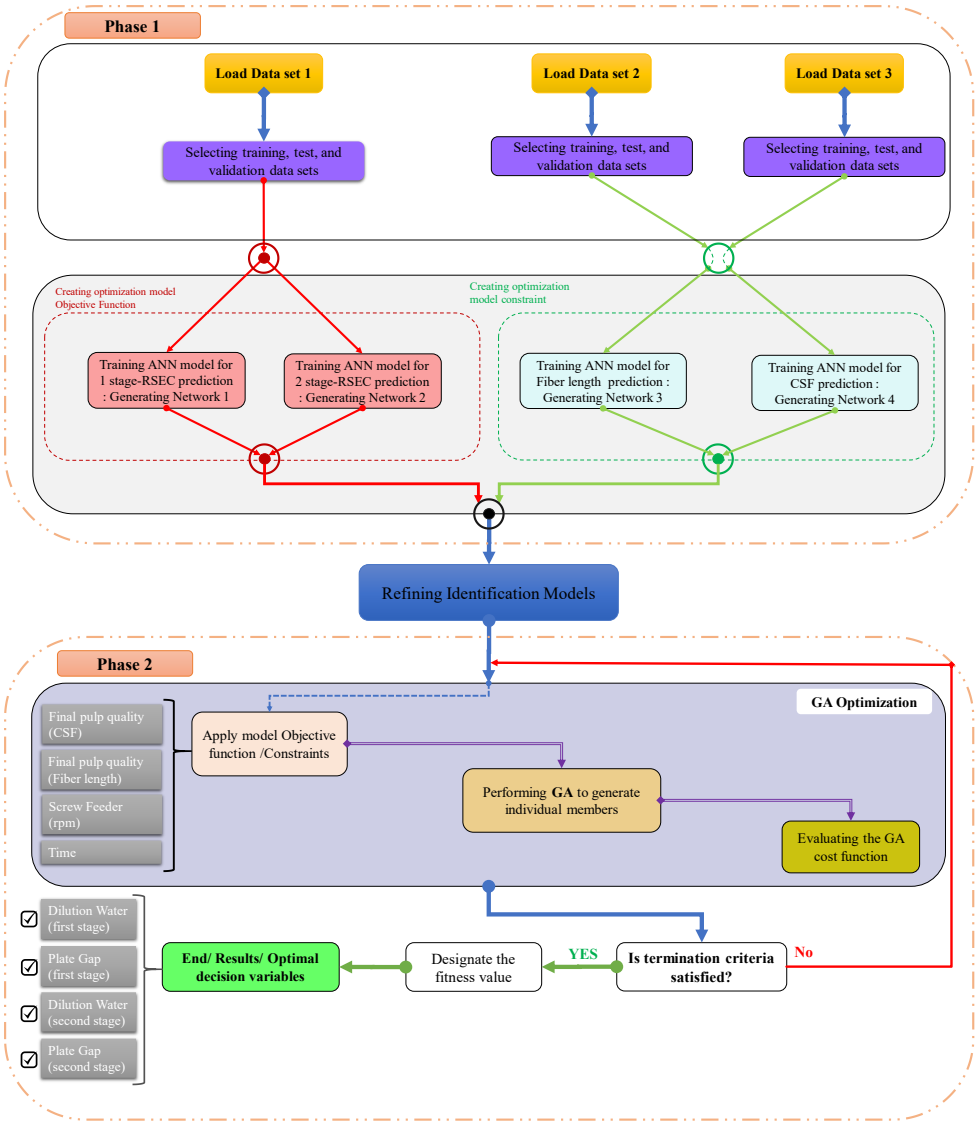



**Fig. 30.** Schematic representation of the proposed advanced energy-saving refining optimization strategy.

*Optimization Model Objective Function:* The optimization objective function given in Eq. 43 is the sum of the primary and secondary refining-specific energy consumptions. Refining is responsible for the most changes in fiber morphology and is the most energy-intensive process in the thermo-mechanical pulping process. In our case study, the average motor load for the secondary refiner (pulp refiner) is 70% of the primary refiner (wood refiner), making the first stage of refining more energy-intensive.

$$ObjFcn: RSEC^1 + RSEC^2 \tag{43}$$

*Optimization Model Constraints:* The optimization model constraints are about the limits on the upper and lower boundaries of the decision variables as well as the expected quality of the paper, including the pulp freeness value and fiber length. Quality constraints are defined as inequality constraints to give more flexibility to the model to avoid causing the probable infeasible optimization model. For this reason, the average neural network error is considered an acceptable interval at each

data point that the CSF value could settle to satisfy the model pulp quality constraint. This fact is also true for fiber length as a measure of pulp quality control. The following equations show the optimization model constraints:

$$FL^{m,j} - FL^{m,j} \times MAPE^{ANN^k} \leq FL^{j} \leq FL^{m,j} + FL^{m,j} \times MAPE^{ANN^k,j} \quad j = 1{:}N; K = 3 \tag{44}$$

$$CSF^{m,j} - CSF^{m,j} \times MAPE^{ANN^k} \leq CSF^{j} \leq CSF^{m,j} + CSF^{m,j} \times MAPE^{ANN^k,j} \quad j = 1{:}N; K = 4 \tag{45}$$

$$DW^{min,i,j} \leq DW^{i,j} \leq DW^{max,i,j} \quad j = 1{:}N; i = 1 \tag{46}$$

$$PG^{min,i,j} \leq PG^{i,j} \leq PG^{max,i,j} \quad j = 1{:}N; i = 1 \tag{47}$$

$$DW^{min,i,j} \leq DW^{i,j} \leq DW^{max,i,j} \quad j = 1{:}N; i = 2 \tag{48}$$

$$PG^{min,i,j} \leq PG^{i,j} \leq PG^{max,i,j} \quad j = 1{:}N; i = 2 \tag{49}$$

Where $i$ indicates the refining stage (i=1,2) and j is the index for each data point. k specifies the neural network number, and $m$ stands for the measured value. $N$ is the total number of data points.

# 7. Advanced energy and cost-efficient design and operation of Energy Hub for Residential and Forest Industry Applications

This section introduces advanced energy and cost-efficient design and operation of the Energy Hub for residential (high-rise building) and industrial (forest industry) applications. The EH advanced cost-efficient design method was first developed for the EH consisting of a CCHP system to supply high-rise residential building heating and cooling demand. Having shown that the proposed method can reduce the total costs of the system effectively, this method was also used to integrate a thermo-mechanical pulp mill and paper machine in another study through heat recovery and the EH concept. The proposed approach is based on the integration of thermo-economic analysis, reliability & availability analysis, and EH load prediction (utilizing advanced machine learning time series models).

## 7.1 Case study 1

The first case study includes the residential application of an Energy Hub (EH) consisting of combined cooling, heating, and power (CCHP) system. On the demand side, the energy consumer is a high-rise residential building that provides its cooling and heating demands through the EH. In the first modeling approach (first scenario), the system is assumed to be always available without failure possibility for the subsystems. However, there is a likelihood of failure of each component in the real operating condition, and the EH operator is required to pay the fee for every kilowatt-hour shortage in supply. In the second modeling approach (second scenario), the failure modes of the subsystems are modeled based on the contribution of the state space and the continuous Markov method. A comparison of the results with the first scenario is conducted to determine the efficiency of the proposed method.

Schematic of the EH equipped with an absorption chiller, a vapor-compressor chiller, a boiler, and a CHP gas engine is illustrated in Fig. 31. The performance of each EH component is as follows: The boiler and CHP systems are being fed by a natural gas distribution network. CHP is always on duty at full capacity and produces electricity and heat. The generated electricity by the CCHP system is sold to the electricity network and in the case of failure in the electricity grid, provides the electricity demand of the vapor compressor chiller in summer. The generated heat by the CHP and boiler units is used to supply the absorption chiller in summer and building heat demand in the winter. The basic cooling load is supplied by the absorption chiller, and if the absorption chiller cannot afford the load on its own, the vapor-compressor chiller will be lit up to help fully meet the cooling demand. As illustrated in Fig. 31, the EH input resources are natural gas and electricity supplied by the distribution network. The ultimate production of the CCHP system is cooling, heating, and electricity.

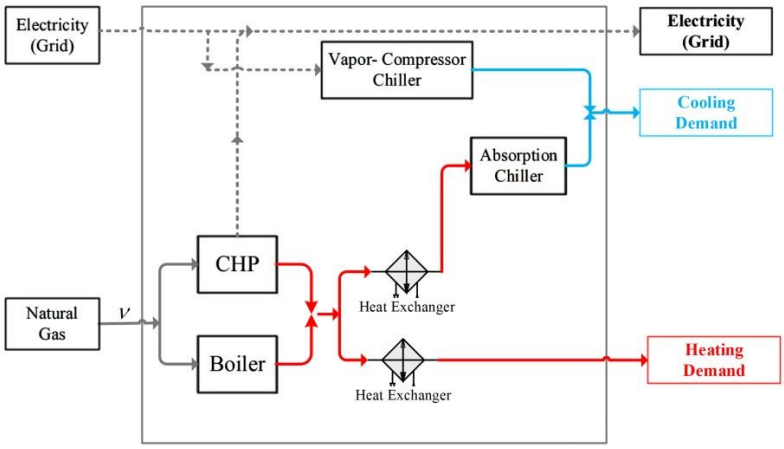



**Fig. 31.** Schematic of the studied Energy Hub (case study 1)

## 7.2 Case study 2

The second case study is the industrial application of EH for the optimal integration of the TMP mill and paper machine through heat recovery. The schematic of the EH, integrated with the paper machine, TMP mill, and electricity grid, is depicted in Fig. 32. The LSTM model predicts the 1-hour ahead TMP heat production and sends the information to the central hub controller (CHC). A central hub controller also receives the hourly heat demand from the paper machine. Heat demand rarely fluctuates at the paper mill, so developing a heat demand prediction model hasn't been considered in this research. At each time step, the CHC receives information about the state of the heat supply system illustrated in the state-space diagram (Fig. 35). The heat supply system consists of the EH and TMP mill. The EH's energy conversion components are the steam generator electric boiler and heat pump, where the heat pump includes a series of five subsystems: compressor, evaporator, condenser, expansion valve, and heat recovery steam generator. As a series system, the entire heat pump fails in case of failure of any subsystems. The electricity grid supplies the electric boiler and heat pump electricity demand at the market price. The EH unit commitment depends on various factors, such as the electricity market price, energy conversion components efficiency, the state of the system (Fig. 35), TMP heat production, and paper machine heat demand. Electrification is an important reason for choosing an electric boiler and heat pump for the hub components.

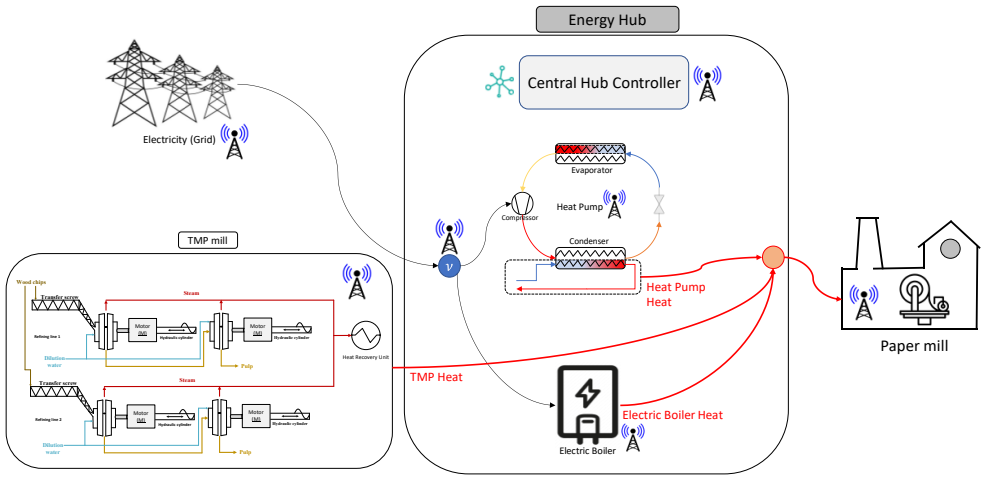



**Fig. 32.** General schematic of the studied system (case study 2).

## 7.3 Reliability and availability analysis

### 7.3.1 General Modeling Concept

Reliability represents the likelihood of an object being healthy under a particular period. The reliability analysis techniques are typically categorized into quantitative and qualitative approaches. Failure modes and effects analysis are the most common qualitative approach. The major quantitative approaches are reliability block diagrams, fault tree analysis, and Markov analysis. The random behavior of repairable systems, such as electric boilers and heat pumps, can be analyzed by the Markov method.

The Markov chain is a stochastic random memoryless method. The probability distribution of the following state is only affected by the present state, and the history of the states is not influential. The Markov chain contains a set of state transitions. For the system reliability and availability analyses, it is necessary to characterize the state transitions for a single step or time interval in a matrix format. A stochastic matrix describes the probability distribution of these state transitions, where each row explains the different system operating conditions. In this study, the state-space method (SSM) is combined with the Markov chain to analyze each subsystem's random failure mode and evaluate the whole heat supply system's availability. The SSM is the best technique for quantitative reliability and availability assessment of repairable and complex systems [94]. The SSM consists of four essential steps: 1-Defining the system boundaries and all the system's possible operating and failed states, 2- Building the state transition diagram, 3- Figuring the ultimate state probabilities during the system lifespan via different methods such as a frequency balance approach.

Fig. 33 shows the state-space diagram of a repairable device with a constant failure and repair rate consisting of two states. The system is referred to as 'up' when the system is healthy and operating as normal. In contrast, the state of a failed system is mentioned as a 'down' state. Important model variables are mentioned below.

$p_0(t)$= *Probability of the operable component at time t*

$p_1(t)$= *Probability of the failed component at time t*

$\mu$= *repair rate*

$\lambda$= *failure rate*

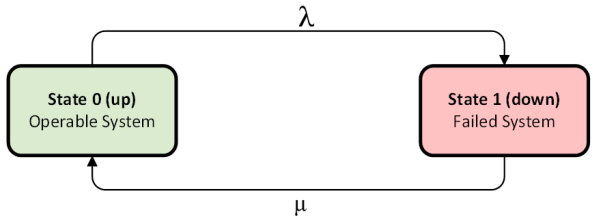



**Fig. 33.** State-Space diagram of a repairable system consisting of one component.

### 7.3.2 Frequency balance approach

The frequency balance approach is an alternative method for obtaining steady-state or limiting-state probabilities values. The concept beyond the frequency balance approach is that in an ergodic system, the leaving and entering expected frequency for each state must be equal. Limiting state prob-

abilities are the unknowns of the frequency balance equations; the summation of limiting state probabilities must be unity. This approach is implemented for a two-state model (Fig. 33). For this case, the frequency balance equations are as follows:

State 0 $-\lambda p_0 + \mu p_1 = 0$ (50)

State 1 $\lambda p_0 - \mu p_1 = 0$ (51)

Eq. 50 and Eq. 51 are not independent. Therefore:

$$p_1 = p_0 \frac{\lambda}{\mu} \tag{52}$$

As mentioned before, the summation of the state probabilities in the state space diagram must be equal to one:

$$p_0 + p_1 = 1 \tag{53}$$

And then:

$$p_0 = \frac{\mu}{\lambda + \mu} \quad \text{and} \quad p_1 = \frac{\lambda}{\lambda + \mu} \tag{54}$$

To streamline the process, the Frequency Balance Method is well suited to use a stochastic transitional probability matrix for the valuation of limiting state probabilities. The approach is to define $\alpha$ as the limiting state probability vector which maintains constant when it is multiplied by the stochastic transitional probability matrix, that is,

$$\alpha p = \alpha \tag{55}$$

For the single repairable component, $\alpha$ is given by $[p_0 \quad p_1]$ and $p$ can be written as follows:

$$P = \begin{bmatrix} 1-\lambda\Delta t & \lambda\Delta t \\ \mu\Delta t & 1-\mu\Delta t \end{bmatrix} \tag{56}$$

Using $\alpha$ and $p$ in Eq. 55:

$$[p_0 \quad p_1] \begin{bmatrix} 1-\lambda\Delta t & \lambda\Delta t \\ \mu\Delta t & 1-\mu\Delta t \end{bmatrix} = [p_0 \quad p_1] \tag{57}$$

Which when written in explicit form, gives

$$-\lambda p_0 + \mu P_1 = 0 \quad \text{and} \quad \lambda p_0 - \mu P_1 = 0 \tag{58}$$

Which are equal to Equations 50 and 51, which again give the same result.

### 7.3.3 Failure and the Repair rate of Series Systems

The repair and failure rate of the system consisting of a series of components are discussed in this section. Each system component's failure and repair rate is essential to formulate the frequency balance equations. Reference [95] introduces the calculation method of the failure and repair rate of the series system with a constant transition rate. The calculation method of failure and repair rate of a system made up of N-series components is provided in the equations below.

$$\lambda_N = \sum_{j=1}^{N} \lambda_j \tag{59}$$

$$\mu_N = \frac{\sum_{j=1}^{N} \lambda_j}{\sum_{j=1}^{N} \lambda_j / \mu_j} \tag{60}$$

Where N is the number of series components, and j determines each subsystem component. The heat pump consists of a series of five components more likely to fail: compressor, evaporator, condenser, expansion valve, and heat recovery steam generator. The failure and repair rate of the whole heat pump can be calculated by substituting each subsystem's failure and repair rate in Eq. 59 and Eq. 60, respectively.

## 7.4 Limiting state probabilities

To simplify using the frequency balance approach for the continuous and discrete Markov process, first, the state-space diagram must be formed by applying the transition rates to the graph. Observing all the potential states in the state-space diagram is essential. There is no primary restriction on the number or the type of states and transitions. Preparing the proper diagram is a crucial step in reliability and availability analysis. The following shows the state-space diagram of the studied systems (case studies 1 and 2).

### 7.4.1 Case study 1

The first case study is an Energy Hub (EH) consisting of combined cooling, heating, and power (CCHP) system to supply high-rise residential building cooling and heating demand. The CCHP system state-space diagram with consideration of five central subsystems that are more likely to fail is presented in Fig. 34. Regarding the low limiting state probabilities of some states, it is conceivable to omit these states from the state-space diagram. For instance, in the studied EH, the possibility of simultaneous failure of three subsystems is rare; so, states with three failed components are neglected and removed from the diagram. This modification reduces the computational cost in a significant way. There are 16 states in the diagram, and there are states with the collapse of four or five components simultaneously that are removed from the graph due to the small limiting probabilities. The boiler, vapor-compressor chiller, absorption chiller, CHP, and power grids are components that are more likely to be failed, and the state-space diagram consists of the cited components. By implementing the frequency balance approach with considering constant failure and repair rate during the time, limiting state probabilities of the system have been acquired. For the CCHP system, $P$ , and $\alpha$ matrixes are provided in the Appendix (see Appendix. A).

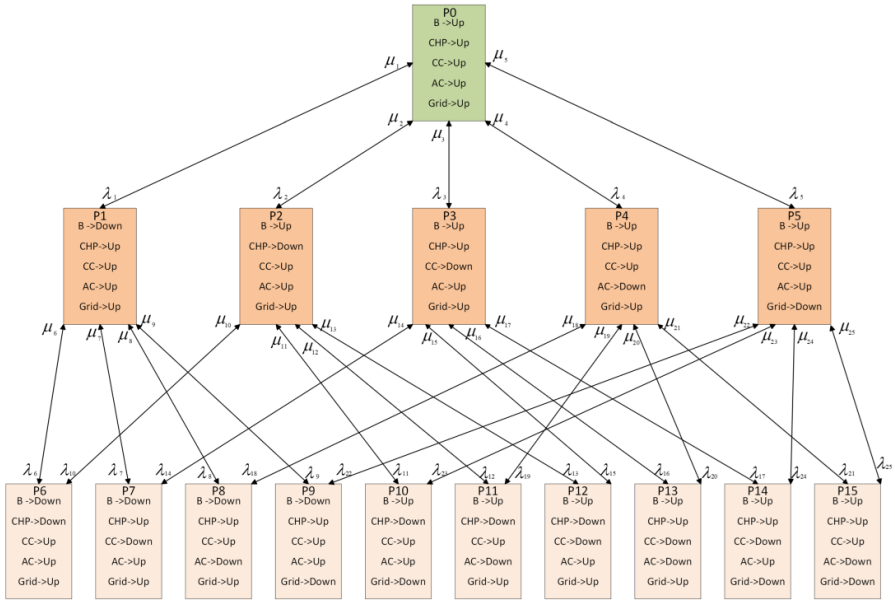



**Fig. 34.** State-space diagram of the EH containing a CCHP system

### 7.4.2 Case study2

The second case study consists of the implementation of the EH in the forest industry for the optimal integration of the TMP mill and paper machine. The system schematic is given in Fig. 32. In Fig. 35, we present the state-space diagram for the studied system, taking into account four central subsystems that are more likely to fail: steam generator heat pump (HP), electric boiler (EB), and refining lines in TMP mill (RL1 and RL2). The computational load can be reduced by omitting states with low limiting state probabilities from the state-space diagram. As an example, in the studied system, it is unlikely that three or more subsystems fail simultaneously; therefore, states with more than two failed components are ignored. There is a significant reduction in computational cost due to this modification. In the diagram, 11 states are included, of which states with more than two collapses at the same time have been removed due to a small limiting state probability. A state-space diagram consists of the most likely to fail components, such as an electric boiler, a heat pump, and refining lines in a TMP mill. As stated earlier, a heat pump is made of five series components where; the system's failure and repair rate calculation is discussed in section 7.3.3. A frequency balance analysis with constant failure and repair rates over time has been implemented to get the system's limiting state probabilities.

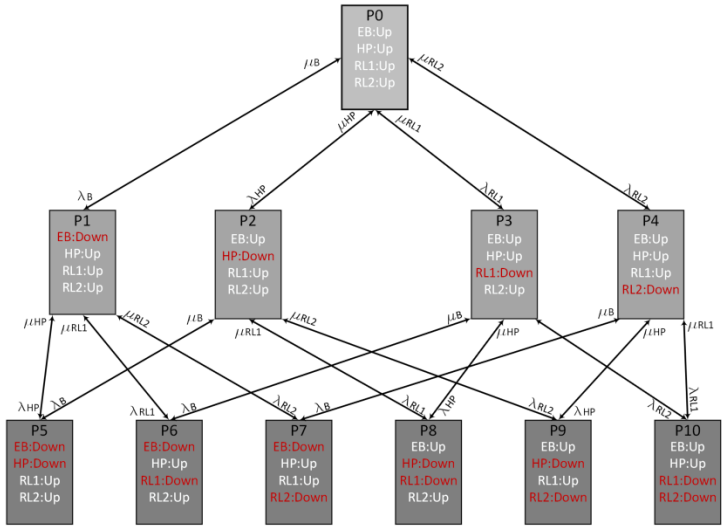



**Fig. 35.** State-space diagram of the heat supply system consisting of Energy Hub and TMP mill.

## 7.5 Economic analysis

Following is an overview of the economic parameters used in this research to design and operate a cost-effective EH system for different case studies. Over the system's lifetime, all annuities will be converted to 2016 in case study 1 and 2020 in case study 2.

### 7.5.1 Net present value

The net present value of the operation and maintenance cost during the plant life can be calculated utilizing the annuity factor (AF). This coefficient calculates the current values of annuities. Using the interest rate i, the annuity factor is defined as:

$$AF = \frac{1-(1+i)^{-m}}{i} \tag{61}$$

Here, i denotes the interest rate, and $m$ specifies the total system operating period which is 20 years. By using $AF$ coefficient all the annual costs including annual operation and maintenance costs (during 20 years) are converted to the base year. This value is 18.04 in our case study considering 0.01 as the interest rate.

### 7.5.2 Capital Costs

An initial capital cost is incurred when a hub designer decides to install a specific component. Every EH component has a capital cost determined by its design parameters. A design variable that influences capital costs the most is capacity. The regression method can roughly represent the capital cost of each EH subsystem as a function of its capacity.

- *Absorption chiller:* In Eq. 62, the relation between the capital cost and the capacity of an absorption chiller is observed [96].

$$M_{AB,chiller}(\$) = 1144.3(\dot{Q}_e)^{0.67} \tag{62}$$

Where $\dot{Q}_e$ is the cooling load of the absorption chiller in kW.

- *Vapor-compression chiller:* The capital cost of a chiller is approximately a function of its cooling load. That can be seen in the equation below [97].

$$M_{VC,chiller}(\$) = 1308(\dot{Q}_e)^{0.77} \quad (63)$$

Where $\dot{Q}_e$ is the cooling load of the vapor-compression chiller in kW.

- *Boiler:* For calculating the cost of the boiler unit, the following expression is used:

$$M_{Boiler}(\$) = 500C_{Boiler}(kW) \quad (64)$$

Where $C_{Boiler}$ is the boiler capacity in kW.

- *CHP:* The Investment costs of a gas-engine CHP plant range from 850-1950 $/kWe depending on the system capacity [98]. According to reference [99], CHP investment cost is estimated by a third-degree polynomial in Eq. 65 as a function of capacity.

$$M_{CHP}(\$) = (8 \times 10^{-5})x^3 - 0.79x^2 + 1645x + 8000 \quad (65)$$

Where $x$ is the CHP capacity in kW.

- *Electric boiler and heat pump:* An initial capital cost is incurred when a hub designer decides to install a specific component. Every EH component has a capital cost determined by its design parameters. A design variable that influences capital costs the most is capacity. Linear regression can roughly represent the capital cost of each EH subsystem as a function of its capacity [100], [101].

$$CC_{EB}(€) = 100\ C_{EB}(kW) \quad (66)$$
$$CC_{HP}(€) = 400\ C_{HP}(kW) \quad (67)$$

Where $C_{EB}$ and $C_{HP}$ are the electric boiler and heat pump capacity in $kW$, respectively.

### 7.5.3 Case study1

*Energy carrier prices:* Tehran's electricity and gas energy prices are considered energy carrier prices [102]. Electricity rates are based on TOU (time of use) pricing; Time-of-use rates are different on the time of electricity usage. There are three periods: Off-peak, When the electricity demand is the lowest; Tehran households and small businesses use the majority of their electricity– during the off-peak hours. Mid-peak, when electricity demand is moderate; These periods are during the daytime, but not the busiest times of the day. On-peak, when demand is the highest; These are the busiest times of day – generally when people are cooking, starting up their computers, and running heaters or air conditioners.

*Penalty costs:* Unsupplied demand penalty cost is encompassed in operation cost as a load loss cost [103]:

$$PC_{i,t} = TLL_{i,t} \times Vll^{TH} \quad (68)$$

Where, $PC_{i,t}$ *and* $TLL_{i,t}$ is the penalty cost and thermal load loss at time t and state i, respectively. $Vll^{TH}$ is the value of thermal unsupplied loads cost ($/kWh).

*Operation costs:* The operation cost ($OC$) is divided into fuel and unsupplied demand penalty costs. The operation cost of the CCHP unit which consists of energy ($EC$) and penalty cost ($PC$) is shown in Eq. 69.

$$OC_{i,t} = \sum_{n=1}^{N} (EC_{n,i,t} + PC_{i,t}) \quad (69)$$

Where $EC_{n,i,t}$ is the energy cost for the component n, at time t and state of the system i.

*Total cost:* In both the first and second scenarios, the total cost is the cost that is imposed on the EH operator. The total cost consists of investment, maintenance, and operation cost by considering the failure probability for system components. Where operation cost is divided into energy cost and unsupplied demand penalty cost. The second term of operation cost is the charge that should be paid by the EH operator as a penalty to the energy consumer for each kilowatt-hour shortage in supply. In this regard, for each kWh shortage in the building thermal demand 3$ is set for the unsupplied demand penalty cost [103]. This cost will be imposed on the EH operator in the case of a shortage in providing the hourly energy demand. The first term of total cost calculates the operation and maintenance cost of the system. the annuity factor is used to transfer all annuities (which is the operation cost in the studied case) to the base year of analysis. Variable $p_i$ is the limiting probability of the ith state and defines the system operating condition. System real operation cost can be calculated by summation of multiplication of the limiting probability of each state by the operation cost of that state for all states in the space-state diagram. It should be noted that in both scenarios, it is supposed that maintenance for different components is carried out at a specified and predetermined interval when there is no need for that component. Eventually, the total cost in mathematical form is:

$$TC = AF \times \left[\sum_{i=0}^{I}\sum_{t=1}^{T} p_i(OC_{i,t}) + \sum_{n=1}^{N} MC_n\right] + \sum_{n=1}^{N} CC_n \quad (70)$$

Where the second term (system capital cost) in Eq. 70 is equal to:

$$\sum_{n=1}^{N} CC_n = M_{AB,chiller} + M_{VC,chiller} + M_{Boiler} + M_{CHP} \quad (71)$$

### 7.5.4 Case study 2

Following is an overview of the economic parameters used in this research to design and operate a cost-effective EH system to supply forest industry heat demand. Over the system's lifetime, all annuities will be converted to 2020.

*Energy Cost:* Energy cost is the cost of electricity consumed by the EH to produce industrial heat through energy conversion units. Electricity is priced according to the market electricity price in this study. The energy cost calculation for subsystem n, at time t, and state i in the state-space diagram is:

$$EC_{i,t} = EP_{i,t} \times MEP_t \quad (72)$$

Where $EC$, $EP$, and $MEP$ are the energy cost, electric power consumption, and market electricity price, respectively.

*Load Loss Cost:* The EH is responsible for providing the heating load determined by the Central Hub Controller. The EH operator is charged for each $MWh$ shortage in the heating supply. For each $MWh$ of unsupplied heat, there is a charge (unsupplied demand penalty cost) that is significantly higher than the heating price. In this research, the unsupplied demand penalty cost ($pc^{heat}$) is considered to be ten times more than the heating price. The heat penalty cost coefficient defines the extent of unsupplied demand penalty cost based on the selling price of the heat. The following equation shows the mathematical expression to calculate the load loss cost ($LLC$):

$$LLC_{i,t} = HLL_{i,t} \times pc_t^{heat} \quad (73)$$

Where $hll_{i,t}(MWh)$ is the heat load loss at time t and state i in the state-space diagram (Fig. 35). $pc^{heat}(\frac{€}{MWh})$ is the unsupplied demand penalty cost for each $MWh$ of unsupplied heat.

*Operation costs:* Energy cost and load loss cost ($llc$) make up the operation cost. System operation cost over the lifespan ($OC_{Total}$) is given by Eq. 74. $p_i$ defines the system operating condition in the state-space diagram and is the limiting state probability of the ith stage. The system operation cost is the summation of the system operation cost at each operating condition state multiplied by the limiting probability of the state. All the annuities have been converted to the base year by annuity factor ($AF$). $T$ is the number of hours in a year (=8765).

$$OC_{total} = AF \times \left[\sum_{i=1}^{I}\sum_{t=1}^{T} p_i \times [\mathrm{EC}_{i,t} + LLC_{i,t}]\right] \tag{74}$$

*System total costs:* The system total cost in this article encompasses the investment, operation, and maintenance costs over the system lifespan. The system state and availability of the subsystems are considered in calculating the system's total cost. As mentioned earlier, system operation costs include unsupplied demand penalty costs and fuel (electricity) costs. System total cost is one of the measures to evaluate the cost-efficiency of the system.
In Eq. 75, we find the mathematical term that can be used to calculate the system's total cost ($TC$). The annual maintenance cost ($MC$) is considered 10% of the system capital cost. Annual maintenance cost is transferred to the base year by annuity factor ($AF$). The last two terms indicate the system's maintenance and capital cost where $n$ indicates the energy conversion component in the EH.

$$TC = OC_{total} + AF \times \left[\sum_{n=1}^{N} MC_n\right] + \sum_{n=1}^{N} CC_n \tag{75}$$

## 7.6 Energy and cost-efficient EH design

### 7.6.1 Case study1

Regarding the non-linear objective function of the model, the system optimization is accomplished by applying the Genetic optimization (GA) algorithm in the MatLab software. Decision variables are the capacities of the different EH components.

*Objective Function:* Objective functions vary according to the modeling accuracy. Here are two different scenarios, in the first scenario, which is common, the components' failure probabilities are not considered in the optimization, and the subsystems' capacities are calculated by following safe components. This assumption is for the model simplification and unreal. Because of simplifications made in this model, the output of this model is not likely to prove the optimal economic design of the system. The related objective function for the first scenario (scenario 1) is provided in Eq. 76.

$$ObjFcn_1 = AF \times \left[\sum_{t=1}^{T}\sum_{n=1}^{N} (EC_{i=0,n,t}) + \sum_{n=1}^{N} MC_n\right] + \sum_{n=1}^{N} CC_n \tag{76}$$

Here, $MC$ is the annual maintenance cost. The zero index (i=0) indicates the zero state in the state-space diagram where all elements are healthy.
In the second scenario (scenario 2), system modeling is based on the real operating condition of the system. Markov method is implemented to observe the effect of the subsystem's failure on the system availability. In this modeling approach, the optimization objective function is considered as the system total cost (Eq. 70).

*Constraints:* Respective constraints in modeling include the hourly relations between the input and output of EH (coupling matrix) and also the constraints regarding the equipment's capacities. According to the schematic of the system depicted in Fig. 31, the EH coupling matrix is given in Eq. 77.

$$\begin{bmatrix} \eta_{VC,chiller} & \left[\upsilon\,\eta_{thermal}^{CHP}\,\eta_{HEX} + (1-\upsilon)\,\eta_{Boiler}\,\eta_{HEX}\right]\eta_{AB,chiller} \\ 0 & \upsilon\eta_{thermal}^{CHP}\,\eta_{HEX} + (1-\upsilon)\,\eta_{Boiler}\,\eta_{HEX} \end{bmatrix} \begin{bmatrix} p_{e,t} \\ p_{g,t} \end{bmatrix} = \begin{bmatrix} l_{cool,t} \\ l_{heat,t} \end{bmatrix} \tag{77}$$

Where the left matrix is called the coupling matrix, which is formed by components' efficiencies and dispatch factor ($\upsilon$). Fixed efficiency coefficients are considered to simplify optimization calculations and reduce nonlinear relations. Other constraints related to the maximum capacity of the system elements are provided in Eq. 78-81.

$$0 \le C_{AB,chiller} \le C_{AB,chiller}^{Max} \tag{78}$$

$$0 \le C_{VC,chiller} \le C_{VC,chiller}^{Max} \tag{79}$$

$$0 \le C_{Boiler} \le C_{Boiler}^{Max} \tag{80}$$

$$0 \le C_{CHP} \le C_{CHP}^{Max} \tag{81}$$

### 7.6.2 Case study 2

This section introduces an optimization model for the EH's cost-efficient design. A cost-efficient optimization model minimizes the system's total cost in the long run by determining the optimal capacity of energy conversion technologies inside the hub. The optimization model decision variables are the capacities of the electric boiler and heat pump. The mathematical model is established by defining linear constraints and the objective function. Therefore, the final optimization model is linear and is solved using the CPLEX algorithm. The model objective function and constraints for the proposed and conventional design approach are discussed in the following.

*Optimization objective function:* In both conventional and proposed designing scenarios, the optimization model minimizes the EH total cost provided in Eq. 82 to achieve the optimum capacities of the EH components. However, The index i=0 should be used in all equations pertaining to the conventional design approach (scenario 1) since state 0 represents the situation where all system components are in healthy condition. The heat load loss cost in the model objective function is zero in scenario 1 as a healthy system is supposed to be able to supply the thermal load. Re-writing the EH total cost for the state 0, Eq. 82 gives the optimization model objective function for the conventional design approach (scenario 1):

$$ObjFcn_{design,case\,1} = min\left\{AF \times \left[\sum_{t=1}^{T}\left(EC_{p_{i=0},t}\right) + \sum_{n=1}^{N} MC_n\right] + \sum_{n=1}^{N} CC_n\right\} \tag{82}$$

$MC, CC, and\ FC$ are the maintenance, capital, and energy costs, respectively. $p$ denotes the state of the system. In the state-space diagram, $p = 0$ represents a zero state in Fig. 35 where all EH components are in healthy condition. $n$ indicates the energy conversion component in the EH.

The second design approach (scenario 2) uses real operating conditions as the basis for system modeling. A reliability and availability assessment is conducted to observe what effect a subsystem's failure would have on the system's performance. In this modeling approach, the optimization objective

function is the total cost by considering all the system states in the state-space diagram. By reorganizing Eq. 75 over all the system's possible states, the optimization objective function for the second optimization scenario (scenario 2) will be:

$$ObjFcn_{design,case\ 2} = min\left\{AF \times \left[\sum_{i=0}^{I}\sum_{t=1}^{T} p_i \times \left[EC_{i,t} + LLC_{i,t}\right] + \sum_{n=1}^{N} MC_n\right] + \sum_{n=1}^{N} CC_n\right\} \quad (83)$$

*Optimization constraints:* In addition to the constraints relating to the EH components' capacities, optimization constraints also include hourly constraints on meeting the EH heat load. A coupling matrix describes how the hub transforms power from the input to the output; it can be derived from a hub's converter's efficiency and structure. Eq. 84 displays the EH input power vector, coupling matrix, and output power vector for the system depicted in Fig. 32.

$$[(\eta_{EB}\vartheta + (1-\vartheta)\eta_{HP})\eta_{HEX} \quad 1]\begin{bmatrix} EP_t \\ TMP_{Heat\ production,t} \end{bmatrix} + HLL_t = PM_{Heat\ demand,t} \quad (84)$$

Where $EP$ is EH's total electric power consumption, and $\vartheta$ is the dispatch factor. $HLL_t$ is the heat load loss at time t in case of subsystem failure. The heat load loss is zero in the state=0 since all the system components are in healthy condition. $\eta_{EB}$, $\eta_{HP}$, and $\eta_{HEX}$ are the efficiencies of the electric boiler, heat pump, and heat exchangers, respectively. $TMP_{Heat\ production,t}$ and $PM_{Heat\ demand,t}$ are the produced heat by the thermomechanical pulp mill and paper machine heat at time t, respectively. The following equations provide additional constraints regarding the maximum capacity of system components. The maximum capacities for the electric boiler and heat pump are considered to be 120 MW.

$$0 \le C_{EB} \le C_{EB}^{max} \quad (85)$$
$$0 \le C_{HP} \le C_{HP}^{max} \quad (86)$$

### 7.7 Energy and cost-efficient EH operation

EH load prediction is key to ensuring the cost-efficient operation of the system. The operation stage takes place after the system design, where the optimal capacity of all system components is determined in the design phase. A central energy hub controller can calculate the EH load based on the heat supply from the TMP mill and the heat demand from the paper machine. Due to the low fluctuation in the heat demand of the paper mill, the development of the heat demand prediction model has not been considered in this research to reduce the complexity. However, the TMP mill's heat production is affected by several time-dependent factors, including seasonal variations in wood chips quality, refining plate erosion, and wood chips moisture content.

Different LSTM architectures studied in this research for TMP heat prediction are single LSTM with dense layer, stacked LSTM with dense layer, and Bidirectional LSTM. Fig. 36 depicts the step-by-step framework to achieve an accurate TMP heat prediction model. The best prediction model will be incorporated with the central hub controller to perform the unit commitment. The central hub controller performs the unit commitment for the next time step by maximizing the hourly profitability of the EH operator. The hourly heat pump and electric boiler power consumption are the optimization decision variables at each time step. The optimization objective function of the linear programming problem is provided in Eq. 87. Each time step must adhere to the same optimization constraint as the coupling matrix provided in Eq. 84. Using the CPLEX algorithm, the linear optimization problem can be solved at each time step.

$$ObjFcn_{operation,t} = min\{FC_t + LLC_t\} \quad (87)$$

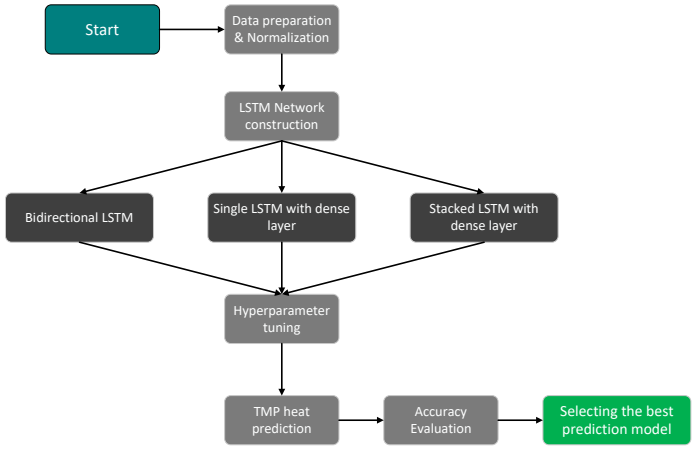



**Fig. 36.** The methodology flow chart to obtain the accurate TMP heat prediction model.

# 8. Results and Discussion

## 8.1 Refining Identification model (Combination of Mechanistic and Empirical methods)

*Basic MATLAB refining Simulink model:* Plate operation time has a minor effect on refining behavior for short-term refining simulation after plate change. Table 7 provides the statistical features of the basic MATLAB Simulink refining identification model for short-term and longer-term analysis. From Table 7, it is clear that the basic MATLAB Simulink model is performing well in predicting refining motor load (when plates are new and in good condition). Nevertheless, for longer-term simulation, the basic MatLab Simulink model is not efficient in predicting refining characteristics because of ignoring the effect of refining plate conditions. Therefore, for the longer-term simulation period, when changing the refiner plate and disc abrasion impose an effect on refining energy behavior, the basic MATLAB Simulink model cannot predict the desired operational variable accurately.

**Table 7.** Basic MATLAB Simulink model results for short- and longer-term prediction of refining motor load.

| Parameter | Short-term analysis | long-term analysis |
|---|---|---|
| $MAPE$ | 1.1% | 8% |
| $R^2$ | 80% | 30% |

Fig. 37 shows the scatter plot of the basic MATLAB Simulink model to predict the refining motor load for the short-term (24 h) and longer-term (200 h) horizon. Red lines are predicted variables obtained from the basic MATLAB simulation model, and black lines are values calculated based on the real measurement from the pulp mill. Fig. 37-b shows the lesser amount for the measured refining motor load than the estimated one because of the plate wear. This is due to the lack of consideration of the refiner plate erosion in the energy simulation of the refining process. Higher plate erosion leads to lower transferred mechanical energy to the pulp and reduces refining energy consumption. Therefore in Figure Fig. 37-b, red points, which are predicted values of the refining motor load (without considering the plate condition), are above the black spots (measured refining motor load).

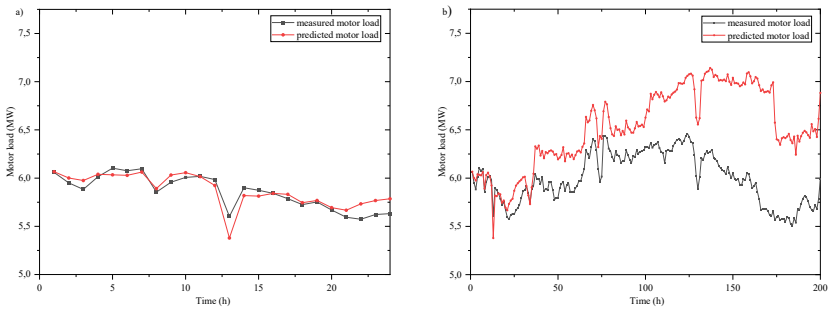



**Fig. 37.** Measured and predicted (basic model) refining characteristics for (a) short-term (b) long-term analysis.

*Developed refining identification model:* Developed refining identification model is a modified version of the basic MATLAB thermo-mechanical pulping toolbox. The main purpose of the developed identification model is to predict the refining energy behavior for longer-term analysis. For long-term analysis, the refining plate condition plays an important role and should be taken into account. As stated earlier, the plate condition is working as a strong disturbance variable for refining energy simulation. As plate operation time increases, plates become more eroded, and there would be more variation in refining motor load for a given production rate. In this study, the simulation box related to the motor load calculation in the MATLAB Simulink model is corrected with three coefficients to model the effect of plate deterioration on refining motor load. Since refining evaporated water is being calculated based on the refining dilution water and refining motor load, correcting the refining motor load leads to the correction of the refining generated steam automatically. Therefore, the developed model can be used for steam prediction as well.

Four optimal model coefficients are presented in Table 8. Due to the differences in the structure of the plate, the model coefficients fall within the given intervals, and no unique coefficient is obtained for all conditions. Even if the plates are of the same type, the slight variation in the quality of the plate manufacturing is an important factor for the differentiation of plates.

**Table 8.** Optimal coefficients for the developed refining simulation model.

| Coefficient | Min | Max |
|---|---|---|
| $C_1$ | -1.48 | -0.65 |
| $C_2$ | 0.92 | 1.25 |
| $C_3$ | -0.00662 | -0,00232 |
| $C_4$ | 0.012 | 0.046 |

Table 9 reports the statistical analysis results regarding the accuracy of the developed MATLAB Simulink model for estimating the refining motor load. The developed model reveals 160% and 78,75% improvement in simulation model determination coefficient and error, respectively. The modified simulation model can be used to predict the two essential parameters (refining generated steam and motor load) to compute $CRSEC$ based on the input variables measured from the mill.

**Table 9.** Accuracy evaluation of the developed Simulink model.

| Parameter | Refining motor load (ML) | Difference% w.r.t basic model |
|---|---|---|
| $MAPE$ | 1.7% | -78.75% |
| $R^2$ | 78% | 160% |

Fig. 38 illustrates the developed MATLAB Simulink model to predict the refining motor load for a longer period. Red lines are predicted variables obtained from the developed MATLAB Simulink model, and black lines are the real measurement of refining motor load from the studied pulp mill. It is clear in the graph that the developed model has a higher accuracy in estimating refining variables, which shows the importance of the developed model to achieve the optimal refining control strategy.

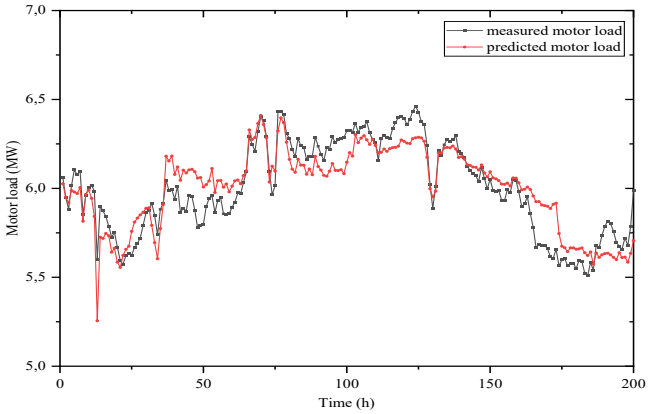



**Fig. 38.** Measured and predicted (developed model) refining characteristics for long-term analysis.

## 8.2 Refining Identification models: ANFIS method

Paper 5 investigates developing a refining identification model by the ANFIS method integrated with 6 different heuristic optimization algorithms to increase the prediction accuracy. Given that different heuristic optimization algorithms have been investigated to heighten the performance of the ANFIS model, good insight into the efficiency of different optimization methods is essential. For this purpose, the performance indicators of different methods are evaluated in the following. Three performance indicators, namely correlation coefficient (R), mean-squared error (MSE), and root-mean-square error (RMSE) are considered as criteria for appraising the precision of the different optimization methods for the best prediction.

Fig. 39 reports the statistic indicators of the six studied models attributing to implementing different heuristic optimization algorithms to the ANFIS method for motor load and Fig. 40 for generated steam prediction. In order to tabulate the data regarding the refining motor load and generated steam models evaluation criteria in Fig. 39 and Fig. 40, Table 10 and Table 11 are provided to address the real values of the mentioned variables. For both motor load and generated steam, the PSO algorithm works best in terms of the correlation coefficient ratio, which is 0.9052 for motor load and 0.9032 for generated steam, respectively. The TLBO algorithm shows the weakest performance in R-value. 0.8472 for motor load and 0.8607 for generated steam prediction are the value of the correlation coefficient (R) resulting from the TLBO optimization algorithm integrated into ANFIS. On the other hand, the results for both root mean squared error and mean squared error show similar behavior as the R criterion. So that the PSO algorithm results in the best and the TLBO algorithm in the worst performance for the error criteria. For instance, the RMSE value in motor load prediction is 0.2712 MW for the PSO algorithm and 0.3488 MW for the TLBO algorithm, which respectively show the best and worst performances in RMSE value. This behavior is also true for the MSE index, as same as RMSE. Such a way that the MSE value for generated steam prediction is 0.0128 $kg^2/s^2$ for the PSO algorithm and 0.0211 $kg^2/s^2$ for the TLBO algorithm, which is approximately two times greater than the PSO one. Thus, the combination of the ANFIS method and the PSO optimization algorithm determines the best model for predicting motor power consumption and produced steam in the refining process. This model results in better model accuracy evaluation criteria, which implies it is the most suitable model for the purpose of energy efficiency analysis and implementing optimal refining control strategy.

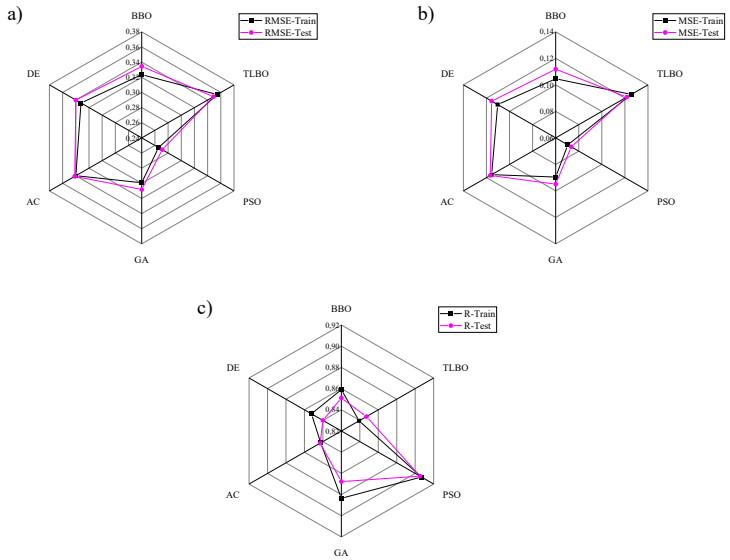



**Fig. 39.** Comparison of the statistical indicators for six evolutionary algorithms to predict the motor load, (a) RMSE, (b) MSE, and (c) R.

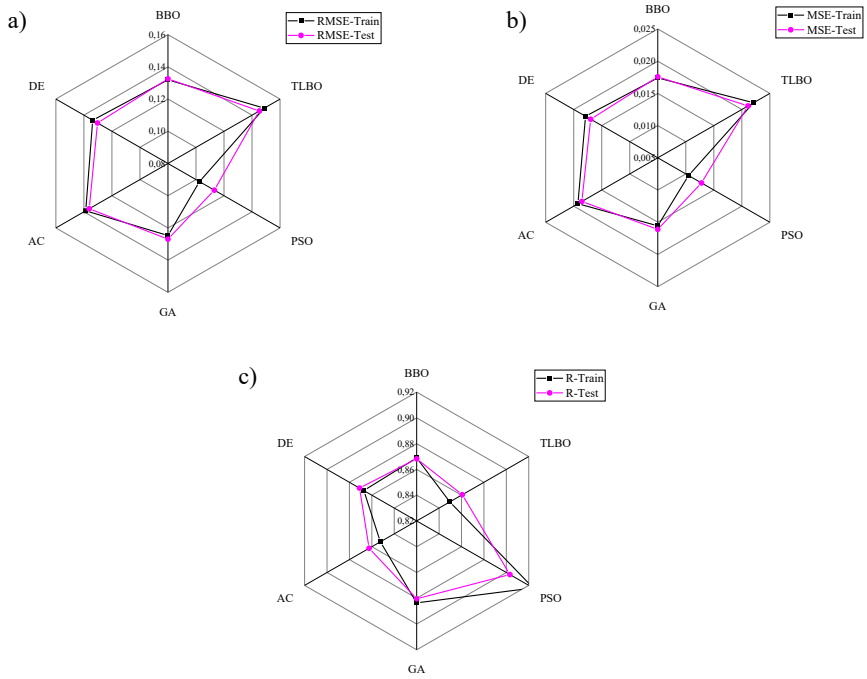



**Fig. 40.** Comparison of the statistical indicators for six evolutionary algorithms to predict the steam generation, (a) RMSE, (b) MSE, and (c) R.

**Table 10.** Model evaluation criteria for six evolutionary algorithms contributing to ANFIS for refining motor load prediction.

| Optimization Algorithm | MSE-Train | MSE-Test | RMSE-Train | RMSE-Test | R-Train | R-Test |
|---|---|---|---|---|---|---|
| TLBO | 0.1256 | 0.1216 | 0.3544 | 0.3488 | 0.8392 | 0.8472 |
| BBO | 0.1046 | 0.1118 | 0.3235 | 0.3344 | 0.859 | 0.8511 |
| DE | 0.1103 | 0.1155 | 0.3321 | 0.3398 | 0.8524 | 0.840 |
| AC | 0.1155 | 0.1167 | 0.3399 | 0.3416 | 0.8424 | 0.843 |
| GA | 0.08954 | 0.09494 | 0.29924 | 0.30813 | 0.8835 | 0.8677 |
| PSO | 0.0705 | 0.0735 | 0.2655 | 0.2712 | 0.9068 | 0.9052 |

**Table 11.** Model evaluation criteria for six evolutionary algorithms contributing to ANFIS to predict the refining steam generation.

| Optimization Algorithm | MSE-Train | MSE-Test | RMSE-Train | RMSE-Test | R-Train | R-Test |
|---|---|---|---|---|---|---|
| TLBO | 0.0221 | 0.0211 | 0.1487 | 0.1452 | 0.8496 | 0.8607 |
| BBO | 0.0174 | 0.01757 | 0.13194 | 0.13256 | 0.8689 | 0.8682 |
| DE | 0.01789 | 0.01697 | 0.1337 | 0.1303 | 0.8672 | 0.871 |
| AC | 0.01925 | 0.01856 | 0.1387 | 0.1362 | 0.8524 | 0.8624 |
| GA | 0.01556 | 0.01611 | 0.12476 | 0.12694 | 0.8836 | 0.8804 |
| PSO | 0.0105 | 0.0128 | 0.1025 | 0.1131 | 0.924 | 0.9032 |

## 8.3 Refining Identification Models and parametric study: MLP method

Table 12 shows the accuracy-test result of the multilayer perceptron neural network for the model input combination of production rate, plate gap, dilution water, and time. The simulation results show the high accuracy of the deep neural network model to predict the refiner motor power so that the correlation coefficient of 98% is accessible. In order to visualize Table 12, Fig. 41 illustrates the prediction model correlation coefficient, determination coefficient, and error for mill 1 and mill 2. Fig. 42 shows the test samples regarding motor load prediction of refiner R1.1 (which is a first-stage refiner) in case study 2 (mill 2). Since data regarding the refiners' power is considered to be confidential by the mills, numbers on the Y-axis scale bar are removed. This was the only way to visualize ANN model efficiency to predict refining motor load.

The simulation results show a slightly better performance of the artificial neural network model to predict the refining energy consumption in mill 1 compared to mill 2. The average obtained correlation coefficient is 98% for the first-stage refiner in mill 1 and 94% in mill 2. However, the average energy prediction correlation coefficient for the second-stage refiner is 95% in mill 1 and 83% in mill 2.

**Table 12.** ANN accuracy test for the subset with all four refining variables

| | | | Model inputs | | | | Model output | R | $R^2$ | MSE | MAPE |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | feeder screw rpm | plate gap | dilution water | Time | | | | | |
| mill 1 | First stage refiner | Refiner R1.1 | ✓ | ✓ | ✓ | ✓ | Refining Motor load | 0.97 | 0.94 | 0.014 | 1.3 |
| | | Refiner R2.1 | ✓ | ✓ | ✓ | ✓ | | 0.98 | 0.96 | 0.01 | 1.5 |
| | Second stage refiner | Refiner R1.2 | ✓ | ✓ | ✓ | ✓ | | 0.94 | 0.88 | 0.0054 | 1.9 |
| | | Refiner R2.2 | ✓ | ✓ | ✓ | ✓ | | 0.95 | 0.90 | 0.038 | 2.36 |
| mill 2 | First stage refiner | Refiner R1.1 | ✓ | ✓ | ✓ | ✓ | | 0.94 | 0.88 | 0.016 | 0.81 |
| | Second stage refiner | Refiner R1.2 | ✓ | ✓ | ✓ | ✓ | | 0.83 | 0.69 | 0.022 | 1.25 |

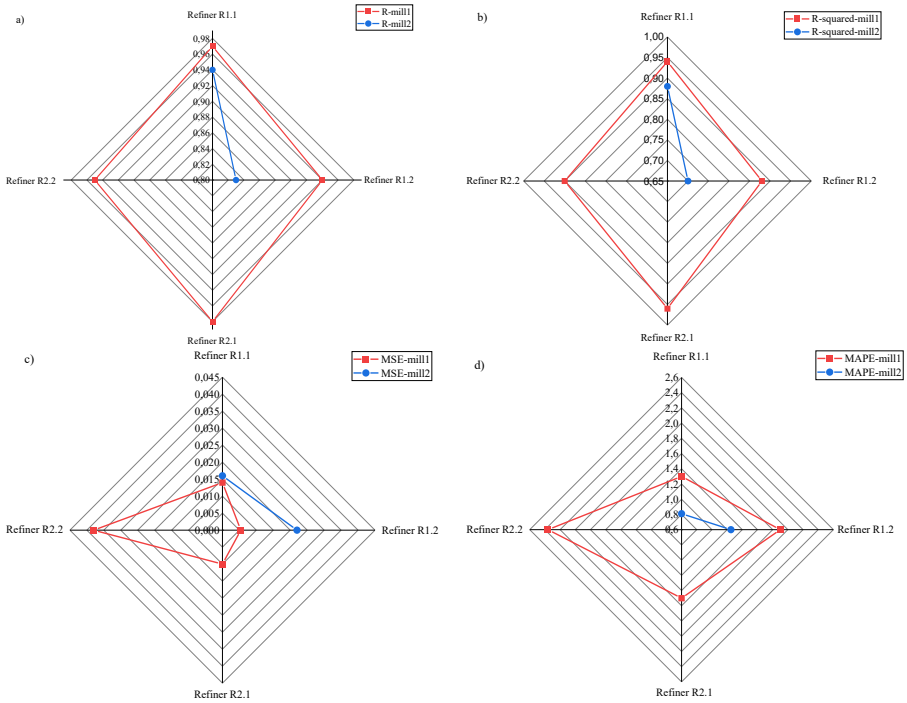



**Fig. 41.** Refining energy identification model performance for two studied mills, a) Correlation coefficient, b) Determination coefficients, c) Mean square Error (MSE), d) Mean absolute percentage error (MAPE)

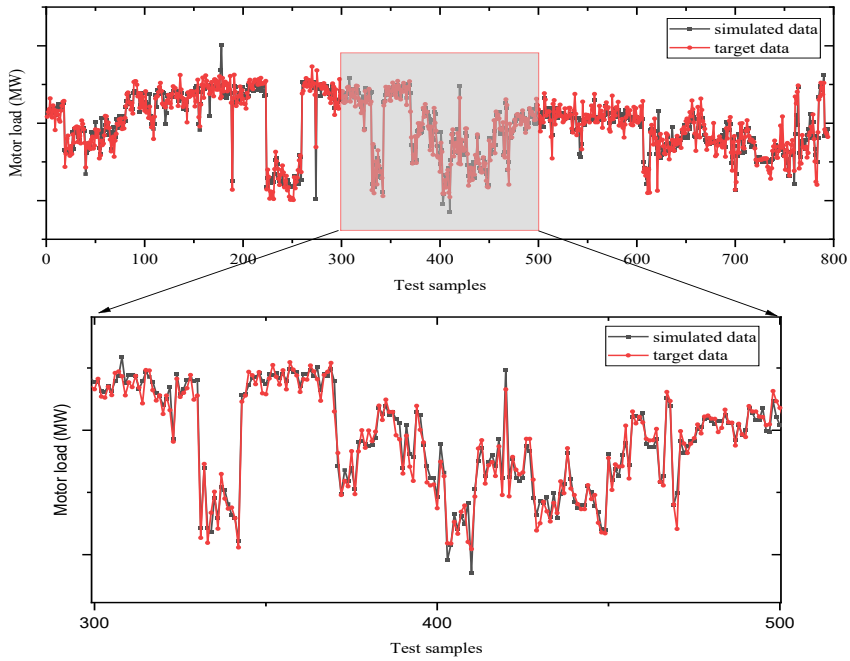



**Fig. 42.** Testing dataset of the MLP refining identification model for predicting refining motor load (refiner R1.1, mill 2)

### 8.3.1 Effect of disturbance variables

Important refining disturbance variables are plate erosion and changes in the quality of input wood chips. Seasonal variations in wood chips quality and density affect the refining production rate; therefore, the correct value of the production rate can not only be calculated through screw speed but also disturbance variables must be involved. On the other hand, the wear of the plates reduces the mechanical energy transferred to the pulp and thus affects the refining motor load. Hence, the effect of disturbance variables on refining energy consumption can be divided into direct (disc erosion) and indirect (changes in pulp production rate) effects. Both phenomena of plate erosion and seasonal changes in the raw wood chips quality are time-dependent, and the simulation results in this paper prove a high correlation between refining disturbance variables and variable time.

Consequently, in this study, disturbance variables' effect on refining energy consumption is included by adding time as a model predictor variable. By removing time in feature subset selection, the effect of disturbance variables on refining energy consumption could be examined. Table 13 examines the importance of disturbance variables in the refining energy consumption prediction. Environmental factors, abrasion of plates, changes in density, and quality of input wood chips are among the influential parameters in the refining process that have a high correlation with time. The variable time has been omitted from predictor variables to observe the changes in the model accuracy. The extent of change indicates the significance of the disturbance variables on refining energy simulation. ANN is generated by dilution water, plate gap, and screw feeder rotational speed as predictor variables to predict refining motor load, and the results of the model have been compared with results in Table 12. Eliminating time as a predictor variable removes the effect of refining disturbance variables on refining throughput and the rate of transferred energy to the pulp per unit of production (because of plate erosion), which leads to a reduction in the accuracy of the ANN simulation. Based on Table 13, the ANN model coefficient of determination reduces by 10 to 20 percent while removing time as a predictor variable. On the other hand, eliminating the effect of refining disturbance variables increases the error of modeling, so that the value of the increase can be up to 140%.

**Table 13.** Effect of Time on identification model results

| | | | Refining variables (model inputs) | | | | Model output | $R$ | $R^2$ | $MSE$ | $MAPE$ | $R^2$ % Differences w.r.t Table 12 | MAPE % Differences w.r.t Table 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | screw feeder rpm | plate gap | dilution water | Time | | | | | | | |
| mill 1 | First stage refiner | Refiner R1.1 | ✓ | ✓ | ✓ | ✗ | Refining Motor load | 0.92 | 0.84 | 0.048 | 2.5 | -10.07 | 92.30 |
| | | Refiner R2.1 | ✓ | ✓ | ✓ | ✗ | | 0.88 | 0.77 | 0.10 | 3.5 | -19.36 | 133.33 |
| | Second stage refiner | Refiner R1.2 | ✓ | ✓ | ✓ | ✗ | | 0.89 | 0.79 | 0.01 | 2.5 | -10.35 | 31.57 |
| | | Refiner R2.2 | ✓ | ✓ | ✓ | ✗ | | 0.89 | 0.79 | 0.08 | 3.4 | -12.23 | 44.06 |
| mill 2 | First stage refiner | Refiner R1.1 | ✓ | ✓ | ✓ | ✗ | | 0.88 | 0.77 | 0.035 | 1.14 | -14.10 | 107.40 |
| | Second stage refiner | Refiner R1.2 | ✓ | ✓ | ✓ | ✗ | | 0.78 | 0.60 | 0.030 | 1.68 | -12.72 | 33.97 |

## 8.4 Advanced Energy-Saving Refining Optimization Strategy

### 8.4.1 Identification models

For the optimization plan, mathematical models (identification models) are required to model the refining specific energy consumption and final pulp quality based on the system's manipulated variables. The models for the refining (first and second stages) specific energy consumption form the optimization objective function, and the models for the final pulp quality prediction contribute to

constructing the optimization constraints. In this research, the concept of artificial neural networks has been employed for refining identification model generation. The identification models are four neural networks, each trained for a particular purpose. $ANN^1$ to $ANN^4$ are developed to predict the first and second stages of refining specific energy consumption, final pulp freeness value, and final pulp fiber length, respectively. The accuracy of the obtained models is examined by the three variables listed in Table 14 and more discussed in Chapter 5.2. In addition, data are surveyed to check the relationship between the measured and predicted variables. Fig. 43–Fig. 46 show the regression plot between predicted and measured data for refining SEC, final pulp CSF, and final pulp fiber length simulation. A higher value for the model correlation coefficient demonstrates a better connection between the target and estimations of response.

**Table 14.** Neural Networks prediction performances on training and testing data sets

| Neural network Model | Metrics | Train | Test | Neural network Model | Metrics | Train | Test |
|---|---|---|---|---|---|---|---|
| $ANN^1$ (First stage-RSEC prediction) | $R$ | 0.98 | 0.97 | $ANN^3$ *(CSF prediction)* | $R$ | 0.72 | 0.67 |
| | $MSE$ | 0.0112 | 0.0124 | | $MSE$ | 3.112 | 4.313 |
| | $MAPE(\%)$ | 0.79 | 0.85 | | $MAPE(\%)$ | 1.69 | 1.95 |
| $ANN^2$ (Second stage-RSEC prediction) | $R$ | 0.95 | 0.94 | $ANN^4$ *(Fiber length prediction)* | $R$ | 0.95 | 0.92 |
| | $MSE$ | 0.0122 | 0.0136 | | $MSE$ | 0.0110 | 0.0117 |
| | $MAPE(\%)$ | 1.06 | 1.18 | | $MAPE(\%)$ | 0.46 | 0.55 |

According to Table 14, the evaluation criteria prove the high efficiency of neural network models in predicting refining operating variables. The obtained mathematical models are reliable to use for building optimization model objective function and constraints. Results confirm lesser accuracy for CSF prediction than other refining operating variables, which could be because of ignoring the effect of some other refining variables affecting CSF. Investigating the cause is beyond the scope of this article and is intended as a research question only. From the performance perspective, the optimum ANN structure for CSF prediction gives a correlation of 0.72 for the train and 0.67 for the test data set, whereas, for fiber length prediction, the ANN model correlation coefficient is 0.95 for the train and 0.92 for the test data set. The best ANN practice for the first and second stage RSEC correlation coefficient is 0.98 and 0.95 for the train and 0.97 and 0.94 for test data sets, respectively. On the other hand, the obtained mean square error ($MSE$) for the best practice is 0.0124, 0.0136, 4.313, and 0.0117 for the first stage RSEC, second stage RSEC, final pulp CSF, and final pulp fiber length (FL) simulation, respectively.

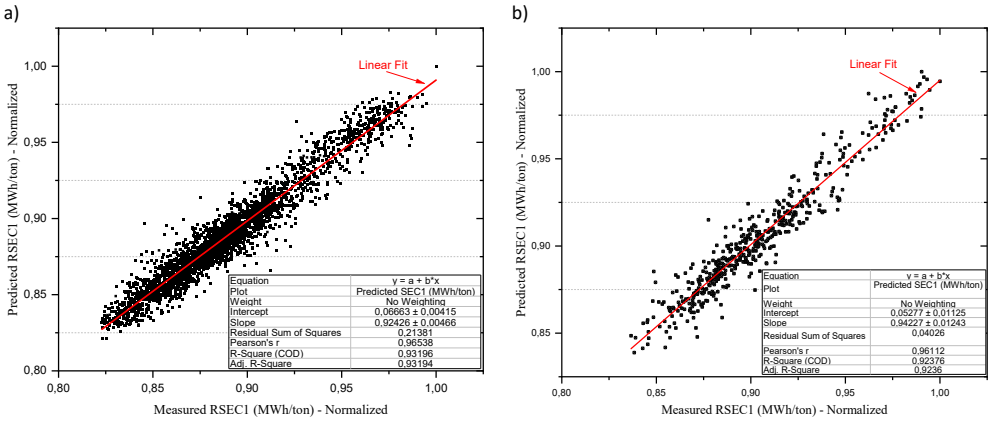



**Fig. 43.** Refining $SEC^1$ ANN model regression plots for the (a) train and (b) test data sets (model: ANN1).

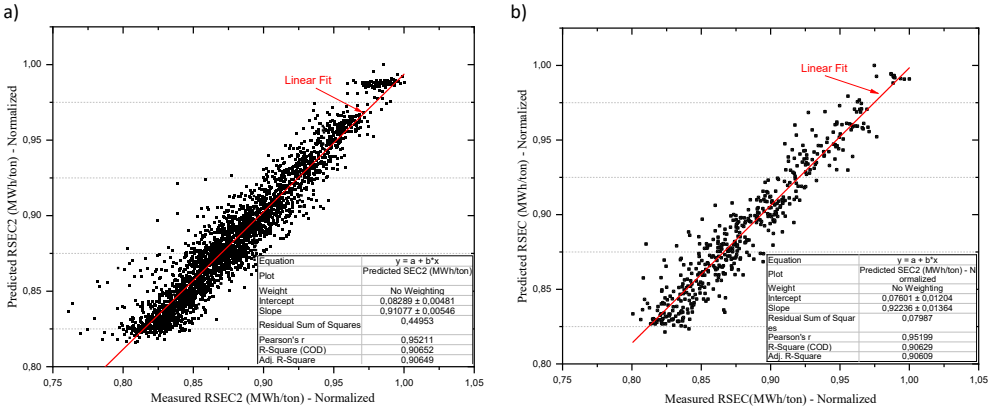



**Fig. 44.** Refining $SEC^2$ ANN model regression plots for the (a) train and (b) test data sets (model: ANN2).

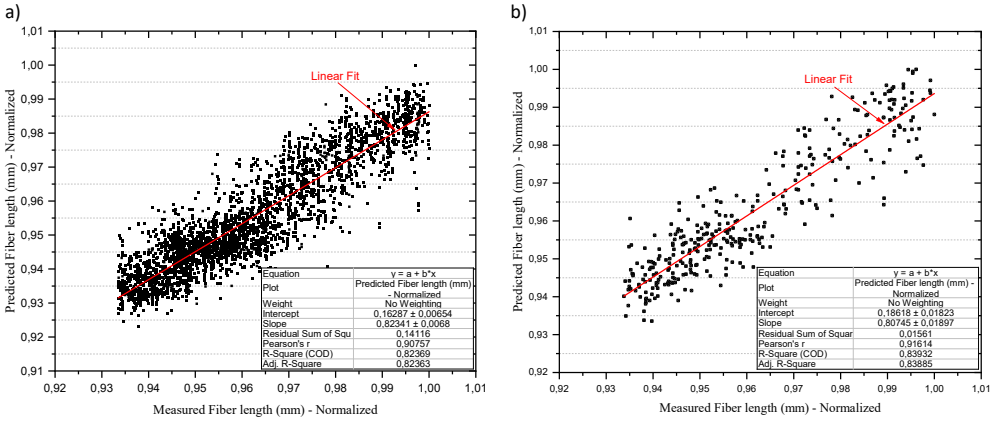



**Fig. 45.** Final pulp Fiber length ANN model regression plots for the (a) train and (b) test data sets (model: ANN3).

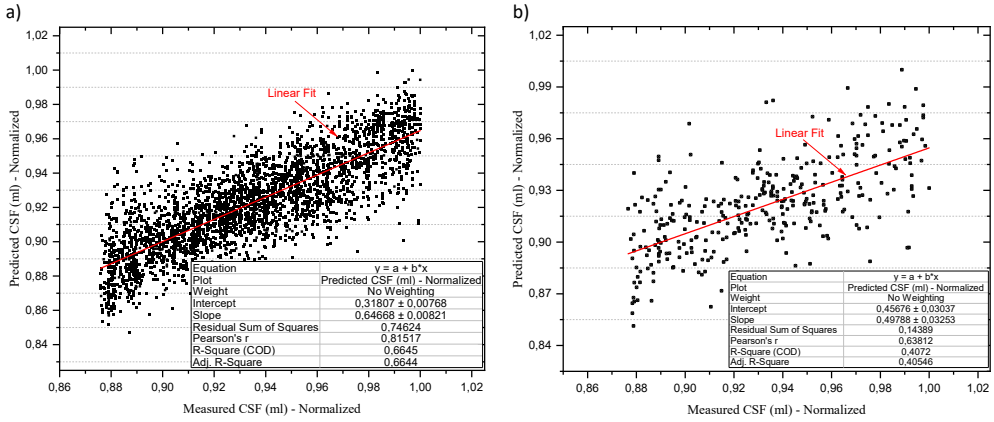



**Fig. 46.** Final pulp freeness ANN model regression plots for the (a) train and (b) test data sets (model: ANN4).

### 8.4.2 Advanced Energy-Saving Refining Optimization plan

*Objective function:* The optimization aims to minimize the total refining-specific energy consumption at the same pulp quality as the current system operating condition by manipulating the refining manipulated variables. The refining manipulated variables which are optimization model decision variables are discussed in the next subsection. The integration of artificial neural networks (to develop a refining identification model) and Genetic algorithm (as an optimization tool) has been proposed for the optimization strategy. The reliability of the proposed method for energy efficiency improvement in the refining process of a thermo-mechanical pulp mill has been examined by comparing the optimization results to the measured values of the total RSEC. Fig. 47 demonstrates measured and optimized total RSEC values of the first 600 representative data points. Measured values that have been pointed out by the red dots are the mill's collected data under the current operating condition where the refining process is handled utilizing the model predictive controller (MPC). Black dots represent the optimized values of total RSEC obtained from the energy-saving optimization plan proposed in this research. As mentioned earlier, these values are normalized to keep our promise to the mill regarding maintaining data security. However, the percentage of changes shown in Fig. 48 is for actual data, not normalized. It is clear from Fig. 47 that almost in all of the data points, black dots that indicate the optimized values are below the red dots, which are the measured ones; meaning the optimization method has reduced the refining specific energy consumption, which implies the fact that refining energy efficiency has been increased. Fig. 48 illustrates the percentage of total refining-specific energy consumption changes using the proposed optimization strategy compared to the current system operating condition. Findings confirm that a specific energy consumption reduction of up to 35% could be possible in some system operating conditions by employing the proposed method, while the average reduction is 14%. Eq. 88 shows the expression for the percentage of change in the refining variables.

$$Change\ (\%) = 100 \times \frac{Optimized\ value - measured\ value}{measured\ value} \tag{88}$$

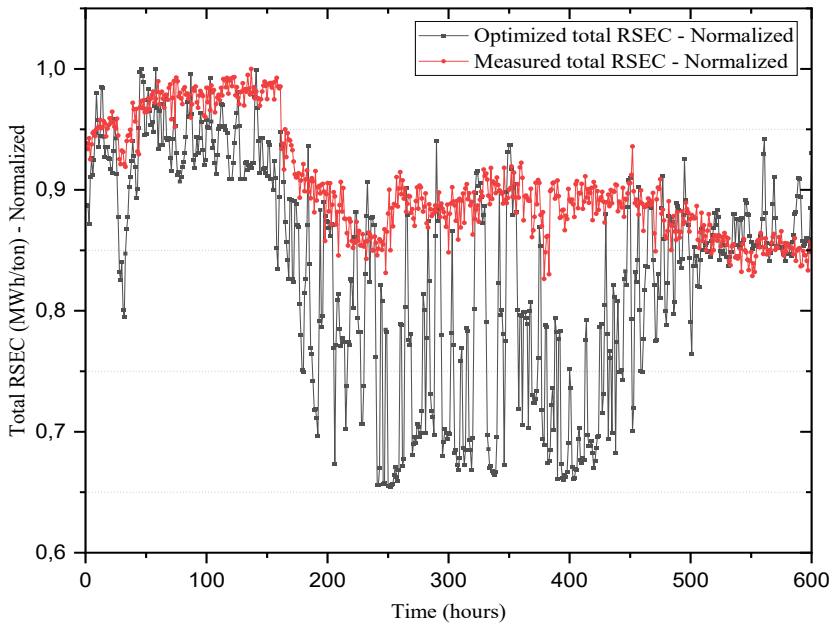



**Fig. 47.** Representative measured and optimized data for the total RSEC.

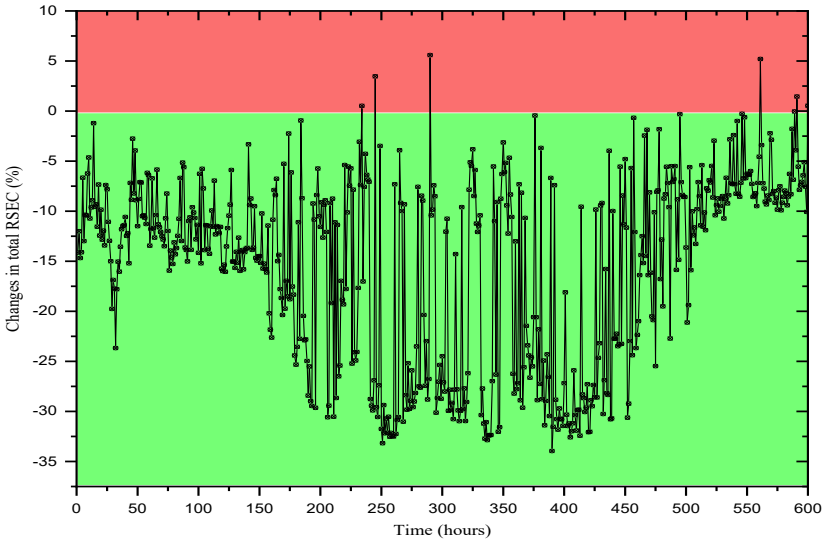



**Fig. 48.** Representative changes in total RSEC while utilizing the proposed optimization plan compared to the current system operating condition (the calculation term is explained in Eq. 88).

*Decision variables (manipulated refining variables):* The optimum combination of refining plate gap and dilution water plays an essential role in constructing the refining optimization strategy from the refining energy consumption and final pulp quality point of view [20], [24]. The optimization model decision variables in the studied refining process are the refining plate gap and dilution water for the first and second refining stages. These variables are considered the most essential manipulated refining variables to optimize the refining energy consumption and control final pulp quality. Fig. 49 and Fig. 50 show the optimum values of the first and second-stage dilution water flow rate and plate gap obtained by the proposed optimization method as well as the measured values under the current system operating condition for 600 hours time period.

Fig. 51 and Fig. 52 show the percentage of changes in refining manipulated variables from the measured values to the optimum values. The expression to calculate the percentage of changes in the refining variables from the optimized values by the proposed optimization strategy and the current system operating is provided in Eq. 88. It is clear from Fig. 51 and Fig. 52 that in the majority of the data points, the optimum value of the first-stage refining plate gap and second-stage dilution water by the proposed optimization strategy is higher than the current system operating condition with an average change of 11.56% and 3.50%, respectively. Meanwhile, the optimum first-stage dilution water and second-stage plate gap are less than the measured values. The average changes in the first-stage dilution water and second-stage plate gap are -7.6% and -27.52%, respectively.

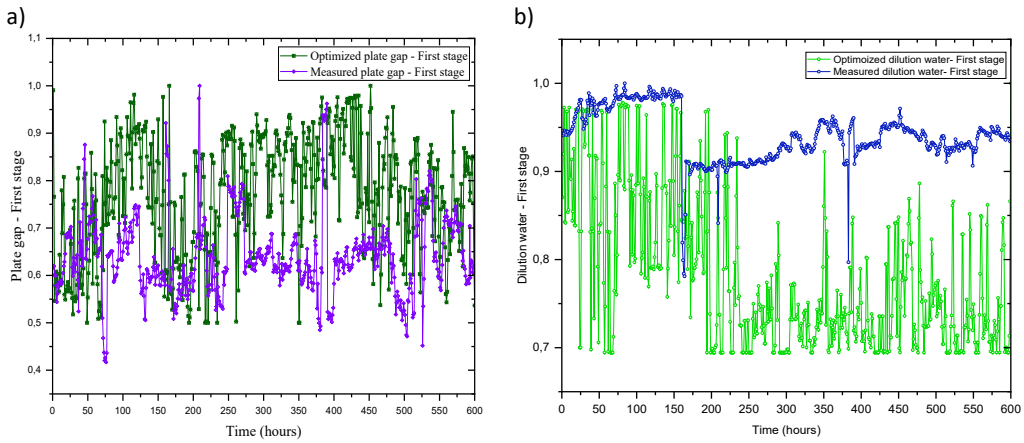



**Fig. 49**. Representative normalized optimum and measured values for first-stage a) refining plate gap, and b) refining dilution water.

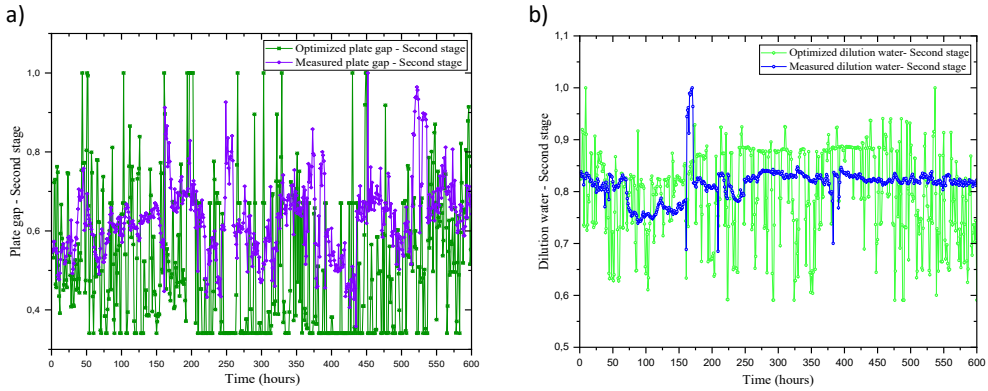



**Fig. 50.** Representative normalized optimum and measured values for second-stage a) refining plate gap, and b) refining dilution water.

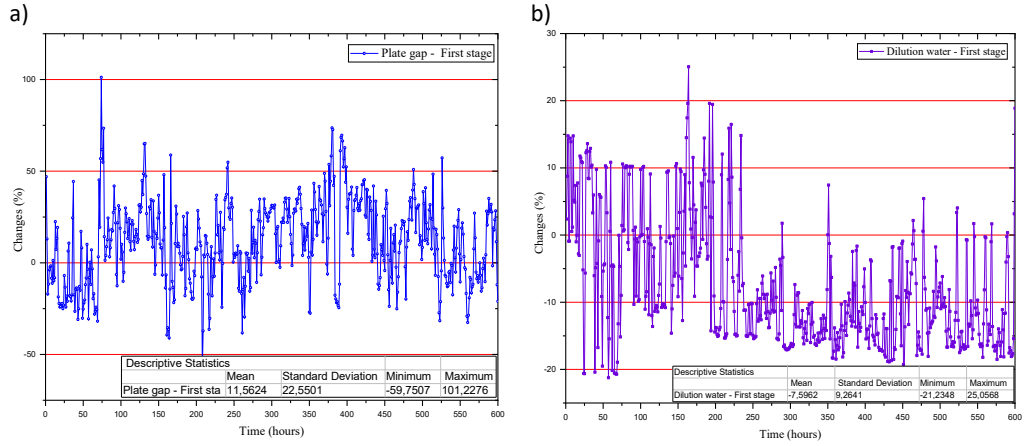



**Fig. 51.** Representative percentages of change in the first-stage a) plate gap and b) dilution water from the measured value to the optimized value (the calculation term is explained in Eq. 88)

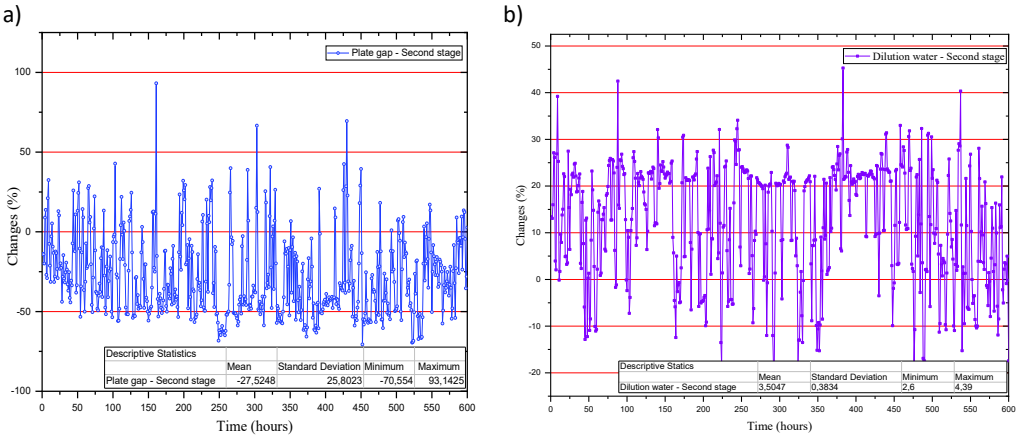



**Fig. 52.** Representative percentages of change in the second-stage a) plate gap and b) dilution water from the measured value to the optimized value (the calculation term is explained in Eq. 88)

*Optimization constraints (Final pulp quality limit):* Energy-saving optimization plan is limited to keep the final pulp quality within a certain range around the pulp quality under current operating conditions. Therefore, process optimization should be designed in a way that the pulp quality is maintained within the desired range, which is close to the current system operating condition. In order to prevent the infeasibility of the optimization model and increase the processing speed, some flexibility is applied to the model constraints. The degree of flexibility is due to the error in the final pulp quality identification model. The ultimate goal of the proposed optimization strategy is to achieve a certain pulp quality following the current system operating conditions with the least total refining-specific energy consumption. Therefore, the closer the estimated variables of pulp quality after energy optimization and the measured variables (current system operating condition) are, the easier it is to evaluate the performance of the proposed optimization strategy. Fig. 53 shows the measured and optimization strategy final pulp fiber length and freeness (CSF). The proposed energy-saving optimization strategy aims to minimize the percentage of changes in paper quality and keep it as close to the measured values because the reduction in refining specific energy consumption must be such that the quality of the paper is maintained. Fewer changes in the pulp quality indicate that the proposed optimization algorithm has a higher power in maintaining pulp quality while reducing energy consumption. The absolute average changes in the final pulp FL and CSF are 1.33% and 1.53%, respectively.

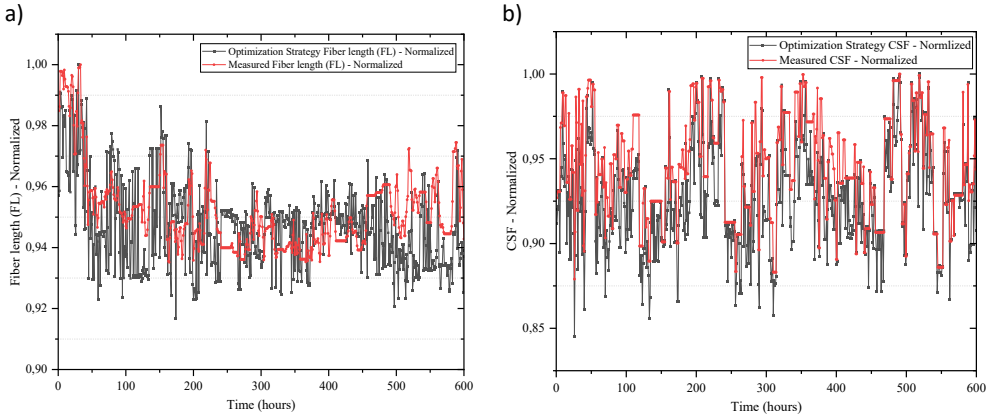



**Fig. 53.** Final pulp quality before (red dots) and after (black dots) applying the proposed optimization strategy: a) Final pulp FL, b) Final pulp CSF.

## 8.5 Advanced energy and cost-efficient design and operation of EH

In this thesis, the advanced design and operation of EH have been developed for a reliable, cost and energy-efficient heat supply system for both residential (high-rise residential building) and industrial (forest industry) sectors. The proposed approach is based on the integration of thermo-economic analysis, reliability & availability analysis, and EH load prediction. For the residential case study (case study 1), the application of EH consisting of a CCHP system is investigated to supply the heating and cooling demand of a high-rise residential building and in the other study (industrial case study: case study 2), the application of EH has been examined for the optimal integration of the TMP mill and paper machine through heat recovery.

### 8.5.1 Energy Hub Optimal Design (case study 1)

In this research, the optimal design of an EH (system schematic in Fig. 31) which supplies heating, cooling, and hot water demand of a high-rise residential building has been investigated under two hypothetical scenarios. In the first scenario, system performance is considered ideal, with no failure possibility of the system components. In this case, repair and maintenance for each component happen at a time when there is no need for that component to operate. In the second scenario, the accurate model, the EH subsystem's failure rates are added to the model to consider the actual operation of the system. Continues Markov method is employed to model the failure modes of the system components. Table 15 presents the limiting state probabilities of the state-space diagram (Fig. 34), utilizing the frequency balance approach. Table 16 shows the optimal capacities of the subsystems, while Table 17 shows the different costs of the system. Whereas under actual operating conditions, in the second scenario (scenario 2), there is a risk of failure for EH elements. In this regard, Table 18 and Table 19 present the optimal capacities and costs of the EH system, respectively. Fig. 54 and Fig. 55 display the system's optimal subsystems capacities and changes in the system costs moving from scenario 1 to scenario 2, respectively.

**Table 15.** limiting states' probabilities of the CCHP system

| States | Probabilities | States | Probabilities |
|---|---|---|---|
| $p_0$ | 0.8153 | $p_8$ | 0.0031 |
| $p_1$ | 0.0311 | $p_9$ | 0.0015 |
| $p_2$ | 0.0592 | $p_{10}$ | 0.0022 |
| $p_3$ | 0.0363 | $p_{11}$ | 0.0018 |
| $p_4$ | 0.0425 | $p_{12}$ | 0.0002 |
| $p_5$ | 0.0019 | $p_{13}$ | 0.0001 |
| $p_6$ | 0.0014 | $p_{14}$ | 9.65E-05 |
| $p_7$ | 0.0027 | $p_{15}$ | 8.25E-05 |

**Table 16.** Optimization results for the first modeling approach (scenario 1)

| Subsystem | | Capacity(kW) |
|---|---|---|
| CHP (gas engine) | Electrical power | 0 |
| | Heating power | 0 |
| Absorption chiller | | 522 |
| Vapor-compressor chiller | | 0 |
| Boiler | | 601 |

**Table 17.** CCHP costs related to the first modeling approach (scenario 1)

| Cost | | Values ($) |
|---|---|---|
| Capital cost | Boiler | 300 k |
| | CHP | 0 |
| | Absorption chiller | 75.8 k |
| | Vapor-compressor chiller | 0 |
| Total Capital cost | | 376 k |
| Total maintenance cost (in plant life) | | 320 k |
| Energy cost | | 122 k |
| Demand penalty cost | | 1.05 M |
| Operation cost (in plant life) | | 1.17 M |
| Total cost (in plant life) | | 1.87 M |

**Table 18.** Optimization results for the second modeling approach (scenario 2)

| Subsystem | | Capacity (kW) |
|---|---|---|
| CHP (gas engine) | Electrical power | 69 |
| | Heating power | 161 |
| Absorption chiller | | 286 |
| Vapor-compressor chiller | | 236 |
| Boiler | | 440 |

**Table 19.** CCHP costs related to the second modeling approach (scenario 2)

| Cost | | Values ($) | % Differences w.r.t scenario 1 |
|---|---|---|---|
| Capital cost | Boiler | 220 k | |
| | CHP | 345 k | |
| | Absorption chiller | 50.62 k | |
| | Vapor-compressor chiller | 70.95 k | |
| Total Capital cost (in plant life) | | 687 k | +82.5 |
| Total maintenance cost (in plant life) | | 584 k | +82.5 |
| Energy cost | | -23.83 k | -119.4 |
| Demand penalty cost | | 322 k | -69.2 |
| Operation cost (in plant life) | | 299 k | -74.5 |
| Total cost (in plant life) | | 1.57 M | -16 |

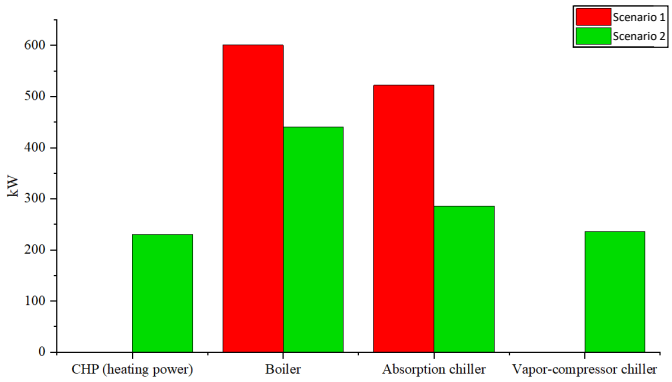



**Fig. 54.** Optimal subsystems' capacities for two scenarios

For the first optimization scenario, Table 16 proves that the optimal sizes for the CHP and vapor-compressor chiller are zero, and the use of these units is not economical. In this case, the absorption chiller provides the whole cooling load in summer. CHP unit provides the heating load in winter and the heat required for the absorption chiller in summer. For the precise modeling (scenario 2), the optimum subsystems capacities are summarized in Table 18. Contrary to the conventional modeling that the use of a boiler and air-compressor chiller was not economical, the results of optimization based on accurate modeling show the importance of using all four types of equipment to reach the optimal point in designing the system. Changes in optimal capacities from the first to second modeling approach are compared in Fig. 54. In this regard, the CHP and vapor-compressor chiller capacities raise from 0 kW (ideal model) to 230 kW and 236 kW (accurate model), respectively. On the other hand, the size of the boiler and absorption chiller is decreased from 601 kW and 522 kW (in scenario 1) to 440 kW and 286 kW (in scenario 2), respectively.

The various system costs for the first and second modeling approach is summarized in Table 17 and Table 19. Based on the statistics in Table 17 and Table 19, Fig. 55 compares the total capital cost, total maintenance cost, energy cost, demand penalty cost, operation cost, and total cost in the plant lifespan for two modeling approaches. As shown in Fig. 55, the investment and maintenance costs are increased from the first scenario to the second scenario while the energy cost, demand penalty cost, and operation cost fall. The model results indicate the profitability of increasing system reliability by diversifying the supply chain which is associated with an 82.5% increase in the cost of investment and maintenance. From Fig. 55, the most significant change in the cost is for the energy cost which varies from 122 k$ (in scenario 1) to -23.83 k$ (in scenario 2). The negative value for the energy cost in scenario 2 is because of the use of CHP in this optimization scenario; where the EH operator not only doesn't have to pay for the electricity and natural gas but also makes a profit of 23.83 k$ over the lifetime of the system by selling the electricity (generated by the CHP unit) to the electricity network. Downsizing the boiler and absorption chiller units by the use of a vapor-compressor chiller and CHP (scenario 2) leads to 119%, 69%, 74%, and 16% reduction in energy cost, demand penalty cost, operation cost, and total cost, respectively.

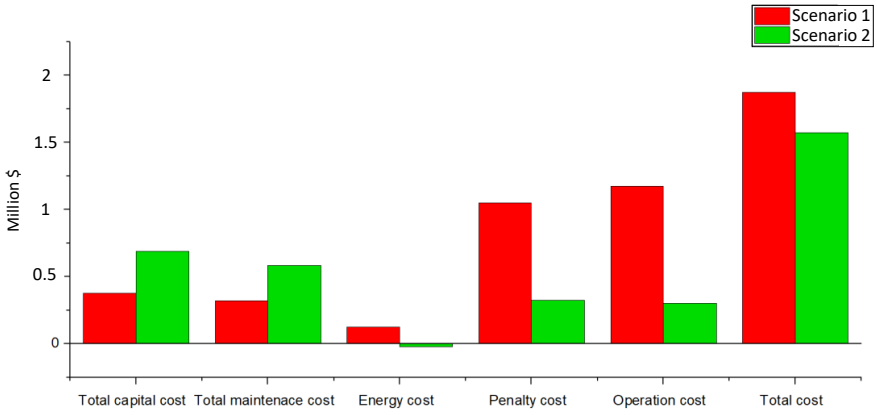



**Fig. 55.** Different system costs for the first and second modeling scenario approach.

### 8.5.2 Energy Hub Optimal Design (case study 2)

This section investigates the energy and cost-efficient design of the EH to supply forest industry heat demand. With a regulatory mechanism, the EH balances energy supply and demand by utilizing the energy conversion technologies inside the hub (system schematic in Fig. 32). The EH's cost-efficiency design requires thermo-economic analysis along with reliability & availability assessment. A reliability assessment factors in the availability of each system component in the system modeling. As each component of the system has a chance of failing in real operating conditions of a system; therefore, reliability assessment is necessary for making a more realistic model. The system optimum (energy and cost-efficient) design approach is introduced in section 7.6.2. The results of the frequency balance technique to calculate the limiting state probabilities of the heat supply system in the state-space diagram (Fig. 35) are presented in Table 20. limiting state probabilities are important values to define the system's limiting states and availability. The heat supply system in the studied case consists of the EH and TMP mill. Limiting state probabilities are important values to define the system's limiting states and availability. In a Markov chain, the limiting state probabilities refer to the long-term probabilities of the system being in each state, as the chain continues indefinitely. These probabilities are reached when the system has reached a state of equilibrium and is no longer changing significantly. The limiting state probabilities provide valuable information about the behavior of the Markov chain over time. In the case study we conducted, the likelihood of a stable system being healthy in the long run is 94%. Table 20 gives the steady-state probabilities for the whole heat supply system in different system operating conditions.

**Table 20.** Limiting state probabilities of the entire heat supply system

| States | Probabilities | States | Probabilities |
|---|---|---|---|
| $S_0$ | 0.9436 | $S_6$ | 0.0001 |
| $S_1$ | 0.0071 | $S_7$ | 0.0001 |
| $S_2$ | 0.0104 | $S_8$ | 0.0002 |
| $S_3$ | 0.0188 | $S_9$ | 0.0002 |
| $S_4$ | 0.0189 | $S_{10}$ | 0.0004 |
| $S_5$ | 0.0001 | | |

This paper investigates the advanced cost-efficient design of the EH based on the conventional (case 1) and proposed (case 2) design approach. The proposed design approach, which is based on the real system operating condition (case 2), is compared with the ideal hypothetical scenario where all the system components are healthy and 100% reliable (case 1). Table 21 presents the results of the optimal capacities for the conventional and proposed EH design approach for different unsupplied demand penalty costs ($pc^{heat}$).

**Table 21.** Optimal Energy Hub subsystem capacities for the different design approach

| | $pc^{heat} = 10\ (\frac{€}{MWh})$ | | $pc^{heat} = 15\ (\frac{€}{MWh})$ | | $pc^{heat} = 20\ (\frac{€}{MWh})$ | |
|---|---|---|---|---|---|---|
| | Capacity (MW) | | Capacity (MW) | | Capacity (MW) | |
| | Electric Boiler | Heat Pump | Electric Boiler | Heat Pump | Electric Boiler | Heat Pump |
| **Case1:** Conventional design approach | 15 | 66 | 16 | 66 | 17 | 66 |
| **Case2:** Proposed design approach | 67 | 65 | 69 | 65 | 71 | 65 |
| % Differences w.r.t case 1 | 346.66% | -1.51% | 331.25% | -1.51% | 317.64% | -1.51% |

For all different unsupplied demand penalty costs, the research demonstrates that the electric boiler capacity is higher in case 2. Nevertheless, there is a slight decrease in the heat pump capacity in case 2 when compared to case 1. As an example, when the penalty cost is set at 10 ($\frac{€}{MWh}$), the electric boiler capacity in case 2 rises to 67 MW, which is a 346.66% rise compared to the conventional design where the capacity is only 15 MW. The optimization model has enabled this increase in capacity to reduce load loss costs. On the other hand, transitioning from case 1 to case 2 only results in a small change of 1.51% reduction in the heat pump capacity. The electric boiler has a higher availability leading to less load loss cost which explains the impressive capacity increase of the electric boiler moving from case 1 to case 2.

Table 22 illustrates the system's costs over the system lifespan for case 1 at various unsupplied demand penalty cost values. Case 1 represents the operating condition where there is no possibility of failure for the system components. Due to this assumption which is not supported by reality, the cost of load loss is significant, amounting to 102.63, 149.75, and 194.22 million € for penalties associated with unsupplied demand at rates of 10, 15, and 20 ($\frac{€}{MWh}$), respectively. Consequently, if failure modes are not taken into account in the optimization model, the EH operator will face a significant penalty cost due to the load loss cost. The system's total cost is equal to 472.63 million € where energy cost and load loss cost contribute to the majority of the system's total cost. Due to the high load loss cost, the possibility of improving the system design to minimize the total cost by considering the real system operating condition is quite evident. Small changes in the system capital cost could result in higher system availability and less load loss cost. This issue has been investigated in the second approach of system design (case 2).

**Table 22.** Energy Hub costs for the first design approach (scenario 1).

| **Costs** (million €) | Unsupplied demand penalty cost $(\frac{€}{MWh})$ | | |
|---|---|---|---|
| | $pc^{heat} = 10$ | $pc^{heat} = 15$ | $pc^{heat} = 20$ |
| Total capital cost | 27.9 | 28 | 28.1 |
| Load loss cost (in lifespan) | 102.63 | 149.75 | 194.22 |
| Energy cost (in lifespan) | 291.88 | 292.38 | 292.87 |
| Maintenance cost (in lifespan) | 50.22 | 50.4 | 50.58 |
| Total cost (in lifespan) | 472.63 | 520.23 | 565.77 |

Table 23 presents a comprehensive analysis of the system costs for case 2 and a comparison with the system costs of case 1. The system's load loss cost experiences a substantial reduction of 98% across all unsupplied demand penalty rates compared to case 1. As an example, when the penalty cost is 10 (€/MWh), the load loss cost drops significantly from 102.63 million € in case 1 to 1.94 million € in case 2. In addition, reductions are also observed in the system's total cost, ranging from 14% to 28%, depending on the unsupplied demand penalty rates. According to the results, the suggested method allows for a higher possibility of decreasing the system's total cost when the penalty rates are increased due. The reason for this phenomenon is that load loss increases rapidly as penalty rates are increased, and the proposed design approach aims to minimize the cost of load loss through the optimal design of the system. The system's total capital cost increases by a range of 17% in case 2 for different unsupplied demand penalty rates. Most of the rise in the capital cost of the system is dedicated to raising the capacity of the electric boiler to provide heat in the event of a heat pump malfunction. This plays an important role in minimizing the system load loss cost which ultimately leads to a decrease in the overall system cost.

In addition, findings indicate that in the second modeling method (case 2), the overall cost of the system throughout its lifespan does not change significantly with variations in unsupplied demand penalty costs. As a result, the proposed approach offers a robust design strategy that is not significantly impacted by unsupplied demand penalty rates, which is a clear advantage of this approach. By subtracting the system total cost in case 1 from the system total cost in case 2, the proposed EH design approach would yield a total benefit of 68.38, 115.11, and 160 million € for penalty cost at the rate of rates of 10, 15, and 20($\frac{€}{MWh}$), respectively.

For an unsupplied demand penalty rate of 10 (€/MWh), Fig. 56 displays the results of a sensitivity analysis conducted on the total cost of the system using various capacities of EH components. The graph indicates that smaller capacities, when deviating from the optimum point, lead to a greater increase in the overall cost of the system compared to larger capacities. The primary explanation for this occurrence is the rise in load loss cost due to inadequate heat supply. Therefore, if the optimization model for an energy system does not take into account the different ways in which components can fail, there could be significant penalties imposed on the energy system operator due to load loss. This is because the system may not be able to meet the energy demands if there is a failure in the system component, resulting in disruptions and outages, and ultimately causing a significant cost to the operator. As a result, it is important to consider failure modes when optimizing the energy system to ensure it operates reliably and efficiently, even in the event of component failures.

**Table 23.** Energy Hub costs for the second design approach (scenario 2).

| **Costs** (million €) | $pc^{heat} = 10\ (\frac{€}{MWh})$ | % Differences w.r.t case 1 | $pc^{heat} = 15(\frac{€}{MWh})$ | % Differences w.r.t case 1 | $pc^{heat} = 20(\frac{€}{MWh})$ | % Differences w.r.t case 1 |
|---|---|---|---|---|---|---|
| Total capital cost | 32.7 | 17.2 | 32.9 | 17.5 | 33.1 | 17.79 |
| Load loss cost (in lifespan) | 1.94 | -98.1 | 2.17 | -98.55 | 2.19 | -98.87 |
| Energy cost (in lifespan) | 310 | 6.46 | 310.83 | 6.31 | 310.89 | 6.15 |
| Maintenance cost (in lifespan) | 58.9 | 17.2 | 59.22 | 17.5 | 59.58 | 17.79 |
| Total cost (in lifespan) | 404.25 | -14.46 | 405.12 | -22.17 | 405.77 | -28.28 |

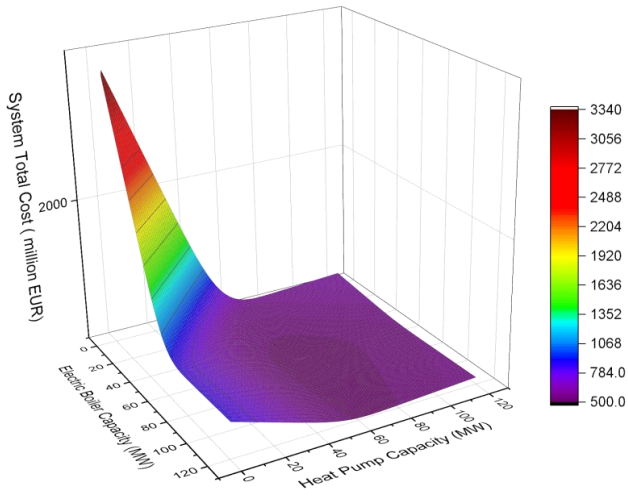



**Fig. 56.** Sensitivity analysis on the system total to examine how changes in the capacities of EH components affect the total cost of the system.

Sensitivity analysis was conducted to determine the influence of the spot electricity price in the reference year on the results of the proposed method, please see Appendix.B. Table B.1 shows the optimal capacities for the EH subsystems for the reference years 2020 and 2021. There would be minor changes in the optimal subsystems capacities while changing the reference year from 2020 to 2021. Table B.2 shows the system costs for the reference years 2020 and 2021. The average Nord Pool spot electricity prices for the years 2020 and 2021 were 28 €/MWh and 73 €/MWh, respectively. The system costs would be influenced by altering the reference year, as is apparent from the results. However, the efficiency of the proposed design method remains unaffected by altering the reference year electricity price, as Table B.2 demonstrates. The results of the proposed method indicate a reduction of nearly 98% in system load loss costs and 11.5% in total costs.

### 8.5.3 TMP refining heat prediction: LSTM method (Case study 2)

The LSTM network performance evaluation to predict TMP heat production is discussed in this section. LSTM network is a state-of-the-art RNN that permits information to remain by sequential network construction. The LSTM learns how to map the sequence of previous observations as input to an output observation. Different LSTM models are analyzed in this research to achieve the best prediction accuracy. Any type of forecasting requires models' performance assessment criteria to evaluate the prediction accuracy. Root mean square error (RMSE), and correlation coefficients ($R$) are the most well-known accuracy metrics to evaluate the prediction model performance. In this study, the prediction model dataset is separated by the first P portion of a percent to the training set and the following (1-P) percent as the testing set. Using a time series forecast model requires dividing training and test data systematically and preserving the data's sequence. Therefore, the division of training and test data cannot be done randomly, and the sequence of the data must be respected. Table 24 shows the accuracy performance of different LSTM structures studied in this research. The employed and tested LSTM structures in this research to predict 1 hour ahead TMP heat production are Single LSTM with a dense layer, Stacked LSTM with a dense layer, and Bidirectional LSTM. All prediction models perform very closely, according to Table 24. However, the Bidirectional model seems to perform better than other models. The correlation coefficient and root mean square error

for the best model, a Bidirectional LSTM structure, are 0.95 and 0.12 for the training dataset and 0.90 and 0.15 for the test samples, respectively. A higher prediction accuracy indicates an improved hour-ahead EH load estimate. This means that if a prediction model is able to accurately forecast the EH load for the upcoming hour, then it is considered to be a better estimator of the EH load. The accuracy of the prediction is an important measure of the model's performance, and a higher accuracy implies that the model is more reliable and robust to heat load fluctuations. Therefore, a model with higher prediction accuracy is more likely to provide an accurate estimate of the EH load, which can help in making informed decisions about energy consumption, distribution, and management.

**Table 24.** Performance evaluation of different studied LSTM structures

| Algorithm | Train | | Test | |
|---|---|---|---|---|
| | $RMSE$ | $R$ | $RMSE$ | $R$ |
| Single LSTM with a dense layer | 0.17 | 0.91 | 0.18 | 0.87 |
| Stacked LSTM with a dense layer | 0.15 | 0.92 | 0.16 | 0.88 |
| Bidirectional LSTM | 0.12 | 0.95 | 0.15 | 0.90 |

Fig. 57-Fig. 59 illustrate the prediction results of the different studied LSTM structures on the training and testing datasets. The blue data points represent the primary measurements, divided into two parts: training and testing dataset. Since the prediction model is a time series, test data are not chosen at random. The test data are placed in the continuation of the training data so that the training data includes the first seventy percent of the total data set and the rest of the data constructs the test samples. The green data are the predicted TMP heat production on the training data set. An initial set of data used to train a machine learning model is called training data (or training dataset). LSTM model prediction on the test dataset is represented in red data. In machine learning, an algorithm is tested using a second (or tertiary) data set after being trained using an initial training data set. The smaller difference between the measured data and the predicted data indicates higher prediction accuracy. Results show that the studied time series models have high accuracy in estimating the heat produced, confirming the feasibility of using these models to optimize EH performance. The data used in this research are measured data from the TMP mill. Due to the contract made with the industrial partner to maintain data security, the y-axis label has been removed from the graph (Fig. 57-Fig. 59) to keep the data secure.

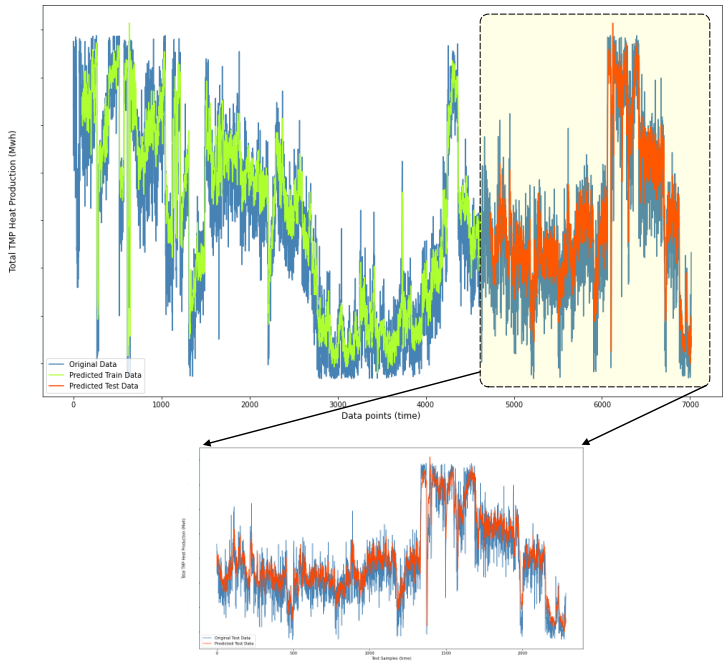



**Fig. 57.** Single LSTM prediction results for train and test datasets.

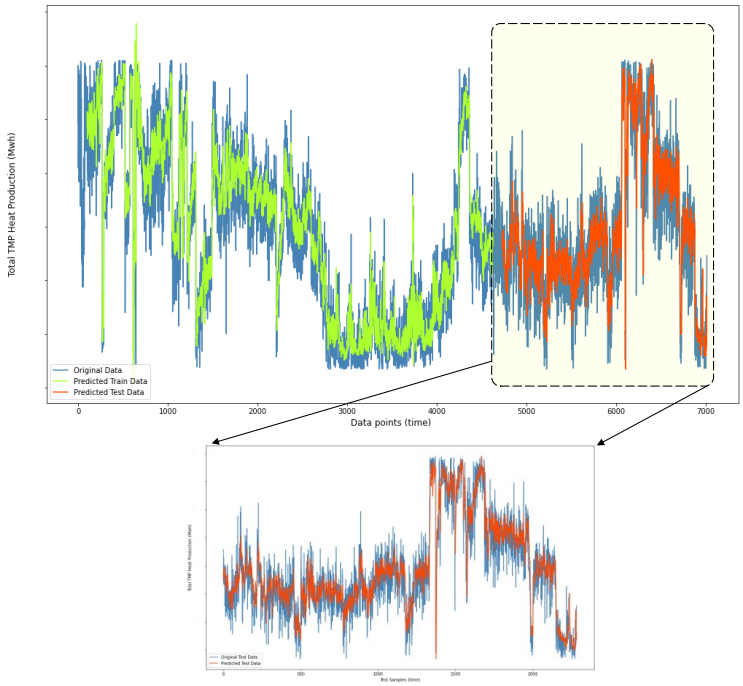



**Fig. 58.** Stacked LSTM prediction results for train and test datasets.

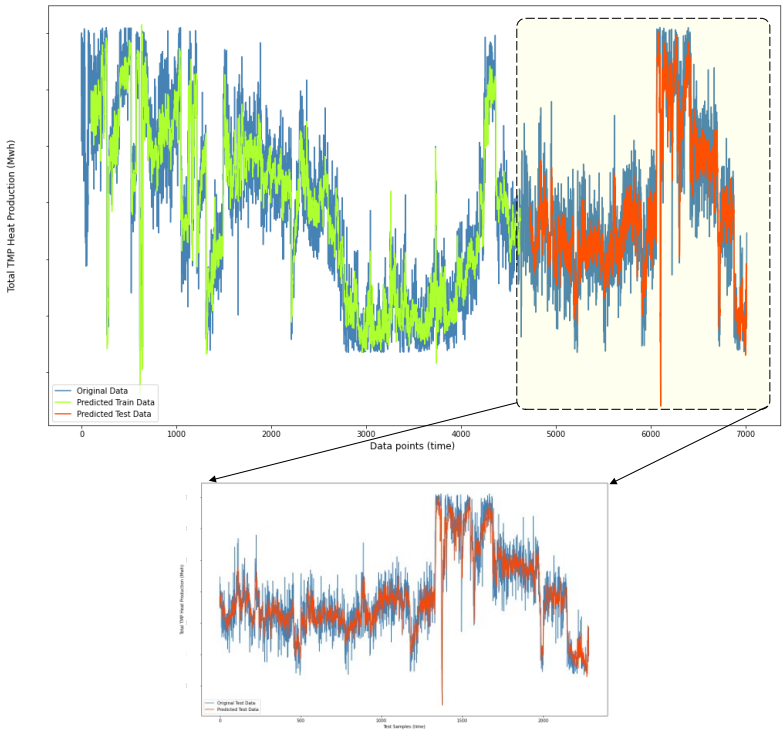



**Fig. 59.** Bidirectional LSTM prediction results for train and test datasets.

## 8.6 Discussion

In the first part of the thesis, different refining energy consumption and final pulp quality identification models were developed through advanced machine learning methods such as MLPNN and ANFIS. However, other refining identification models that are based on the combination of mechanistic and data-driven approaches such as the MATLAB thermo-mechanical pulping Simulink library were also examined. Results show that machine learning algorithms are powerful methods to develop refining energy and quality identification models. Finally, the generated identification models were utilized to develop an advanced energy-saving optimization strategy in the refining line of the thermo-mechanical pulping process. The advanced energy-saving optimization strategy is based on the integration of neural networks and heuristic optimization methods such as a genetic algorithm (GA).

In one study (paper 1), the refining process identification model is developed through the hybrid ANFIS method. The efficiency of the six different evolutionary optimization algorithms has been investigated to optimize the network parameters. The PSO and the TLBO algorithm have the best and weakest results in terms of efficiency for optimizing ANFIS parameters both for the objective refining motor load and generated steam prediction, respectively. The model correlation coefficient and root-mean-square error of the optimized model with the PSO algorithm are 0.9052, 0.2712 MW for motor load prediction, and 0.9032, 0.1131 kg/s for generated steam prediction.

In the second research (paper 2), the refining identification model development was studied based on a combination of mechanistic and empirical methods. The basic MATLAB Simulink refining identification model is fairly efficient for short-term analysis when plate condition does not affect refining

condition. Therefore, the plate deterioration function is added to the Simulink model to consider the effect of plate erosion on refining conditions. In this research, the refining plate erosion is modeled based on the operation time of using the same plate for the refining process by modifying the basic MATLAB Simulink model. The developed model is a modified version of the basic MATLAB thermo-mechanical pulping toolbox, where the effect of plate deterioration on refining behavior is considered. The developed model gives a correlation coefficient of 78% and MAPE of 1.7% for estimating the refining motor load for 200 hours time period.

Next (paper 3), the refining identification model was developed through a deep supervised learning method. A multilayer perceptron neural network with four hidden layers and 50 neurons is used for data training. A comprehensive parametric study has been made to investigate the effect of refining disturbance variables on the energy consumption of the refining process. Simulation results confirm the high capability of AI methods to develop refining energy identification models. The prediction model correlation coefficient of 98% is achievable by the use of a deep supervised learning method. Findings confirm a high correlation between disturbance variables and variable time. Disturbance variables share in refining energy identification model development is 15% in the model determination coefficient. This effect includes the sum of the effect of the disturbance variables on the changes in the refining production rate due to the change in the quality of the wood chips (indirect effect of disturbance variables), and the energy transferred to the pulp due to the wear of the refiner plates (direct effect of disturbance variables).

Finally (paper 4), in the fourth research, an advanced energy-saving optimization strategy in the TMP mill was developed by integration of a machine learning algorithm (artificial neural networks) and a heuristic optimization method (Genetic algorithm). In the beginning, the refining identification models were created using artificial neural networks to model the two-stage high-consistency refining process from the refining-specific energy consumption and final pulp quality point of view. The results indicate the high accuracy of artificial neural networks in modeling the refining process, confirming the license to use these models as refining identification models to develop the refining optimization strategy. While the generated neural networks are utilized as system identification models, the Genetic Algorithm (GA) is employed as an optimization solver to minimize the total refining specific energy consumption. The optimization objective function is the sum of the first and second stage refining specific energy consumption where the optimization is subjected to keep the final pulp quality at the current system operating condition. Canadian Standard of Freeness (CSF) and pulp fiber length (FL) are considered a measure to evaluate the final pulp quality. Optimization decision variables are the first and second-stage refining dilution water and refining plate gap. The proposed refining optimization strategy results are compared with the current system operating condition to evaluate the energy-saving potential. Findings confirm that an energy consumption reduction of up to 35% could be possible in some system operating conditions by employing the proposed method, while the average reduction is 14%, equal to 305 kWh per ton of production.

The second part of the thesis (paper 5 and paper 6) is about the energy and cost-efficient design and operation of EH for the optimal integration of the pulp mill and paper machine. The concept of EH has been used for this purpose. The proposed approach for the energy and cost-efficient design and operation of EH is based on the integration of thermo-economic analysis, reliability & availability analysis, and EH load prediction.

The proposed approach was first introduced and evaluated for the energy and cost-efficient design of a CCHP system that provides the hourly thermal demand of the high-rise residential building. The effect of the system failure and repair rates on the system's optimal design and operation are consid-

ered through the proposed method (referred to in the text as the second scenario). Results are compared to the conventional method (first scenario) of designing the CCHP systems where reliability assessment is not performed. Results prove the 16% reduction in system total cost by increasing the model accuracy from the first scenario to the second scenario. The results of the first scenario modeling designate that the use of CHP and vapor-compressor chiller is not economical. While reducing the share of absorption chiller and boiler by adding vapor-compressor chiller and CHP, leads to 119%, 69%, 74%, and 16% reductions in energy cost, demand penalty cost, operation cost, and total cost, respectively.

As the presented method has shown to be effective in residential EH applications, this method was examined in a second case study (the forest industry) to determine the optimal integration of TMP mill and paper machines. In the second study, an energy and cost-efficient design and operation of an EH in interaction with a TMP mill and paper machine have been studied. The studied system is an EH consisting of a steam generator heat pump and electric boiler integrated with the TMP mill to provide the paper machine heating demand. Paper mill heat demand is given to the Central Hub controller on an hourly basis. The paper mill's heat demand prediction is not necessary as it is mostly constant and stable. However, the TMP mill heat production depends on various factors and is predicted using the LSTM model. The Central Hub controller receives information regarding the paper mill heat demand, TMP mill heat production, electricity market price, and the availability of each subsystem to perform the unit commitment.

The Markov chain integrated with a state-space diagram was employed to model the reliability and availability of the system. The proposed EH design approach (referred to in the text as the second scenario) was compared with the conventional design (first scenario) to evaluate the superiority of the proposed method. The proposed design approach indicates a 98.1% and 14.46% reduction in system load loss cost and total cost, respectively. For the best time series prediction model, different long-short-term memory (LSTM) structures have been developed and tested on the TMP heat production data. Our experiments demonstrate that all studied LSTM structures are effective at predicting TMP heat. However, Bidirectional LSTM appears to perform better than other models, with correlation coefficient and RMSE of 0.9 and 0.15, respectively.

# 9. Conclusion

This thesis is based on industrial projects with pulp & paper industries in a Nordic country. The focus of the research is on the energy-efficient development of the thermo-mechanical pulping process and the optimal integration of the TMP mill and paper machine through heat recovery by the Energy Hub concept. A new energy-saving refining optimization strategy for the TMP process was developed using advanced statistical approaches and machine learning methods. The results demonstrate that machine learning techniques could develop an accurate refining identification model. Using refining identification models to predict refining energy (such as specific energy consumption) and final pulp quality (such as freeness and fiber length) can further lead to developing a refining control and optimization strategy. According to the developed optimization strategy in this research, which incorporates ML methods and genetic optimization algorithms, total refining-specific energy consumption can be reduced by 14 %.

In the second part of the thesis, the EH concept was used to investigate the optimal energy integration of TMP mills and paper machines through heat recovery. A cost-effective and energy-efficient approach to designing and operating EH is based on integrating thermo-economic analysis, reliability & availability analysis, and load prediction. the proposed approach was implemented and evaluated for: residential applications (high-rise residential buildings-case study 1) and industrial applications (forest industry-case study 2). Using the proposed method, the average total cost of the system can be reduced by 16% for the residential application. Furthermore, the proposed method was examined in a second case study (the forest industry) to determine the optimal energy integration of TMP mills and paper machines. The suggested design approach offers a strong and economical design that isn't influenced by unsupplied penalty rates. The results demonstrate that the suggested method allows for a bigger possibility of decreasing the total cost when the penalty rates are increased. Depending on the penalty rates, the system's overall cost could decrease by 14% - 28% if the suggested design approach is put into practice. As an example, the suggested design approach enables a decrease of 98.10% in the system load loss cost and a reduction of 14.46% in the system total cost for unsupplied demand penalty rates of 10 (€/MWh).

To sum up, the main achievements of this thesis are developing new methods for cost and energy-efficient pulp and paper production. The forest industry could benefit from the developed methods and concepts in this thesis for more cost and energy-efficient production. The machine learning models for process identification models developed in this study provide accurate simulations for the thermomechanical pulping process. Advanced refining optimization strategy developed in this study could reduce thermomechanical pulping energy consumption without requiring major changes to the production process. In addition, the industrial and residential sectors could benefit from the advanced method developed in this thesis for the optimal design and operation of the Energy Hub. The developed method showed that the availability and reliability of energy system components play a crucial role in the design and operation of an optimal energy system. A cost- and energy-efficient

integration of the TMP mill and the paper machine is possible through the advanced design and operation method of the Energy Hub developed in this thesis.

In future works, other refining identification models to predict the further pulp quality variables such as strength properties: tensile strength, burst strength, and tear strength can be developed to be incorporated with the suggested advanced refining optimization strategy in this thesis. In addition, other optimization algorithms rather than Genetic algorithms could be applied and tested to develop the advanced refining optimization strategy. Minimizing the optimization processing time at each time step is vital to develop an online refining control strategy. Regarding the application of an Energy Hub in the forest industry, uncertainty and variations in electricity market price could be studied for the optimal design and operation of the EH. In addition, time-variant failure & repair rates of energy systems components could be considered (rather than constant failure & repair rates) for the accurate reliability & availability assessment.

# Appendix

## A.

$$P=\begin{bmatrix}
1-\left(\sum_{i=1}^{5}\lambda_i\right)\lambda_1 & \lambda_2 & \lambda_3 & \lambda_4 & \lambda_5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\mu_1 & 1-\left(\sum_{i=6}^{9}\lambda_i\right)0 & 0 & 0 & 0 & \lambda_6 & \lambda_7 & \lambda_8 & \lambda_9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\mu_2 & 0 & 1-\left(\sum_{i=10}^{13}\lambda_i\right)0 & 0 & 0 & \lambda_{10} & 0 & 0 & 0 & \lambda_{11} & \lambda_{12} & \lambda_{13} & 0 & 0 & 0 & 0 \\
\mu_3 & 0 & 0 & 1-\left(\sum_{i=14}^{17}\lambda_i\right)0 & 0 & 0 & \lambda_{14} & 0 & 0 & 0 & 0 & \lambda_{15} & \lambda_{16} & \lambda_{17} & 0 & 0 \\
\mu_4 & 0 & 0 & 0 & 1-\left(\sum_{i=18}^{21}\lambda_i\right)0 & 0 & 0 & \lambda_{18} & 0 & 0 & \lambda_{19} & 0 & \lambda_{20} & 0 & \lambda_{21} & 0 \\
\mu_5 & 0 & 0 & 0 & 0 & 1-\left(\sum_{i=22}^{25}\lambda_i\right)0 & 0 & 0 & \lambda_{22} & \lambda_{23} & 0 & 0 & 0 & \lambda_{24} & \lambda_{25} & 0 \\
0 & \mu_6 & \mu_{10} & 0 & 0 & 0 & 1-(\mu_6+\mu_{10})0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \mu_7 & 0 & \mu_{14} & 0 & 0 & 0 & 1-(\mu_7+\mu_{14})0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \mu_8 & 0 & 0 & \mu_{18} & 0 & 0 & 0 & 1-(\mu_8+\mu_{18})0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \mu_9 & 0 & 0 & 0 & \mu_{22} & 0 & 0 & 0 & 1-(\mu_9+\mu_{22})0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \mu_{11} & 0 & 0 & \mu_{23} & 0 & 0 & 0 & 0 & 1-(\mu_{11}+\mu_{23})0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \mu_{12} & 0 & \mu_{19} & 0 & 0 & 0 & 0 & 0 & 0 & 1-(\mu_{12}+\mu_{19})0 & 0 & 0 & 0 & 0 \\
0 & 0 & \mu_{13} & \mu_{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1-(\mu_{13}+\mu_{15})0 & 0 & 0 & 0 \\
0 & 0 & 0 & \mu_{16} & \mu_{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1-(\mu_{16}+\mu_{20})0 & 0 & 0 \\
0 & 0 & 0 & \mu_{17} & 0 & \mu_{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1-(\mu_{17}+\mu_{24})0 & 0 \\
0 & 0 & 0 & 0 & \mu_{21} & \mu_{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1-(\mu_{21}+\mu_{25}) \\
\end{bmatrix}$$

$$\alpha=[p_0,p_1,p_2,p_3,p_4,p_5,p_6,p_7,p_8,p_9,p_{10},p_{11},p_{12},p_{13},p_{14},p_{15}]$$

# B.

**Table B.1.** Optimal Energy Hub subsystem capacities for the different design approaches (reference Year: 2020 & 2021).

| $pc^{heat} = 10\ (\frac{€}{MWh})$ | Reference Year: 2021 | | Reference Year: 2020 | |
|---|---|---|---|---|
| | Capacity (MW) | | Capacity (MW) | |
| | Electric Boiler | Heat Pump | Electric Boiler | Heat Pump |
| **Case1:** Conventional design approach | 14 | 67 | 15 | 66 |
| **Case2:** Proposed design approach | 67 | 66 | 67 | 66 |
| **%** Differences w.r.t **case 1** | 378.57 % | -1.49 % | 346.66 % | -1.51 % |

**Table B.2.** System costs for different modeling approaches (reference year: 2020 & 2021).

| | Reference Year: 2021 | | | Reference Year: 2020 | | |
|---|---|---|---|---|---|---|
| $pc^{heat} = 10\ (\frac{€}{MWh})$ | Values (million €) - case 1 | Values (million €) - case 2 | % Differences w.r.t case 1 | Values (million €) - case 1 | Values (million €) - case 2 | % Differences w.r.t case 1 |
| Total capital cost | 28.2 | 33.1 | 17.37 | 27.9 | 32.7 | 17.2 |
| Load loss cost (in life span) | 104.81 | 1.84 | **-98.24** | 102.63 | 1.94 | **-98.10** |
| Energy cost (in life span) | 384.31 | 409.24 | 6.48 | 291.88 | 310 | 6.46 |
| Maintenance cost (in life span) | 50.76 | 59.58 | 17.37 | 50.22 | 58.9 | 17.2 |
| Total cost (in life span) | 568.08 | 503.76 | **-11.32** | 472.63 | 404.25 | **-14.46** |

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